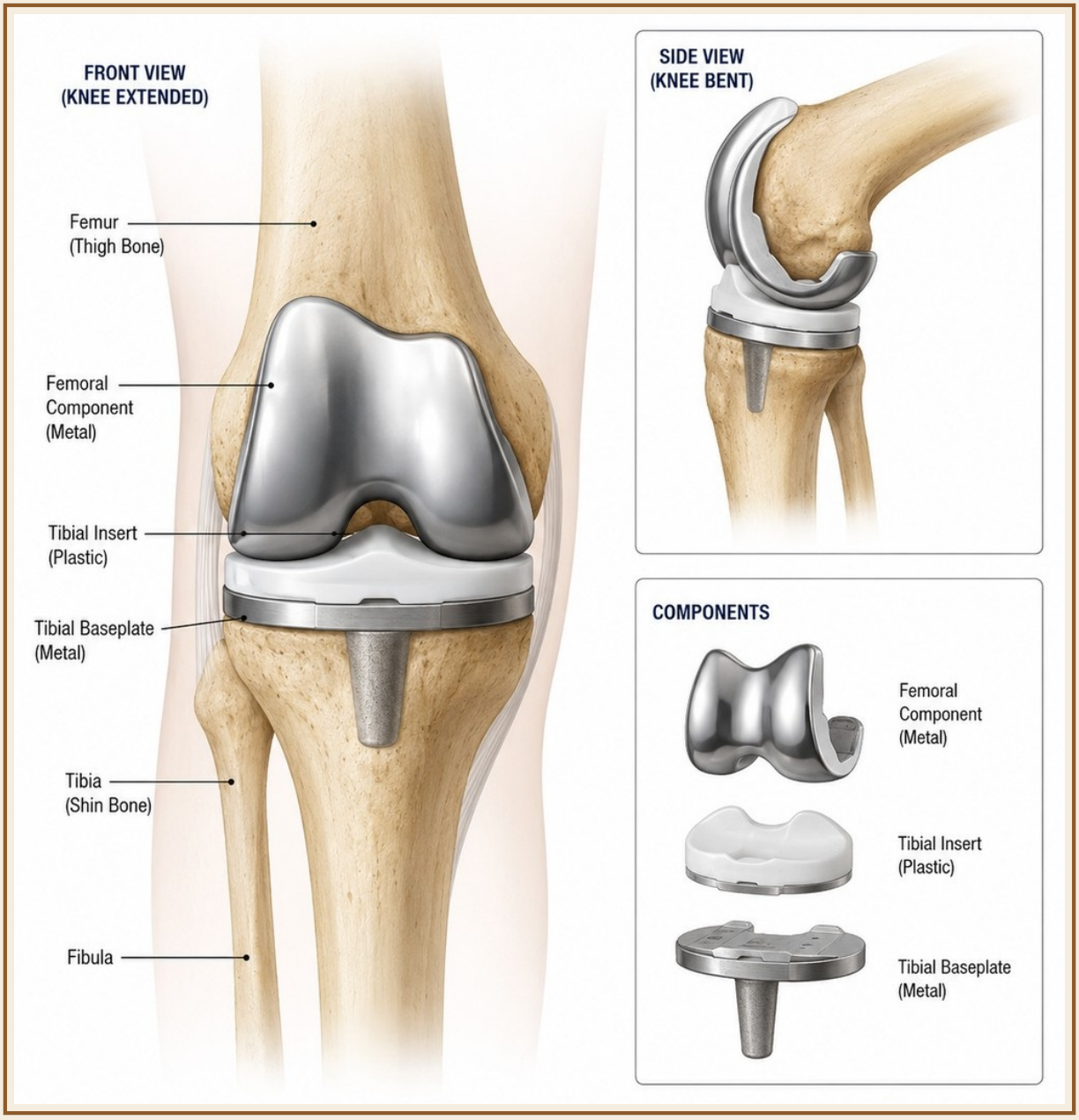
COURSE

TKA - Miscellaneous

7 lectures



15 October 2026

TRANS PRO
Developed by FAM

Lecturers

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IMPORTANT

The world is changing very fast, especially with the rapid development of Artificial Intelligence (AI). We believe we should keep up with these changes and use them to make learning easier and more accessible.

This work was created using a computer program developed by us through Vibe Coding, together with several other AI-assisted tools. Using these tools, we were able to create transcripts of lectures, including lectures delivered in Egyptian Arabic, and organize them into a more academic, book-like format.

The aim is to help the listener focus on listening without having to write notes, while having an organized transcript they can easily return to for quick revision.

This transcript is not a replacement for the original lecture or the relevant textbooks and scientific references. The accuracy is generally very good, but AI-assisted transcription can still contain errors. If you are unsure about any point, please check the original lecture or a reliable reference.

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How to use this book

- Click any line in the Contents to jump to that lecture.
- The time at the bottom right of each page (» 04:57) opens the original lecture on YouTube at that exact moment.
- "↑ Chapter" returns to the lecture's contents; "↑ Book" returns to the book's contents.
- Click any section heading to return to the lecture's contents.
- "(see Fig.)" jumps to the figure; "↑ back to text" under the figure brings you back to where it was cited.
- "Ask AI" beside each heading opens a page that turns that section into a ready question for an AI: simplify it, clarify what the recording left unclear, or check for newer guidelines.
- Orange notes such as [C1] and [Q1] are AI corrections and doubts - mostly where the recording was hard to transcribe. Each lecture lists them after its index of figures.

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Knee OA treatment options

(From Conservative Care to Arthroplasty in Knee Osteoarthritis)

Prof Dr Sherief Khaled

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 17) ACR 2019 strongly recommends against vitamin D for OA; not conditionally recommended.

Q2 (p. 21) Contradicts chapter: routine NSAIDs smallest dose ~2 weeks; maximum dose only flare-up for one week.

Index Case Presentation Ask AI

- Lecture 1 → general introduction to **osteoarthritis (OA)**
- Topic particularly confusing for those working outside Egypt
- Index case **referred by a rheumatologist**
- 31-year-old woman → referred from rheumatology for **knee replacement**
  - Left knee: advanced OA → aim **total knee replacement (TKR)**
  - Right knee: very mild OA
- Her question → what to do for the **right knee**?
- Problem posed: right knee options beyond current **rheumatology** treatment

Nature of Osteoarthritis Ask AI

- OA = **degeneration** of everything in the joint:
  - Cartilage
  - Bone
  - Synovium
- Accompanied by **secondary inflammation** → NSAIDs a major part of treatment
- **Pain** → what affects the patient's life

High Yield

- Indication for total knee replacement → **pain**

Radiological Classification Ask AI

- Many **radiological classifications** exist

Kellgren–Lawrence (KL) Ask AI

- Little reliance on classifications in **orthopaedics**; favoured by others
- On real radiographs:
  - Grade 2 vs 3 → slight difference
  - Grade 3 vs 4 → **clear difference**

Ahlbäck Classification Ask AI

Ahlbäck grade	Finding
1	Joint space narrowing (< 3 mm)
2	Obliteration of joint space
3	Minor bone attrition
4	Moderate bone attrition: 5–10 mm
5	Attrition > 10 mm

- Higher grades common in Egypt → late presentation
  - Large part of the **articular surface** already eroded
- All Ahlbäck grades → **KL grades 3–4**
- Ahlbäck merely **subdivides** the KL 3–4 range

Clinical Diagnosis of Osteoarthritis Ask AI

- **Plain X-ray** alone → not diagnostic
  - Ahlbäck type 4 radiograph ≠ need for **total knee**
- 2021 paper → diagnosis relies on multiple elements

Diagnostic Elements Ask AI

- Full **history**
- Clinical features:
  - **Morning stiffness** → short duration; shorter than in inflammatory arthritis
  - **Crepitus**
  - Evident leg **deformity**
- Radiology → last:
  - **Osteophytes**
  - **Joint space narrowing**
  - **Subchondral cysts**
  - **Subchondral sclerosis**

The X-ray Trap Ask AI

- X-ray sent on **WhatsApp** → "joint replacement" reply → never do this
- Example: father's X-ray shows OA in a joint he never complained of
  - **Incidental finding** while complaining of something else
  - Friend advises "total knee" → foolish advice
  - Unseen: gait, lifestyle, **pain severity** and disability
- Also unseen without examination:
  - Ability to perform **activities of daily living**
  - **Genu varum** or genu valgum
- Frequent at **conferences**: colleague replies by phone → "joint replacement; paperwork on Saturday"
- Unseen comorbidities, e.g.:
  - **Super-obese** → 250 kg, never weighed
  - **Diabetic**
  - **Hypertensive**
  - **Open-heart surgery** two months ago
  - **History of DVT** in that leg three times
- Patient must be seen; **history** taken personally
- X-ray → only one part of the picture
- "Diagnosis of OA" from **plain X-ray** alone → does not exist

Pitfall

- Never decide **joint replacement** from an X-ray sent by phone
- Diagnosis of OA → **clinical** together with radiological

Treatment Pyramid Ask AI

- **Management pyramid** → used throughout the world
- Abroad → few patients reach the **surgeon**
- Lower tiers → **primary care physician** / family physician
- Surgeon receives patients from the **red (upper)** tier

Primary Care Measures Ask AI

- Lower two tiers → family / **primary care physician** advises:
  - **Exercise**
  - **Weight reduction**
  - **Comfortable footwear**
  - **Topical agents**
  - **Simple analgesics** → Panadol (paracetamol) or OTC products
  - **Nutraceuticals**

Patient Pathway in Egypt Ask AI

- Long journey before reaching the **surgeon**:
  - Traditional healer → pharmacist → nearby clinic → district-town doctor (if from a village) → Cairo

OTC Products Abroad Ask AI

- Very popular in USA, Canada, Europe → all in **small doses**

Drug	Proper adult dose	OTC tablet
Ibuprofen	600 mg four times daily	200 mg
Naproxen	1000 mg	50, 150 or 200 mg
Panadol	—	e.g. 300 mg → even two tablets remain low dose

Topical Agents and Intra-articular Injections Ask AI

Topical Agents Ask AI

- **Local gels**:
  - Anti-inflammatory → e.g. **diclofenac**
  - Warming/cooling → **capsaicin**

Escalation in Egypt Ask AI

- Next → **nutraceuticals** and NSAIDs
- Simple level → NSAIDs ± **physiotherapy** (very important)

Intra-articular Corticosteroids Ask AI

- Medications failed → **intra-articular injections**
- Permitted in most **disease stages**
- Especially patients unable to take **oral medication**
  - Polypharmacy → "I already take fifty pills"
- **Corticosteroid** → injectable at any point of time
- Many **other uses** beyond the familiar ones

Hyaluronic Acid, PRP and Stem Cells Ask AI

- **Intra-articular hyaluronic acid** → not useful for everyone; not limited to the young
- Key condition for use → **joint space narrowing**
  - Narrowed joint space → **hyaluronic acid** to lubricate? No
- **KL grade 3, 4** or more → not treated by injection
- Egypt: such knees injected anyway → **goal** unclear
- Failure → **PRP**
  - PRP failure → **stem cells**
  - Analogy: letter to a dead person → no reply

Mechanical Problems Ask AI

- Address the **lateral** compartment and patellofemoral joint
- Injection alone fails → problem **mechanical**, not inflammatory
- Underlying causes:
  - **Hamstring and calf tightness**
  - **Flexion deformity**
  - **Extension lag**
  - → **patellofemoral** OA
- Injection failure → patient blames doctor
  - Real miss → unexamined **flexion deformity**
- Tell patient honestly → injection alone insufficient; **physiotherapy** needed

Clinical Pearl

- Patellofemoral OA from tight hamstrings/calf or **flexion deformity** → mechanical
- Mechanical problem → physiotherapy, not injection alone

Evidence Grading of OA Treatments Ask AI

Recommended Ask AI

- Accepted worldwide as effective in OA → **no dispute**
  - **Exercise** → aerobic or stretching
  - **Physiotherapy** → TENS, thermotherapy
  - **Paracetamol**
  - **NSAIDs** → topical or oral
  - **Walking sticks**
  - **Arthroplasty**

Inconclusive Ask AI

- **Braces** → unclear effect
- **Lateral wedge insoles**
- **Opioids** → minimised abroad
  - → placed in the inconclusive group
- **Pain patches**
- **Intra-articular corticosteroids** → effective, but last only ~a month and a half
  - Indications → **inflammatory** (rheumatological) conditions
  - Our two indications → **acute gouty arthritis**; acute flare-up
  - Also → **pes anserine bursitis**
- **Hyaluronic acid**, PRP, nutraceuticals → inconclusive in all papers up to 2025

Nutraceuticals Ask AI

- Combinations that give results:
  - **Glucosamine sulfate (pharmaceutical grade)** + chondroitin
  - Glucosamine sulfate + **risedronate** (osteoporosis drug)
- **Collagen** and oral hyaluronic acid → no proven benefit
- Other products, e.g. **Boswellia** → effect unknown
  - All → unfortunately **inconclusive**
- Papers → roughly **half and half** (results vs none)
- **Systematic reviews** → "promising" in some cases; no effect in others
- Safest approach → **one-month trial**; no relief → stop

Placebo Response Ask AI

- ~40% of patients → respond to **placebo**
  - Improve on anything, e.g. collagen → "I am better
  - … this medicine is great"
- **Non-responders** → no improvement however hard you try
- Lecturer's observation → **28 years** of practice

Not Recommended Ask AI

- **Acupuncture**
- **Arthroscopic lavage** → formerly performed
  - Prohibited abroad since ~2000 → surgeon loses **licence**
  - Egypt → performed until recently

OARSI Guidelines Ask AI

- **Osteoarthritis Research Society (OARSI)** recommendations

Core Treatments Ask AI

OARSI

- **Education**
- **Structured exercise programmes**
- **Weight management**
- **Physiotherapy**
- Core treatments → apply regardless of **patient box**
- Patients → **four boxes**, each considered separately

Four Patient Boxes Ask AI

OARSI

- **Knee OA without comorbidities** → box with the most effective options:
  - **Paracetamol**
  - **Selective COX-2 inhibitors** (oral coxibs)
  - **Oral and topical NSAIDs**
  - **Topical capsaicin**
    - In all warming ointments → e.g. Algesal, Moov, Salonpas, Deep Heat
  - **Intra-articular steroids**
  - **Duloxetine**
  - **Walking cane**

Box	Removed	Remaining options
Multi-joint OA	Cane, braces, topical agents (cannot rub whole body with diclofenac)	Same as knee-only box otherwise
With comorbidities	Oral NSAIDs, paracetamol, duloxetine	Intra-articular corticosteroids, topical NSAIDs, walking cane, biomechanical intervention

- **Multi-joint OA + multiple comorbidities** → no medical treatment either

Duloxetine and Central Sensitisation Ask AI

OARSI

- Multi-joint patients → **duloxetine**
  - Chronic pain → **cycle of depression**
- **Neuropathic pain** → central sensitisation develops
- Pain improves on a drug → brain does not **register** the improvement
- Brain keeps signalling pain → focused on that **joint**
- Needs a drug shutting down this **"pain station"** → guidelines recommend duloxetine

ACR 2019 Guidelines Ask AI

- **American College of Rheumatology (ACR)** Osteoarthritis Guidelines 2019

Agent	Hand	Knee	Hip
Oral NSAIDs	Green	Green	Green
Topical NSAIDs	Green	Green	Not recommended

- Red = **strongly recommended against**:
  - **Bisphosphonates**
  - **Glucosamine**
  - **Chondroitin**
  - **Hydroxychloroquine**
  - **Methotrexate**
  - **Intra-articular hyaluronic acid**
  - **Botulinum toxin**
- Rationale → **insurance**
  - Insurers pay only for guideline-recommended treatment

  - All red items → **not reimbursed**
- **Conditionally recommended** for patients who need them:
  - Vitamin D *[Q1 · ACR 2019 strongly recommends against vitamin D for OA; not conditionally recommended.]*
  - Opioids → specifically **tramadol**

Exam Pearl

- ACR 2019 → **oral NSAIDs** green for hand, knee and hip
- ACR red list → glucosamine, chondroitin, **intra-articular hyaluronic acid**

NSAIDs, SYSADOAs and DMOADs Ask AI

NSAIDs Ask AI

- **Oral NSAIDs** → effective for all patients → first line
- Typical prescription → NSAIDs (topical or oral) + **nutraceutical** / chondroprotective / SYSADOA

SYSADOAs Ask AI

- **SYSADOA** = symptomatic slow-acting drugs for osteoarthritis
- Members:
  - **Collagen**
  - **Glucosamine**
  - **Hyaluronic acid**
  - **Boswellia**
  - **Avocado–soybean unsaponifiables**

NSAID Duration Rule Ask AI

- **Smallest possible dose for the shortest possible duration** → ~two weeks
- Repeating beyond two weeks → **not acceptable**
  - → something else needed: **physiotherapy**, injection, or another measure
- Improvement 60–70% after two weeks → another **two weeks** allowed
  - **Change NSAID family** → e.g. diclofenac → coxibs
- Main problem → **stomach**
  - → most prefer a **selective COX-2 inhibitor**

Mild to Moderate OA Ask AI

- Mainstay → **NSAIDs**
- Plus **topical NSAIDs**
- Plus **SYSADOAs**

DMOADs Ask AI

- **DMOADs** = disease-modifying osteoarthritis drugs
- Some books → glucosamine etc. classed as DMOADs → **disease modifying**

Exam Pearl

- NSAIDs → **smallest possible dose for the shortest possible duration**
- Duration → ~two weeks; repeat only if 60–70% improved
- Switch **NSAID family** on repeat course

Index Case Resolution and Age Limits Ask AI

- More advanced cases → **combination** of treatments:
  - NSAIDs + topical + DMOADs + SYSADOAs + physiotherapy
- → answer for her **right leg**
- Left leg → **rheumatologist** already referred her for it

Age Limits for Replacement Ask AI

- **Age < 50** → very wrong management
  - Exception → **rheumatological** cases (next lecture)
- **Above 90** → also considered wrong management
- Decision → many factors, e.g. patient > 90 who:
  - Walks two hours daily
  - Works for an hour
  - Runs errands
  - Lives alone
  - Drives a car
  - → **operate**
- Example in coming lectures' cases → patient in the **eighties**

Acute Flare-up of Osteoarthritis Ask AI

- **Acute flare-up of osteoarthritis** → seen in most clinics
- Patient circulates first → pharmacy → traditional healer → **dispensary** → you

Pain Pattern Ask AI

Flare-up

- OA pain → goes **up and down**

VAS (Visual Analogue Scale)	Patient behaviour
0–2	Manages on his own
3–4	Starts medication or asks someone
6–7	Comes to your clinic

Patient	Flare-up frequency
Elderly	~once a year
Somewhat younger	Once every five years
Forties	Once every ten years

- Trigger → often identifiable, e.g. **exertion**
  - Moving house two months ago → place with **stairs**
  - Living on sixth floor → frequent **stair climbing**

KOFUS Score Ask AI

Flare-up

- **KOFUS** = Knee Osteoarthritis Flare-Up Score
- **Maximum score** → 14

Item	Points	Assessment
Morning stiffness	1	Leg stiff in morning, loosens with movement
Nocturnal awakening	2	Ask whether knee pain wakes him at night
Effusion	2	Usually suprapatellar
Limp	3	Watch patient walk in
Swelling	3	Look at the leg
Warmth	3	Hand on knee; compare with other knee for temperature difference

- Score > 7 out of 14 → **acute flare-up**

Must-Know

- Acute flare-up → diagnosed by **KOFUS**
- KOFUS > 7 out of 14 → **acute flare-up**
- Limp, swelling, warmth → 3 points each

Treatment Ask AI

Flare-up

- Same drugs → **different doses**
- **Celecoxib** → normally 200 mg
  - First dose → two capsules together (400 mg) on the **first night**
  - Patient does not leave your clinic without a dose
  - From second day → **twice daily**
  - **Maximum dose** 400 mg → hence two capsules at once at night

Drug	Flare-up dose	Note
Diclofenac	150 mg	Management dose; long-acting (as is celecoxib)
Etoricoxib	60 mg (American guidance)	Europe: 120 mg in acute flare-up
Naproxen	1000 mg	Europe and America alike
Ibuprofen	2400 mg	600 mg four times daily or 800 mg three times daily

- Difference from routine treatment → **higher NSAID dose**, only one week
- After one week → reduce to **safe doses**
  - Flare-up doses very high → **maximum dose** of everything
- Step-down after one week of diclofenac 150 mg:
  - Not 150 mg for two weeks straight
  - → **75 mg twice daily**
  - Or → switch to **coxibs**
- Shorter NSAID duration → fewer **complications**

	Routine treatment	Acute flare-up
Dose	Smallest possible	Maximum dose
Duration	~two weeks (+ two if 60–70% improved)	One week → then safe doses

Index Case: Left Total Knee Replacement Ask AI

- Left side → **total knee**
- **Valgus**, not marked → standard approach

Operative Details Ask AI

Index case

- **Standard bone cuts**
- **Cystic defect** medially
  - Rheumatoid patients → cystic area sometimes found
  - Filled with bone from the **femoral condyle cuts**
- **Osteoporosis** visible beneath the tibial tray
- **Periarticular local infiltration** cocktail:
  - **Marcaine** (bupivacaine)
  - **Xylocaine** (lidocaine)
  - **Dexamethasone**
  - **Ketorolac**
  - **Saline**
- **Morphine** formerly added → acts centrally → no value locally → removed

Follow-up Ask AI

Index case

- Images → three months and two years
- Two years → **bone quality** beneath the joint back to normal
  - Normal weight bearing → **localized osteoporosis** of first image gone

Summary Ask AI

- OA = **degeneration and repair** with secondary inflammation
- Diagnosis → **clinical together with radiological**; never X-rays alone
- Images sent by phone → "total knee" reply = **wrong**; patient seen first
- Medical treatment → NSAIDs at **maximum dose** for two weeks → further two weeks *[Q2 · Contradicts chapter: routine NSAIDs smallest dose ~2 weeks; maximum dose only flare-up for one week.]*
  - Three months → ineffective and wrong
- **Patellofemoral** involvement + flexion deformity:
  - → **physiotherapy**; injections give no result
  - Some problems → may need **surgery**
- **SYSADOAs** and DMOADs → controversial
- **COX-2 inhibitors** → best GIT tolerance → used
- Acute flare-up → diagnosed by **KOFUS**
- Advanced cases with **persistent pain** → arthroplasty
- Indication for arthroplasty → **pain**

Biomechanics of TKR

(Soft-Tissue Balance, Kinematics and Implant Design)

Prof Dr Mahmoud Abd Elkarim

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 32) Lecturer gave no direction; meta-analyses show PS knees, not CR, gain slightly more flexion (~2-4°).

Biomechanics in TKA Ask AI

- Biomechanics and **total knee arthroplasty (TKA)** → closely linked
  - Many surgeons able to perform TKA
  - Complication → surgeon who knows **underlying anatomy** and biomechanics → corrects himself
- First rule: TKA = **soft-tissue procedure**, not a bony-cuts procedure
  - Re-emphasised in a later lecture of this meeting
  - Not an easy task
  - Opening the company instrument set ≠ guaranteed good result

The "6, 3, 7" Numbers and Technical Goals Ask AI

- "6, 3, 7" → **take-home message** and objectives of TKA
- Name precisely: **distal femoral valgus angle**, not "valgus angle"
  - Precise name → examiner satisfied, no further pressing
  - Vague name → answer level drops
- Numbers relate to femur and tibia → ordered systematically, proximal → distal

Number	Meaning
6	Distal femoral valgus angle
3	External rotation of femoral component
7	Posterior tibial slope

- Numbers chosen because each has **variations**
  - Distal femoral valgus angle → varies
  - 7 → also varies
- Many other numbers exist; these are the essential ones

High Yield
- 6 → distal femoral valgus angle
- 3 → external rotation of femoral component
- 7 → posterior tibial slope

Objectives Ask AI

TKA

- **Kinematics**
- Anatomical and mechanical axes
- **Patellofemoral joint**
- Further points + biomechanics → next lecture, Dr. Samir
- **PCL-retaining** vs PCL-substituting designs → also addressed
- TKA → one of the most successful surgeries
  - Best results at 10–15 years, if performed correctly

Technical Goals Ask AI

TKA

- Understood and applied intra-op, not memorised
- **Restore mechanical alignment**
- **Joint line** → neither elevate nor lower
  - Varus vs valgus knee handling → later lecture
- **Restore Q angle** → patellofemoral joint and patellar tracking
  - Applies even without patellar resurfacing
  - Patella → integral part of knee → affected by every step

Preoperative Planning and Imaging Ask AI

History and Patient Selection Ask AI

- Proper **history** → every patient
- From the first word → can I deal with this patient?
- "Perfectly well" patients → may still end with problems
- Never-satisfied patient → encountered by all
  - Problems often from **poor selection**, not surgical skill alone
- Assess and study the patient well before operating

Imaging Ask AI

- **Full-length AP view**, hip → ankle
  - Europe / developed countries → all patients
  - Locally → rarely routine
- Mandatory indications:
  - Suspected **deformity**
  - **Very short** → 152 cm or less (dwarfs)
  - **Very tall** → taller than about 190 cm (~6 ft 3 in)
- Very short / very tall → standard **6° valgus** rule no longer applies
- Differ in **femoral entry point** and canal → different valgus degree

Patient	Height	Distal femoral valgus
Normal	—	5–7° (6° middle)
Very tall	> ~190 cm	Reduce → 4–5°
Very short	≤ 152 cm	Increase → 7–8°

- Investigations → each with a purpose:
  - **Standing films** → weight-bearing views
  - **Stress views**
    - Varus and valgus stress views
  - **Lateral views** → in extension and in flexion
  - **Long cassette** films → very important
- Purpose → **template component size**
  - e.g. looks too large → smaller size
- Discuss investigations with patient → involve patient in decisions

Must-Know

- Full-length AP film → suspected deformity
- Full-length AP film → very short patients, 152 cm or less
- Full-length AP film → very tall patients, above ~190 cm

Kinematics of the Knee Ask AI

- Knee → **3 degrees of freedom**
  - **Flexion–extension** → main movement
  - Rotation
  - Abduction–adduction
- **Tri-axial motion**, not uniplanar

Mechanical and Anatomical Axes Ask AI

Mechanical Axis Ask AI

Limb

- Three points:
  - **Geometric center of femoral head**
  - Center of knee joint
  - Center of ankle joint
- Line through these points → **mechanical axis** of lower limb
- Mechanically neutral → passes through knee center
- Full-standing X-ray:
  - Lateral to knee center → **mechanically valgus** alignment
  - Medial to knee center → **mechanically varus** alignment
- Preferred over anatomical axis → **weight-bearing line**
  - Represents load transmitted through lower limb
  - Femoral head center → ankle center, through knee center
- Equal load distribution → neutral knee; deviation → varus or valgus

Anatomical Axes Ask AI

- Two: **femoral anatomical axis** and tibial anatomical axis
  - Differ from mechanical axis → as shown below
- Femoral anatomical axis:
  - Runs along **middle of shaft**
  - Two cortical points at two levels → bisect each level → join midpoints
  - Joined bisectors → **line of symmetry** of shaft
- Femoral mechanical axis → femoral head center → knee center
- Angle between the two → next section

Distal Femoral Valgus Angle — "6" Ask AI

- Approximately 6°
  - Never an absolute figure → **mean and standard deviation**

  - Range 5–7°, middle 6°
- First cut in TKA (femur-first) → **distal femoral valgus cut**, 5–7° by height

Exam Pearl
- First cut in TKA (femur-first) → distal femoral valgus cut
- 5–7°, depending on patient height

- Angle between **anatomical axis of femur (AAF)** and mechanical axis → varies with height
- Tall → longer femur → lines further apart → **smaller** angle
- Short → angle **increases**
- Apply when approaching the cut, even before stem decision
  - Tall → 4–5°
  - Short → 7–8°

Tibial Axes and Bone Cuts Ask AI

Tibial Axes Ask AI

- **Tibial anatomical axis** → two midpoints → intramedullary line
- Tibial vs mechanical axis → **normally coincide**
  - Childhood **malunited fracture** → do not coincide
  - **Congenital tibial bowing** → do not coincide
  - Post-traumatic, other deformities → differ
- Tibial mechanical axis → knee center → ankle center

Tibial Alignment Methods Ask AI

- **Intramedullary** → rod inside tibia guides the cut
- **Extramedullary** → identify ankle center and knee center
  - **Extramedullary alignment rod** → placed externally
  - Proximal → **tibial tuberosity** in thirds → junction medial □ / lateral □
  - Distal → ankle center → toward **second metatarsal**
    - Alternative → space between 2nd and 3rd metatarsals; 2nd metatarsal used here
- Foot alone → not reliable; foot may itself be deformed
  - Ankle center → bisect medial–lateral **malleoli** → move 6–7 mm medially
- Tibial tuberosity → **one of the most consistent lines**, with experience
- Abnormalities common → try all, use several reference lines together

Bone Cuts Ask AI

Cuts

Bone	Cut	Reason
Femur	6°	Anatomical ≠ mechanical axis
Tibia	90° proximal cut	Anatomical = mechanical axis

- Normal situation
- Knee flexed:
  - Tibia natively in **3° varus**, yet cut at 90°
  - Compensation → **medial femoral condyle** projects more distal (and posterior) than lateral
  - Result → **trapezoidal flexion space**, not rectangular
  - Goal → **rectangular** flexion space
- Balancing the medial condylar excess → second number

Femoral Component External Rotation — "3" Ask AI

- **External rotation of femoral component** → compensates 3° tibial varus
- Referenced to **posterior femoral condyles**, not tibia
  - Medial femoral condyle → naturally larger
- Posterior condylar line vs horizontal → 3°
- 3° rotation → **rectangular flexion space**
- Direction → upward = **external**, not internal
- 3° external rotation → position of femoral component
- Cutting block → "3" settings built in → still understand the principle
- **Posterior referencing**:
  - Very important for balanced cuts
  - More bone removed from **medial posterior condyle** than lateral, by those degrees

Posterior Tibial Slope — "7" Ask AI

- **Proximal coronal cut** of tibia, viewed from side → 0°, 3° or 7° posterior
- Most companies → slope built into **tibial cutting jig** (workshop)
- Benefit → better flexion
- **Anterior slope** → worst error
  - Anterior tibial bone quality → weaker
  - Implant **sinking** → possible failure
- Always neutral or posterior, per company

Company	Posterior slope
Johnson & Johnson (DePuy)	3° or 5°
Zimmer	7°

- Overdoing the slope:
  - Cutting from posterior → unintended slope > 7°
  - Posterior breach → **PCL** insertion compromised

  - **Flexion gap ↑**
- Knee "closed well, stable" at the end ≠ success
  - Later → review yourself and instruments ("jig misled me")
  - **Every single step** → pays off in final result

Pitfall

- Never make an anterior tibial slope → weak anterior bone → implant sinking
- Excess posterior slope → PCL insertion compromised
- Excess posterior slope → flexion gap ↑

Summary of the Numbers Ask AI

6-3-7

- Relative to **mechanical axis** → distal femoral valgus cut 5–7°
- 3° external rotation → compensates tibial varus → aligns with **transepicondylar axis**
- 7° → posterior slope, with proximal coronal tibial cut

Rotational Axes of the Femur Ask AI

- Four axes → must be known very well in TKA

Posterior Condylar Axis Ask AI

- Line touching back of **lateral and medial femoral condyles**
- Medial condyle → projects more distal (posterior) than lateral
- Compared with neutral line → **transepicondylar axis**
- Angle between them → 3° (audience estimate ~2°)
  - Posterior referencing → component **internally rotated** relative to lateral condyle
  - Goal → neutral, **parallel** to transepicondylar axis
  - Rotation needed → 3°

Transepicondylar Axis Ask AI

- Main reference → **neutral alignment**
- Better than posterior condylar axis:
  - **Epicondyles** → palpable by hand
  - Deformity → posterior condyles may be abnormal
  - Valgus knee, other condylar disease → landmarks not preserved → posterior condylar axis unreliable
  - Valgus, **rheumatoid** or severely affected knee → mark transepicondylar line (pen)
  - Condyles not intact → preferred over posterior condylar axis

Whiteside Line Ask AI

AP axis

- Through **trochlear groove** (patellofemoral sulcus) + **intercondylar notch** behind it
- Identified and seen intra-op
- Perpendicular → almost identical to **transepicondylar axis**
- Angle usually 90–91°, but **inter-individual variance** wide
  - Whiteside alone → deviation up to 10° above or below 90°
- Take-home → **use all axes together**:

1. Transepicondylar axis
2. Whiteside line, even as imaginary line
3. Posterior condylar axis → only if condyles intact
    - **Lateral femoral condyle hypoplasia** → lateral referencing unreliable

- Whiteside line = **AP axis** → perpendicular to transepicondylar axis

Cut Surface of the Tibia Ask AI

- Fourth axis → some avoid relying on it
- Tibia cut first at 90° → all subsequent cuts built on it
  - Right or wrong → requires skilled **balancing**
- Last reference → wrong tibial cut → everything built on it wrong
- Used more with **LCS (Low Contact Stress)** prosthesis, mobile-bearing, J&J → specific tools
- Relying on tibial cut surface → **very well-balanced knee** required
  - Otherwise → all cuts deviated from normal

Axis	Definition	Limitation
Transepicondylar	Line through epicondyles; neutral reference	— palpable, preferred
AP axis (Whiteside)	Trochlear groove → intercondylar notch	Perpendicular varies up to ±10°
Posterior condylar	Line touching both posterior condyles	Unreliable if condyles abnormal / hypoplastic
Cut surface of tibia	Proximal tibial cut at 90°	Wrong cut or poor balancing → all cuts wrong

Surgical Techniques Ask AI

- Depends on **ligament balancing** and flexion–extension gap technique; trials (workshop)
- Two techniques → practised in workshop

	Measured resection	Flexion–extension gap
Sequence	Femur → tibia	Tibial cut first → build on it
Method	Reference lines → axis → trial	Balance flexion gap with extension gap
Femoral resection	By reference lines	By amounts removed
First step	Soft-tissue balancing	Soft-tissue balancing

Choosing a Technique Ask AI

- Each school → pros and cons
- Early starters / beginners → **measured resection**
- More experienced → flexion–extension gap
- Either without **soft-tissue balancing** → knee fails early (~3 months)
- Before expertise → start with the easy option

> **Clinical Pearl**
> • Beginners → measured resection technique
> • Experienced → flexion–extension gap technique
> • No soft-tissue balancing → failure, whatever the technique

Design Choice Ask AI

PCL

- Most surgeons → **PCL-substituting** knees
  - Despite evidence → ~4° difference favouring retaining; possibly not statistically significant *[Q1 · Lecturer gave no direction; meta-analyses show PS knees, not CR, gain slightly more flexion (~2-4°).]*
  - Reason → **balancing** in retaining mode much harder
  - Early career → avoid mistakes → substituting design
- PCL-retaining design → Dr. Samir's lecture
- Retaining → the difficult option → requires tension balancing

Navigation Ask AI

- Another method of determining the axes
- Not widely used locally at present
- During cuts → feedback on hand alignment
- Wrong reference **registration** at start → everything wrong

Q Angle and Component Positioning Ask AI

Q Angle Ask AI

- **ASIS** → patella center → quadriceps / extensor mechanism line
- **Patellar tendon** → patella → tibial tuberosity
- **Abnormal patellar tracking** → most common, overlooked complication
- Any Q angle ↑ → risk of **patellar dislocation**
- Component positioning → key **exam answer**

Femoral Component Ask AI

Femoral

- Rotation → **external**
- Internal rotation → patella pulled → Q angle ↑ → dislocation → **patellar maltracking**
- **No-thumb test** → required
  - Surgeons hold patella in place by hand while testing
  - = cheating → remove hand → patella dislocates
- Steps built on one another:
  - External rotation → rectangular flexion space + **centrally tracking patella**
  - Internal rotation → patellar problems

> **Pitfall**
> • Holding the patella by hand during trial → masks maltracking
> • Perform the no-thumb test → hand removed

Tibial Component Ask AI

Tibial

- Rotation → **external**, like femur
- Avoid internal rotation → careful positioning
- Trial marks → workshop
- Align → **intermalleolar technique** or tibial tuberosity medial □ / lateral □ junction
- **Lateralise** → on lateral cortex of tibial cut surface → adjust component
- Internal rotation → **tibial tuberosity** relatively lateral
  - → Q angle ↑
  - → **lateral driving force** on patella → patellofemoral problems
- Tibia → central or external; external better than internal
- Normal gait → feet in external, not internal, rotation (clinical illustration)

Mediolateral and Vertical Position Ask AI

- Everything lateral, except **patella** → medial
- Femoral component → central
  - Between sizes → lateral → avoids ↑ **valgus directing force**
- Tibial component → **lateralise** (workshop)
- Patella → only medial component
  - Counteracts **valgus directing force** → prevents Q angle ↑ → prevents maltracking
- Patella height → superior or inferior?
  - **Central**; may be superior, never inferior

Component	Rotation	Mediolateral	Vertical
Femoral	External; never internal	Central; lateral if between sizes	—
Tibial	Central or external; never internal	Lateralise	—
Patella	—	Medial (only medial component)	Central or superior; not inferior

- Further detail → Dr. Samir

Role of the PCL Ask AI

- PCL-substituting vs PCL-retaining → Dr. Samir
- Normal knee on flexion → **not pure rolling**
  - Contact points → move progressively backwards
  - = **combined rolling and gliding**
  - Without it → deep flexion → femur rolls off back of tibia → **impingement** → failure
- Controlled by the **PCL**
  - Resected (substituting) → function substituted by implant
  - Retained without proper balancing → problem on flexion
- **Femoral rollback** → femur moves backwards on flexion, PCL-controlled
  - PCL → holds femur and tibia posteriorly
- Retaining without very good balancing or correct **flexion space**:
  - PCL → tight posterior structure → tightens
  - Every flexion → **stresses** underneath implants
  - Over time → component **lift-off** → seesaw rocking
  - → **loosening** ↑
- Retaining knees → not wrong
  - Requires correct **balancing**
  - Retaining → harder than substituting (sacrificing) knee

	PCL-retaining	PCL-substituting
PCL function	Native PCL → femoral rollback	Substituted by implant
Balancing	Very good balancing mandatory → much harder	Easier
Poor balancing	Tight PCL → stresses → lift-off → loosening	—
Use	Evidence ~4° difference in favour	Most surgeons; early career

Final Summary Ask AI

- **6** → distal femoral valgus angle, within 5–7°
- **3** → external rotation of femoral component, relative to posterior femoral condyles
  - Compensates 3° tibial varus → tibia cut at 90°
- **7** → posterior slope up to 7°, per company, built into jig
  - **Never anterior slope**
- **Patellofemoral tracking** → femoral component external rotation
  - Tibial component → central or external rotation
- **Never internal rotation**
- Femoral component → **lateralise**
- **Patella** → only medial (or central) component → superior, not inferior
- Testing → **no-thumb test**
- Correct result → **balancing** mandatory → topic of next two lectures

Alignment and Preoperative Templating in Total Knee Arthroplasty

(Mechanical Alignment, Gap Balancing and Radiographic Planning)

Prof Dr Mahmoud Abd Elkarim

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 41) "height about 191 or 193 degrees" → "height about 191 or 193 cm"

Queried - kept as spoken, but looks wrong

Q1 (p. 62) as spoken; 2 minutes saved per operation gives 200, not 400, minutes per 100 operations

Technical Goals of TKA and the 6–3–7 Code Ask AI

Technical Goals Ask AI

- **Templating** and planning → require understanding of alignment in TKA
- Properly functioning TKA → defined technical goals
- Technical goals
  - Restore **mechanical alignment**
  - Restore alignment of components (cut surfaces)
  - **Ligament balancing** in flexion and extension
  - Proper **patellofemoral tracking**
  - **Joint line** restoration
- TKA ≠ bony cuts alone → ligament balancing among most important end-tasks
- **Patellofemoral joint** → often forgotten, especially without patellar resurfacing
- Attention to all components → knee functions well as a whole
- Alignment types
  - **Coronal**
  - **Sagittal**
  - **Rotational**
- **Learning outcomes**
  - Coronal, sagittal and rotational alignment
  - Patellofemoral joint tracking
  - Joint line

The 6–3–7 Code Ask AI

- Mnemonic code (also in biomechanics lecture) → **6–3–7**
- Six, three, seven → each number explained in sequence

Coronal Alignment Ask AI

Mechanical Axis of the Lower Limb Ask AI

- Simplified for TKA; deformity-correction complexities excluded
- Two axes → **mechanical** and anatomical
- **Mechanical axis** → line on standing long-leg AP radiograph
  - Centre of femoral head → centre of talar dome
- Ankle centre → usually taken as **talar dome**
- Inclined toward midline vs vertical axis → $3° \pm 1.5°$ on average
- Ideal → passes through centre of knee joint → **neutral mechanical axis**
  - Medial to knee centre → **varus mechanical alignment**
  - Lateral to knee centre → **valgus mechanical alignment**
- "Centre of the knee joint" → defined in planning section

Femoral Mechanical and Anatomical Axes Ask AI

Femur

- Knee → two bones + **patella**
- Femur → mechanical axis ≠ anatomical axis; tibia differs
- **Femoral mechanical axis** → centre of femoral head → centre of knee
- TKA → series of **bony cuts**
- First cut (unless tibia-first) → **distal femoral valgus cut**
- Valgus set at 5, 6 or 7° → basis below
- Distal femoral valgus cut → **perpendicular** to femoral mechanical axis
  - → even mechanical load distribution across knee and implant
  - → improved implant **survivorship**
- Mechanical axis (orange line) → cut perpendicular to it
- **Anatomical axis of femur** (green line) → axis of medullary canal
  - Two pairs of points across canal, upper and lower
  - Pairs about 10 cm apart → bisect each → line through midpoints
  - Trajectory extended → centre of knee at distal femur
- Anatomical axis ≠ mechanical axis → **angle** between them

Femoral Valgus Angle (the "6") Ask AI

- Angle between femoral mechanical and anatomical axes → 5–7°
  - Average 6 → the "6" of **6–3–7**
- Termed **valgus angle**
- Intraoperative axis
  - Option → marker over femoral head before anaesthesia (e.g., **diathermy lead**)
  - Usual → **intramedullary rod**
- Intramedullary rod/nail in canal → runs along **anatomical axis**
- Cut perpendicular to mechanical axis → compensate for inter-axis angle
  - Cut at calculated angle (6, 7, 8 or 4°) → perpendicular to mechanical axis
- Anatomical axis extension at distal femur → determines **entry point** of femoral IM guide rod

Femoral Entry Point Ask AI

Femur

- Common practice → midline, 0.5–1 cm in front of **notch** → not correct in all cases
- **Ideal entry point** → femoral anatomical axis intersecting distal femoral articular surface
  - Depends on variable angle and axis trajectory
- Wrong foundation → wrong result → entry point critical from start
- Analogy → **guide wire** in DHS
  - Perfect placement at start → whole operation and measurements follow

Variability of the Valgus Angle Ask AI

- Average 5–7° → **not fixed**

Patient	Height (books)	Valgus angle	Cut
Very tall	~191 cm or 193 cm	Lower	4–5° instead of 6–7° *[C1 · transcript: "height about 191 or 193 degrees"]*
Short / dwarf	~152 cm (5 feet)	Higher	Up to 8° or more

- Note whether patient very tall or **short**

Bony Cuts and Gaps Ask AI

- Depth of distal femoral or tibial cut → no effect on component **rotation**
- These two cuts → affect the **gaps**
  - Distal femoral valgus cut → **extension gap**
  - Tibial cut → **both** flexion and extension gaps
- Up to this point → rotation irrelevant
- Further detail → **soft tissue balancing**, surgical technique and deformity lectures

Valgus Cut According to Deformity Ask AI

Insall

- Valgus knee → **lesser valgus cut**, not 6–7°
- Planning → measure angle; **extremes of angles** → implant position problems

Knee (Insall)	Distal femoral valgus cut
Non-deformed	6° (middle course; lecturer's preference)
Varus	7°
Valgus	5°
Obese	5°

- Obese → 5° → avoid **knees rubbing together**
- Scientific name for exam → **distal femoral valgus** cut, average 6°

Exam Pearl

- Distal femoral valgus cut → perpendicular to femoral mechanical axis
- Femoral valgus angle → 5–7°, average 6°
- Insall → non-deformed 6°, varus 7°, valgus 5°, obese 5°
- Very tall → 4–5°; short → up to 8° or more

Tibial Mechanical and Anatomical Axes Ask AI

Tibia

- Tibia → mechanical and anatomical axes distinct but **parallel**
  - Almost coincident → lie next to or exactly on each other
- **Tibial mechanical axis** → centre of proximal tibia → centre of ankle
- Tibial cut → **perpendicular** to mechanical axis
  - → even mechanical load distribution on implants
- **Tibial anatomical axis** → bisects medullary canal → determines entry point
  - Intramedullary entry point → for **intramedullary technique** on tibial side
- Two axes → coincident, separate but parallel
- Considered one → used interchangeably
  - No angle between them → except with **deformity**
- **Tibial cut angle** → perpendicular to mechanical axis → 0°

Extra-Articular Tibial Deformity Ask AI

- Axes **divergent** → extra-articular deformity
  - Mechanical axis → centre of proximal tibia → centre of ankle
  - Anatomical axis (green) → drawn by bisecting points
    - Split into proximal and distal parts
- **Very careful analysis** → correction so cut ends perpendicular to mechanical axis
  - Beyond scope → separate lecture on extra-articular deformities
- Tibia → mechanical axis coincident and parallel with anatomical → cut perpendicular to **both**
  - Only exception → **extra-articular** or tibial deformity

	Femur	Tibia
Mechanical axis	Femoral head centre → knee centre	Proximal tibia centre → ankle centre
Anatomical axis	Medullary canal axis	Bisects medullary canal
Relation	Valgus angle 5–7°	Parallel, almost coincident
Cut	Perpendicular to mechanical → valgus cut	Perpendicular to both → 0°
Anatomical axis role	Entry point of IM rod	Entry point (IM technique)

Rotational Alignment: Rationale and Gaps Ask AI

Orientation of the Figure Ask AI

- Second alignment → **rotational**
- Common figure → knee viewed from below; flexed or extended?
- Knee in figure → **flexed**
  - Tibia; posterior (not distal) aspect of femur; anterior distal femur

Normal Proximal Tibial Varus (3°) Ask AI

- Proximal tibial articular surface → not 0°
  - Normally 3° **varus** relative to mechanical axis
- Natural compensation → **medial femoral condyle** projects more
  - Posteriorly in flexion; distally in extension

Posterior Condylar Axis (3°) Ask AI

PCA

- Flexed knee → medial femoral condyle projects more **posteriorly** than lateral
- Tangent line → **posterior condylar axis** (PCA; posterior femoral condylar axis)
  - Slightly inclined → 3° to horizontal
- Tibia 3° varus; PCA 3° to horizontal
- Tibial cut → does not follow native 3°
- After distal femoral cut → **rotational alignment** of component
  - Knee in flexion → component rotation judged against reference axes

Flexion and Extension Gaps Ask AI

- **Extension gap** → knee extended
- **Flexion gap** → knee flexed
- Both → symmetric and equal
  - Mismatch → tightness or **instability**
- Ideal TKA → gaps equal, symmetric → **rectangular**
  - One side wider or narrower → **trapezoidal**
- Rectangular, symmetric gap in flexion and extension → **basis of cuts**

Three Degrees of External Rotation (the "3") Ask AI

- "3°" requested from company representative
- = **external rotation** of femoral component
  - Relative to **PCA**
- Reason → posterior medial femoral condyle projects more posteriorly (distally) than lateral
- Tibial cut at 90° + femoral cut parallel to PCA → 3° inclination → **trapezoidal** flexion gap
- 3° ER relative to PCA → femoral cut horizontal, parallel to tibia
  - → **rectangular** flexion gap, equal and symmetric to extension gap

High Yield

- Proximal tibial articular surface → 3° varus to mechanical axis
- PCA → 3° to horizontal (posterior medial condyle projects more)
- Femoral component → 3° ER relative to PCA
- 3° ER → rectangular flexion gap equal to extension gap

Femoral Rotational Reference Axes Ask AI

Four Reference Axes Ask AI

- Femoral rotation references
  - **Trans-epicondylar axis** (TEA)
  - Posterior condylar axis
  - Anteroposterior axis → **Whiteside's line**
  - **Cut surface of proximal tibia** → cuts parallel, after extension soft tissue balancing

Trans-Epicondylar Axis Ask AI

TEA

- **Most commonly used** reference
- Palpate medial and lateral femoral epicondyles → most prominent points → TEA
- Slight **external rotation** relative to posterior condyles
  - About 3°
- Cut parallel to TEA → desired **rotation** achieved
- Identified intraoperatively → line drawn on cut end of femur
  - Most prominent point of medial → lateral **epicondyle**

Anteroposterior Axis (Whiteside's Line) Ask AI

- Bottom of **trochlear sulcus** (groove) → top of intercondylar notch
- Drawn intraoperatively → palpate sulcus and **notch**, connect
- Nearly **perpendicular** to TEA → about 91°
  - Line perpendicular to AP line → parallel to **epicondylar axis**
  - → rotation reference for femoral component
- Average 91° → range 80–102°
- **No single reference** 100% accurate → use several together
  - Every TKA → draw TEA, AP line and its perpendicular, plus PCA → compare

Posterior Condylar Axis and Referencing Jig Ask AI

PCA

- PCA → 3° **internal rotation** relative to horizontal
- TEA → 3° external rotation relative to PCA
- Cause → posterior medial condyle projects more
- Target → 3° ER to PCA
  - → aligned with horizontal, TEA, perpendicular to **Whiteside's line**
- **Posterior condylar angle** → about 3°
- Templating points → reproduced intraoperatively (as in total hip)
- **Posterior referencing jig** (company device) → determines rotation
  - Two foot pieces → on posterior condyles
  - **Zimmer NexGen** jig → marked 0 and 3 → pins fixed at 3° to compensate
  - **Anterior stylus** → anterior femur, anterolateral or midpoint
    - About 2 cm proximal to **trochlear groove**
- PCA → used intraoperatively in most cases
  - Another jig type → explained later

Variability of the Posterior Condylar Angle Ask AI

- Not always exactly 3°
- Usually 3° → **not in all cases**
- **Hypoplasia of lateral femoral condyle** (valgus, rheumatoid) → angle up to 10°
- Rheumatoid patient → 3° not fixed → great caution
  - Hypoplasia → may be both distal and **posterior**
- Posterior referencing alone + fixed 3° → **big error**
- Angle varies with → rheumatoid hypoplasia, any **defect**
- Varus knee → same issue → posterior **medial femoral condyle** more eroded
  - Foot pieces placed posteriorly → **more external rotation**
- Watch valgus/lateral distal hypoplasia AND varus with posteromedial erosion
  - PCA alone → unintended more ER of component
- More ER → **better tolerated** than more IR → avoid both
- Valgus → eroded **lateral femoral condyle** unnoticed → more IR of femoral component → alignment problems

Knee	Eroded condyle	Error with PCA referencing
Valgus / rheumatoid	Lateral (hypoplasia)	More internal rotation
Varus	Medial (posterior erosion)	More external rotation

Pitfall

- Fixed 3° on posterior referencing → big error with condylar hypoplasia or erosion
- Lateral condylar hypoplasia → posterior condylar angle up to 10°
- Unnoticed lateral erosion (valgus) → femoral component internally rotated
- Posteromedial erosion (varus) → femoral component excessively externally rotated

Femoral Rotation in Valgus Knees Ask AI

- Valgus knee → 3° may not apply; separate talk

Valgus deformity	Femoral component ER
Non-valgus (standard)	3°
10–20°	5°
>20° (severe; often with patellar subluxation)	7°

- Some company jigs → **3, 5 and 7** options

Tibial Cut Reference and Hybrid Technique Ask AI

- First three references → TEA, Whiteside's line (AP axis), PCA
- Fourth → **cut proximal tibia first** (perpendicular to mechanical axis)
  - Femoral cuts parallel to it → after extension soft tissue balancing
  - Detail → soft tissue balancing / **flexion–extension gap** lecture
- **Take-home** → several variables → use more than one reference axis
  - Single reference → component **malrotation**
  - AP axis ~91° but range 80–102 → current case may be an outlier
  - Few degrees of range in each → use all three axes
  - **Gap-balancing technique** → tibial cut surface as additional reference
- Use all → **hybrid technique** → especially in complex cases

Reference	Landmarks	Relation
TEA	Most prominent points of medial and lateral epicondyles	~3° ER to PCA → cut parallel
Whiteside's line	Bottom of trochlear sulcus → top of intercondylar notch	~91° (80–102) to TEA → cut perpendicular
PCA	Tangent to posterior condyles	3° IR to horizontal → cut at 3° ER
Tibial cut surface	Proximal tibial cut perpendicular to mechanical axis	Femoral cuts parallel, after extension balancing

Grand Piano Sign Ask AI

- Frequently asked by lecturer to trainees
- After **anterior femoral cut** → cut surface shaped like grand piano
- Source → American Journal of Orthopedics, 2011
- Cut surface → **bimodal peak** on anterior distal femur
  - Correct rotation → lateral peak ≈ twice medial peak
- Bimodal peak → good correlation with proper **femoral rotation**

Recap of the Code Ask AI

- 6 → **distal femoral valgus cut** on average
- 3 → **external rotation** of femoral component
  - Angle between PCA and transverse plane
    - Transverse plane → TEA, perpendicular to Whiteside's line, or tibial cut surface
  - Valgus knees → sometimes 5 or 7
- Posterior lateral femoral condyle **hypoplasia** (rheumatoid) → account for it in posterior referencing

Tibial Rotational Alignment Ask AI

Tibial Tuberosity Ask AI

- Several references → use together → reduce error
- **Tibial tuberosity** → divided into thirds
  - Reference → junction of medial one-third and lateral two-thirds (or middle third)
- Following this landmark → **neutral rotation** of tibial component
  - Tuberosity → just lateral to tibial midline
- Imaginary line → drawn or palpated along it
- From above → line in the middle; also marked on implant
  - Tuberosity → **lateral to midline** of tibia
- Intraoperative → jig **alignment rod**
  - Index finger on medial border of tuberosity → align rod with medial-third/lateral-two-thirds junction
- **Single landmark** → not enough

Intermalleolar Axis Ask AI

- Alignment rod → relative to **medial and lateral malleoli**
  - Reference → about 5 mm from exact intermalleolar midpoint
  - Ankle centre → possibly ~15 mm relative to intermalleolar distance
  - Rod mark → placed about 5 mm from intermalleolar **midpoint**

Second Ray and Tibialis Anterior Tendon Ask AI

- Distal end of rod / cutting jig → in line with **second ray**
  - Unreliable → external rotation deformity, rheumatoid foot, any foot problem
  - Then → intermalleolar axis, and tibial tuberosity first
- **Tibialis anterior tendon** (more than one paper)
  - Ankle dorsiflexion → tendon visible
  - Reference → just lateral to tendon
  - Helps rotation → mark before **OpSite** drape
  - Or palpate intraoperatively with **dorsiflexion**

Tibial rotation landmark	Reference	Caveat
Tibial tuberosity	Junction medial 1/3–lateral 2/3	—
Intermalleolar axis	~5 mm from midpoint	—
Second ray	Rod distal end in line	Unreliable with foot deformity (ER/IR) or rheumatoid foot
Tibialis anterior tendon	Just lateral to tendon	Palpate in dorsiflexion

- Landmarks established → **diathermy** mark on tibia

Posteromedial Tibial Uncoverage (Fifth Landmark) Ask AI

- **Medial tibial condyle** → projects more posteriorly, larger than lateral
- Tibial base plate (trial) → symmetrical
  - → more **posteromedial uncoverage**
- More posteromedial than lateral exposure → rotation correct
- Not alone → **additional indication** only
- Rotation → more than one reference (five)
- Seen from above → **posteromedial tibia** more uncovered than lateral
- **Curve on curve** → tray first covers lateral tibial condyle → then medial
  - Medial condyle projects more → uncoverage at posterior aspect of tibial component

Never Internally Rotate the Tibial Component Ask AI

- **Never internally rotate** tibial component
  - → increased Q angle → patellar dislocation or subluxation
- Rotation take-home → more than one **reference** for femoral and tibial components
- Avoid internal rotation → both femoral and tibial components

Sagittal Alignment Ask AI

Posterior Tibial Slope Ask AI

- Sagittal alignment → **posterior tibial slope**, then femur
- Native posterior slope → varies; average 7–10°
  - Line perpendicular to **tibial axis** vs articular surface line
- Ideal → reproduce **native slope**
- Company slope options (e.g., 7° or 3°) → per **manufacturer's recommendations** for design

Read the Manual and Learn the System Ask AI

- First use of a prosthesis → read its **PDF manual**
- Request **workshop** (e.g., saw bone); attend cases with an experienced user
  - Each design → own specs → learn before operating on patients
- Theatre time precious → no trial-and-error **intraoperatively**
- Read repeatedly; ask **senior colleague** or representative
- Saw bone session; instruments brought by representative

Built-in Slope Ask AI

- Some systems → **built-in slope** in tibial component or plastic insert → ask
- Zimmer → slope from the cut, about 7°
- **Most systems** → 3–7°
- Calculate patient's slope → replicate
  - Avoid **excessive** or inadequate slope
- Insert thicker anteriorly than posteriorly
  - → **built-in slope**
- Equal anterior and posterior thickness → no built-in slope → create intraoperatively
  - **Zimmer NexGen** → about 7° posterior slope

Slope Errors Ask AI

- Slope may increase **unnoticed**
- **Excessive slope** (>7–10°; cut much higher anteriorly)
  - Wide flexion gap
  - Limited / incomplete extension
  - **Fixed flexion deformity** → unsatisfactory outcome
- **Reversed slope** → downward slope posteriorly (anteriorly sloped)
  - Limited flexion
  - **Recurvatum**
- Anterior tibial bone → weaker
  - → component **subsidence** and failure

	Excessive posterior slope	Reversed slope
Cut	>7–10°, much higher anteriorly	Anteriorly sloped
Gap	Wide (excessive) flexion gap	—
Motion	Limited / incomplete extension	Limited flexion
Deformity	Fixed flexion deformity	Recurvatum
Implant	Unsatisfactory outcome	Subsidence, failure (weak anterior bone)

The "7" in the Code Ask AI

- Third number of the "Da Vinci code" **6–3–7**
- 7 → **posterior tibial slope** (variable)
  - Companies → 3, 7 or 10 → middle course 7
  - Neither increase nor decrease → reproduce patient's slope
- Set slope modifiable (e.g., 3 → 5)
  - **Alignment guide** moved away from / closer to tibial shin → add 1–2° to cutting guide slope

Must-Know

- 6 → distal femoral valgus cut (average)
- 3 → femoral component ER relative to PCA
- 7 → posterior tibial slope

Sagittal Alignment of the Femoral Component Ask AI

- Femoral component → may be placed in **flexion** or extension
- Lateral view → draw **anatomical axis**
  - Two point pairs about 10 cm apart → bisect
- Ideal postoperative X-ray → implant perpendicular to this axis
  - + maintenance of **posterior offset**
- Determined by **entry point** on lateral aspect
  - Wrong start → everything wrong (as in DHS)
- Entry too anterior → **extended component**
  - → less flexion, recurvatum deformity
- Entry too posterior (e.g., 2 mm instead of measured ~8 mm or 1 cm from notch) → **flexed component**
  - → incomplete / limited extension, fixed flexion deformity
- Ideal → implant axis perpendicular to femoral **anatomical axis**

Femoral entry point	Component	Result
Too anterior	Extended	Less flexion, recurvatum
Too posterior	Flexed	Incomplete extension, fixed flexion deformity

- Sagittal care → **tibial slope** + femoral component position

Patellofemoral Tracking and Joint Line Ask AI

Q Angle Ask AI

- **Patellofemoral joint** → always forgotten
- **Q angle** (quadriceps angle)
  - Axis of extensor mechanism (quadriceps above patella) vs patellar tendon axis
  - Proximal limb → quadriceps → centre of patella
  - Distal limb → centre of patella → **tibial tuberosity**
- Average → 13–18°
- **Males** smaller than females
  - Female **gynecoid pelvis** → wider → about 18°
  - Standard deviation → smaller or larger possible

Component Position and Patellofemoral Tracking Ask AI

- Malposition of patellar, femoral or tibial component → affects **PF tracking**
- Flexed position → prosthetic patella placed **medially**
  - Lateral position → more lateral subluxation
- Medial position → avoid **lateral patellar subluxation**
- Avoid **inferior** position
- Aim → slightly anterior not inferior; slightly medial not lateral
- More IR of tibial or femoral component → **lateral patellar subluxation**
- Femoral component IR → ↑ **Q angle** → lateral patellar subluxation
- Femoral rotation → parallel to TEA, or ~3° ER to posterior condyles (+ other axes)
- Trial femoral component → cover lateral aspect
- **Do not medialize** femoral component
  - Medialization → relative lateral patellar subluxation
- **Do not internally rotate** tibial component
  - IR → ↑ Q angle; tuberosity more lateral to tray midpoint → lateral subluxation
- Violations → improper **PF tracking**

Must-Know

- Patellar component → medial position
- Avoid inferior patellar position
- Femoral component → no internal rotation, no medialization
- Tibial component → no internal rotation (↑ Q angle)

Joint Line Landmarks Ask AI

- Especially relevant → complex primary and **revision**
- Values → from revision textbooks and articles
- **Lower pole of patella** → native joint line about 10 mm (1 cm)
- Medial femoral epicondyle → medial side projects more distally
  - Joint line → 30 mm distal to **medial epicondyle**
- Joint line → 25 mm distal to **lateral epicondyle**
- Epicondyles → unaffected by femoral defects
  - Except **major revision**
- **Fibular head** → joint line 10–15 mm proximal

Joint line landmark	Distance
Lower pole of patella	~10 mm
Medial femoral epicondyle	30 mm distal
Lateral femoral epicondyle	25 mm distal
Fibular head	10–15 mm proximal
Peripheral meniscal remnants (revision)	At native joint line

Moving the Joint Line Ask AI

- Raising / lowering → **patella alta** or baja
- Limit → not >8 mm on average; some studies → not >4 mm
  - → considerably reduced **range of movement**
  - → impaired patellar function and stability
- Accurate joint line → requires accurate cuts
- Accurate **components** → essential
- Revision → **peripheral meniscal remnants** → native joint line (rarely found)

Alignment: Clinical Relevance and Take-Home Ask AI

Clinical Relevance Ask AI

- Many reports → **malalignment** linked to revision
  - Within ~3° → acceptable; still aim **neutral** (error inevitable)
  - Beyond 3° → unsatisfactory outcome, decreased implant **survivorship**

Take-Home of Alignment Ask AI

- Reproduce coronal and sagittal **mechanical alignment**
  - Distal femoral valgus cut → 6° average and its origin
  - 3° ER → and its origin
  - Tibial slope; femoral component **sagittal alignment**
  - Femoral and tibial rotational alignment → several references
    - → use more than one
- Component **malrotation** → mechanics affected + PF maltracking
- Malalignment over time → **stress concentration**, abnormal range of movement

Templating: Goals, Steps and Radiographs Ask AI

Rationale Ask AI

- Alignment principles → applied in **planning**
- Template every knee until experienced → advantages (as in **total hip**)

Goals and Advantages of Templating Ask AI

- **Templating** → avoids surprises during TKA steps; simpler than hip
- Predict implant size → not the most important goal
- Juniors → often unaware of alignment components in operations attended
- Goals
  - Verify restoration/correction of coronal, sagittal and rotational **alignment**
  - Implant size
    - **Extreme size** (very small / large) not brought by company → problems
    - Before anaesthesia → confirm sizes and **instruments** ready
  - Unexpected **defects** → prepare augments per defect size (separate lecture)
  - Detect **hypoplasia**
  - All sizes incl. extremes + required **instrumentation** available
- Example → mild extra-medullary femoral deformity
  - Not needing correction → **short medullary rod** required
  - Large angle → **osteotomy** + stem, simultaneous or staged
- → less **operative time**
- → improved **precision**
- More experience → more complex cases → templating more important

Steps of Templating Ask AI

- Predicted size (e.g., size 5 tibia) → need for larger/smaller up to 100% → company prepares **all sizes**
- Steps (as total hip)
  - Obtain appropriate **radiographs**
  - Check scale / **magnification** between templates and radiographs
  - **Analyse radiographs** → axes, angles, cuts (most important)
  - Template **femoral** component, then tibial component

Pre-operative Planning Starts in the Clinic Ask AI

- Starts at first meeting → **history** and clinical examination (beyond scope)
- Treat a **human being**, not an X-ray

Pre-operative Radiographs Ask AI

- **Consistent clinical results** → proper pre-operative radiographs + planning + examination
- Review coronal, sagittal and rotational alignment on them
- Predict **corrections**; identify defects and extra-articular deformities
- **Full-length film** (scanogram) → mandatory before every TKA
  - Patella → **facing forward**
  - Rule out **extra-articular deformity** and possible bone loss
- **Weight-bearing** standing view → coronal laxity
  - Predict need for non-constrained vs more **constrained** implant
- Unprepared, no constrained prosthesis → incomplete operation or **residual instability**
- Requested views
  - AP weight-bearing
  - Lateral
  - Full-length → hip to ankle
  - Probable **stress views**
    - → assess laxity

Axial View Ask AI

- **Axial view** → patellar dysplasia / trochlear hypoplasia
  - → avoid PF maltracking

View	Purpose
AP weight-bearing (standing)	Coronal laxity
Lateral	Lateral-view planning
Full-length hip to ankle (patella forward)	Extra-articular deformity
Stress views (probable)	Laxity
Axial	Patellar dysplasia, trochlear hypoplasia

Magnification Ask AI

- Standard magnification → about 20%
  - **Most templates** account for it
- **Magnification marker** → at level of knee joint
  - E.g., 10 mm marker measuring 12 mm → 20%
  - Alternative → radiograph with **ruler** (cm marks)
- **Digital templating** software → easier; marker still required

Radiographic Analysis: Axes and Centres Ask AI

Axes on the Radiograph Ask AI

- First → **mechanical axes** of femur and tibia
  - Femoral mechanical axis → centre of femoral head → centre of knee
  - **Femoral anatomical axis** → two point pairs on shaft, ~10 cm apart, bisected
    - Trajectory → predicts **entry point**
  - Angle between them → **distal femoral valgus** angle
- Tibia → mechanical and anatomical axes parallel, alike
  - **Tibial mechanical axis** → centre of proximal tibia → centre of ankle
  - Tibial anatomical axis → two point pairs bisected → coincident with mechanical

Centre of the Femoral Head Ask AI

- **Mose circles** → concentric circles with central point
  - Best-fit circle on femoral head → central point = centre
  - Also on some protractors / **goniometers**
- Without device
  - Highest and lowest head points → diameter → bisect → **radius**
  - Measure radius from peak of **medial femoral head margin**
  - Intersection → centre

Centre of the Knee (Distal Femur) Ask AI

- **Apex of femoral notch**
- Midpoint of femoral condyles
- Usually nearly **coincident** → either acceptable

Centre of the Proximal Tibia Ask AI

- Centre between medial and lateral **tibial spines** → easiest
- Midpoint of **soft-tissue outline** → difficult; only if clear
- Midpoint of **tibial plateau** → bisect medial–lateral plateau line

Centre of the Ankle Ask AI

- Sources → websites; **Dror Paley** deformity correction book (recommended)
- Most used → centre of **talus** / talar dome
  - Bisect line along proximal talar surface
- Midpoint between medial articular surface of lateral malleolus and lateral articular surface of **medial malleolus**
- Centre of **soft tissue**
- Centre between the **two bones** → range about 5 mm
- Lecturer's order → talus centre, then malleolar articular midpoint
  - Soft tissue centre → if clear on X-ray
  - Malleolar articular midpoint ≈ **talus centre**

Centre	Landmarks
Femoral head	Mose circles / goniometer; or radius from medial head margin peak
Knee (distal femur)	Apex of femoral notch; midpoint of condyles (nearly coincident)
Proximal tibia	Between tibial spines (easiest); soft-tissue midpoint; plateau midpoint
Ankle	Talar dome (most used); malleolar articular midpoint; soft tissue centre; centre of two bones (~5 mm)

Step One: Mechanical Axis of the Limb (Exercise) Ask AI

- For **precision**
- Draw mechanical axis → femoral head centre → ankle centre
  - Neutral → through knee centre
  - Example → passes medial → **varus mechanical axis**

Hip–Knee–Ankle Angle Ask AI

- Femoral mechanical axis vs tibial mechanical (= anatomical) axis → **intersect** at an angle
  - Example → varus about 15° (JAAOS)
- Angle → **coronal deformity** / hip–knee–ankle angle

Femoral Anatomical Axis and Valgus Angle (Exercise) Ask AI

- Two point pairs 10 cm apart → bisect → **anatomical axis**
- Femoral head centre → knee centre line; angle measured → 5°
  - → **distal femoral valgus cut** of 5°

Planning the Bony Cuts and Entry Point Ask AI

Distal Femoral Cut Ask AI

Femur

- Line perpendicular to mechanical axis → **distal femoral valgus cut** (5, 6 or 7°)
- **Implant thickness** → about 9 mm on average (e.g., Zimmer NexGen)
  - Confirmed from manufacturer's recommendations / manual
- Resect 9 mm from **less involved** (more normal) condyle
  - Jig foot piece → touches more prominent, less diseased condyle
- Adjustable → **+2 / −2** mm steps from 0
- Draw line perpendicular to mechanical axis → 9 mm from condyle
- Keep resected bone → unworn condyle piece ≈ **implant thickness** (~9 mm)
- Radiograph → 9 mm after **magnification** correction

Proximal Tibial Cut Ask AI

Tibia

- Line perpendicular to tibial mechanical (= anatomical) axis → **proximal tibial cut**
- Tibial **stylus** → one side 10 or 8 mm; other side 2
  - 10 → on less involved side (varus → **lateral plateau**)
  - 2 mm → on **more involved** side
- Draw about 10 mm from less involved side
- Both sides defective → **never below fibular head level**
  - Joint line → 10–15 mm above fibular head
  - Metal base plate + insert → upper insert edge = **joint line**
  - Excess resection → lowers joint line
- Measure 10 mm or 2 mm → per **side** cut

Major Tibial Defects Ask AI

- 10 mm line misses medial side → never cut lower
  - Near fibular head → **absolutely wrong**
  - Refresh defect → prepare **augmentation**
- Preserve bone → nothing equals **native bone**
- Cut 8–10 mm → then decide (separate lecture)

Femoral Entry Point: AP View Ask AI

- Entry point → where **anatomical axis** exits
  - Usually slightly medial to knee centre

Femoral Entry Point: Lateral View (Blumensaat's Line) Ask AI

- Landmark → **notch**; distance in front of or behind it
  - Too anterior / posterior → extended / flexed component
- Entry point → usually about 8 mm (7 mm–1 cm) from notch
- Radiograph → draw **Blumensaat's line**
- Draw anatomical axis on lateral view → two point pairs, bisected
- Measure axis exit → notch (e.g., 8, 7 or 10 mm) → **entry point**
- Entry point depends on
  - **Mediolateral position** → AP view
  - Distance anterior to notch → lateral view
- Replicate **intraoperatively**

Measuring the Posterior Tibial Slope Ask AI

- **JAAOS** method
  - Line A → tangent to anterior tibial cortex (shin)
  - Line B → perpendicular to line A
  - Articular line → tangent to articular surface
    - 3–4 most prominent points → **anterior tibial corner**
  - Slope → angle between line B and articular line
- Typical → 5, 7 or 10
  - Average 7–10°
- Native 7 → reproduce 7
- Native 5 → **fine-tune** jig → 5
- 5 vs 7 → small difference; avoid excessive increase or decrease
- Templating → know **patient's normal** → stay close
  - Native 3 → not 10 or 7
  - Native 5 → 7 or 3 per availability / **jig**

Osteophytes and Defects on the Lateral View Ask AI

- Defects / large **osteophytes** → very clear on lateral view
- Missed **posterior osteophyte** (tibia or femur) → abnormal balancing, tightness
  - → flexion gap problem → must be removed
- Remove intraoperatively; assess **defects** on lateral and AP

Implant Sizing Ask AI

Why Size Matters Ask AI

- After entry point and slope → determine **implant size**

Undersizing	Oversizing
Iatrogenic femoral fracture on impaction (notching, cortical breach)	Pain from overhang (most common; esp. two sides)
Tibial subsidence on impaction	Reduced postoperative flexion (tight gaps)
Postoperative instability (unbalanced flexion–extension gaps)	PF maltracking; ↑ PF forces and shear

- Size errors → altered **biomechanics**

Femoral Component Templating Ask AI

- Final planning step → **femoral component size**
- **Four measurements** → femur AP + lateral; tibia AP + lateral
- Femoral AP → component perpendicular to mechanical axis
  - Cover **medial and lateral condyles**
  - In between → **overhang** much worse → avoid; aim for exact fit
- Lateral → **maximum coverage** of condyles
  - Avoid flexion / extension of component
  - Along femoral anatomical axis, exactly perpendicular
  - Entry point in front of **Blumensaat's line** → mid-femur trajectory

Tibial Component Templating Ask AI

- AP → perpendicular to **tibial mechanical axis**
  - Cover medial and lateral plateau; no overhang or undersizing
- Lateral → parallel to **anterior tibial cortex**; adjusted for slope
  - 7 or 5, or patient's slope → best match to **own anatomy**

Acetate Templates Ask AI

- Transparent **acetate templates** → request from company
  - Size marked (e.g., E, F) → AP and lateral sheets
  - Insufficient femoral AP coverage → next size
  - → until best component
- Lateral → check **anterior notching** → adjust size

Stem Size Ask AI

- Revision → plan **stem size** (tibia or femur); often unavailable
- Primary → stem possibly for osteoporosis or **obesity** (debated)
- Unanticipated, unavailable stem → **non-ideal** component
- Tibial AP and lateral → same approach
  - Defect → required component / **augment** size
  - → increased precision; fewer complications
- More experience → **complex cases** → better preparation

Femoral Sizing in Practice Ask AI

- **Software** or acetate template
- Medial femoral condyle larger (distal, posterior) → cover **both condyles**
  - Example size too small → upsize
  - Larger size → tangent to anterior femoral cortex → no **notching**
- **In-between sizes** → reason for gender-specific implants
- Size increments → formerly 6–8 mm → now 2 mm → smaller **margin of error**
- In between → choose **larger size** → can downsize

Posterior Condylar Offset Ask AI

- Check **posterior condylar offset** + avoid notching (JAAOS)
- Tangent to **posterior femoral cortex** vs posterior condylar cortex → distance = offset
- Ideal → removed = replaced
- Preserved offset + alignment → reproduces **patient's anatomy**
  - Compare pre- and postoperative radiographs
- Alternative → **ratio** from anterior femoral cortex → check unchanged
- AP → **mediolateral position**
- Femoral size → lateral view **more accurate**
- Too small → next size; consider downsizing or **upsizing**

Tibial Sizing in Practice Ask AI

- AP template / software → coverage without undersizing or **overhang**
  - Overhang → **postoperative pain** → avoid

Templating Accuracy, Time and Closing Message Ask AI

Personal Study: Templating vs Navigation Ask AI

- UK study → **electromagnetic computer-assisted navigation**
- 30 cases
  - Pre-op templating → femur AP + lateral; tibia AP + lateral
  - Intraoperative **gold standard** → navigation size
  - Templating vs intraoperative size → **agreement** rate
- Commonest → size E femur ×12; size E tibia ×10
  - Possibly 9 of sizes F or G
- Femur → **lateral view** most accurate (83%); other papers agree
  - AP view → 70%
  - ±1 size (e.g., E → D or F) → 100%
- Size errors → many problems → reduce with **tips and tricks**
- Tibia → **AP view** more accurate → 80%
  - 24 of 30 cases; lateral view → 56%
  - ±1 size → 93%

Component	View	Exact size	±1 size
Femur	Lateral	83%	100%
Femur	AP	70%	—
Tibia	AP	80% (24/30)	93%
Tibia	Lateral	56%	—

- Take-home → **lateral view of femur** + AP view of tibia for templating

Exam Pearl

- Femoral size → lateral view (83%; ±1 size 100%)
- Tibial size → AP view (80%; ±1 size 93%)

Time Saved by Templating Ask AI

- Study (total hip) → delay fetching each missing component
- Templating → saves ≥2 minutes → 400 minutes per 100 operations *[Q1 · as spoken; 2 minutes saved per operation gives 200, not 400, minutes per 100 operations]*

Templating Software Ask AI

- No acetate templates → hospital **software**
  - OrthoSize (password from Zimmer), OrthoView, TraumaCAD
  - Very good results

Closing: History, Examination, Soft Tissue Balancing Ask AI

- Lecture → alignment and planning on **X-ray**
- History and examination first → treat a **human being**
- **Soft tissue balancing** → most important
  - As important as alignment and **bony cuts**
  - Separate lecture → varus, valgus, fixed flexion deformity
- TKA → **soft tissue procedure** with some bony cuts
- Template every case → compare with intraoperative choice → track **accuracy**
- Each case → next one more **accurate**

Summary of the Second Part Ask AI

- Templating **goals** → predict and anticipate problems
- Draw lines and **axes**
- Predict **implant size**
- Femur → lateral view; tibia → **AP view** more accurate
- Soft tissue balancing → grows with **training** and experience
- Learn from seniors, **manual**, workshops
- More experience → less operative time, fewer **complications**

Questions and Answers Ask AI

Why "Valgus Angle"? Ask AI

- **Anatomical axis** → valgus relative to mechanical axis
  - → named **valgus angle**

Patellar Pole for the Joint Line Ask AI

- **Distal pole of patella** → in extension

Role of 3D CT Ask AI

- Extra imaging → useful, especially in **revision**
  - CT with 3D reconstruction → metal **artefacts**
  - Shows defect size, **loosening** signs, periprosthetic fracture extent
    - Fracture line beyond **prosthetic flange**
- Primary TKA → not routine; only **complex primary** with suspected defect

Template Size and the Planned Cuts Ask AI

- Template → on **pre-operative X-rays**
- Cuts imagined or drawn → **distal femoral cut** runs with implant thickness on template
- Draw distal femoral and proximal tibial cuts → place **template** on them

Patellar Template Ask AI

- Lecturer never used one → patella by intraoperative **trial** only

Curve-on-Curve Tibial Rotation Ask AI

- **Lateral tibial plateau** → smaller than medial
- Tibial component → no internal rotation, no **medialization** (as femoral)
  - → increased Q angle → **lateral patellar subluxation**
- **Curve on curve** → rounded anterior tray curve on rounded lateral plateau curve
  - → proper **base plate** coverage of lateral plateau
- Detail → workshop and surgical technique

Slope Without a Full-Length Tibia Ask AI

- Full-length tibia → **not needed**
- Long-leg **lateral film** → mentioned in studies, rarely used
- Tangent to proximal half of anterior tibial cortex → perpendicular → articular line → angle
- **Proximal tibia** 10–15 cm → sufficient

Basics of Total Knee Alignment goals

(Distal Femoral Valgus Cut, Component Rotation and Alignment Targets)

Prof Dr Sherief Khaled

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Introduction and Lecture Scope Ask AI

Opening Questions Ask AI

- Opening questions to the audience
  - Why a **valgus cut angle** of 5–7° in the distal femur?
  - Difference between **mechanical** and anatomical axis?
  - Why cut the femur in 3°? → audience: **external rotation**; cut made within those 3 degrees
- Munir answered the first question → tools and concepts shown during the lecture
- Hence another 3° adjustment needed elsewhere → explained in detail in this lecture

Acknowledgement and Background Ask AI

- Credit: Fouad Zamel, Mahmoud Abdel-Karim, Ahmed Hazem Abdel-Azim, rest of the group
  - Search for existing **total knee arthroplasty (TKA)** lectures → only material found
  - That material → this group's course at **Kasr Al-Ainy**, ~6–7 years earlier
  - Only course covering the subject thoroughly
  - Essentially no complete courses on total knee
- Preparation difficult → no platform, no established framework, no systematic thinking
  - → very hard to prepare a lecture from zero

Why the Subject Confuses Ask AI

- This lecture → **alignment** only
- Many speakers mix alignment with **bone cuts**, gaps and so on
  - Halfway through → **measured resection** and gap balancing → listener lost
- Plan of the series
  - This lecture → **alignment goals** only
  - **Bone cuts** → separately
  - Measured resection and **gap balancing** → separately
  - **Soft tissue management** → varus knee and valgus knee, each on its own
- Separating the topics → easiest way to understand; this is the part that gets lost
- Typical lecture → surgeon describes his operative steps
  - "If varus → one, two, three; if valgus → one, two, three"
  - "Take care that the posterior …; here rotate three degrees …"
  - "Valgus → rotate a little more or less; take this line, the other one confuses"
  - Everything said together in operative context → listener mixed up; not everyone follows
- Personal experience: started total knees in the **late 1990s**
  - Real problem understanding the 3 degrees → why rotate by 3°?
  - Every explanation → "so that you have a **rectangular flexion gap**"
  - Open question → why no rectangular flexion gap without rotating 3°?
  - Understood only after studying it for a long time

Topics Covered Ask AI

- **Concepts of anatomic and mechanical axis**
- **Alignment of the components** → coronal, sagittal and rotational planes
- **Posterior femoral offset**
- **Patellofemoral tracking**

Goals of Total Knee Arthroplasty Ask AI

The Six Goals Ask AI

- Goals → easy, comfortable, good knee
  - **Prosthetic position relative to the mechanical axis** → limb back on mechanical axis → normal movement
  - **Restore the joint line** → original level, neither higher nor lower
    - Patella accustomed to that joint line
    - Raised → **patella baja**
    - Lowered → **patella alta**
  - Third goal → the audience's answer; written in italics as most important
    - **Ligament and soft tissue balancing** → *TKA is a surgery of soft tissue balancing*
    - Most important step
  - **Balanced, equal and symmetrical flexion and extension gaps**
  - **Correct rotation**
  - **Adequate patellofemoral tracking**
- Knowing these goals → present yourself well when travelling and working abroad

High Yield

- *TKA is a surgery of soft tissue balancing* → most important step
- Joint line raised → patella baja
- Joint line lowered → patella alta

The Operating-Room Team Abroad Ask AI

- Abroad → **scrub nurse** performs many of these steps
  - No company technician → the nurse has studied the subject
- Team → surgeon and nurse only
- Company **representative** → does not scrub
  - Only stands there and opens requested sizes ("open femur 4, open tibia …")
- Assistant → the nurse, actually working with the instruments
- Possible because instrument sets and implants are **on-shelf**
  - Same on-shelf implants used by the nurse for 5 or 10 years
  - Very good training beforehand → understands the steps and the underlying concept
  - Studied and took courses, e.g. **AO** course for ORP (operating room personnel)
- Trauma nurses → know instrument names
  - Understand **relative stability** and absolute stability
  - Understand what is being done
- System abroad → a big difference

Mechanical and Anatomical Axes of the Lower Limb Ask AI

Limb Mechanical Axis Ask AI

- **Single limb stance** (standing on one leg) → ankle under the body's center of gravity
- Center of gravity in midline → limb runs slightly inward, hip → midline
- **Mechanical axis** → 3° of valgus from the midline (vertical); the first step
  - Two legs → wide pelvis above, feet close together below
  - Limb runs outside (above) → inside (below)
  - → mechanical axis inclined 3°, outside → inside
  - Memory aid → "3° of valgus from the midline"

Joint Line Orientation Ask AI

- Tibial articular surface → **not at 90°** to the transverse axis; at 87°
  - Tibial plateau → ~3° of **varus** relative to the transverse axis; normal
  - One side 93°, other side 87°
  - Femur → opposite (93° on its side)
  - → **joint line** parallel to the ground

Femoral and Tibial Axes Ask AI

- **Tibia** → mechanical axis = anatomical axis
- **Anatomical axis of the femur** → bisects femoral shaft; 6° from mechanical axis
- Degrees from the midline?
  - → 9° (6° + 3°); figures of 9 or 6 both quoted
  - → corresponding angles 99° and 81°
- Munir's earlier point → seen in a later slide
- **Femoral axes** → femoral mechanical axis + 6° = femoral anatomical axis
- **Tibial axes** → tibial mechanical and anatomical axis along the same line
  - Strictly, books → not exactly the same line
  - Difference ~1–2° → ignored for simplicity

Deviation of the Mechanical Axis Ask AI

- Limb **mechanical axis** line → runs straight down
  - Deviates laterally → **valgus** knee
  - Deviates medially → **varus** knee

Coronal Bone-Cut Angles Ask AI

Femoral Axes Ask AI

Femur

- **Mechanical** → center of femoral head → center of knee
- **Anatomical** → bisecting the femur (shaft)

Valgus Cut Angle Ask AI

Femur

- Key point (Ahmed Munir's) → **femoral component** perpendicular to mechanical axis
  - → **distal femoral cut** perpendicular to mechanical axis
- Problem → distal cut made with an **intramedullary rod**
  - Rod → lies along anatomical axis
  - Cut on the rod → wrong; cut must reference mechanical, not anatomical axis
- → cut adjusted ~5–7° = **valgus cut angle**
- Same value in every case (5, 6 or 7)? → commonly 6°
- **Long films** (long-leg standing radiographs) → all cases; not difficult, no reason to skip
  - Draw anatomical and mechanical axes → measure the difference
  - 5° → ask nurse/representative for a 5° valgus cut
  - 6° → 6° valgus cut; 7° → 7° valgus cut
- → valgus cut **planned on long films**

What the Distal Femoral Cut Controls Ask AI

Femur

- Valgus cut controls →
  - **Extension gap**
  - Flexion gap? → no
  - Rotation? →
  - → no; distal femoral cut affects **extension gap** only
- Anyone not following → stop the lecturer mid-way
  - Falling behind at one point → cannot follow the rest

Variation with Stature Ask AI

Femur

- **Valgus cut angle** fixed in all people? → no → plan on long films
  - **Tall** person → valgus cut angle decreases
  - Very short person (**dwarf**) → valgus cut angle increases
    - Dwarfs → some **coxa vara** or femoral bowing → valgus angle increases

Tibial Cut Angle Ask AI

Tibia

- **Tibial mechanical axis** → center of proximal tibia → center of ankle
- **Tibial anatomical axis** → bisecting the medulla
- Roughly coincide → considered coincident for simplicity
- **Tibial cut angle** = angle between tibial anatomical and mechanical axis → 0
  - Books → "tibial cut angle is 0"
  - 0 with a 90° cut → tibia cut as it is; no axis difference
  - Femur by contrast → 5–7°, two femoral axes differ
- Zero except when? → audience: **tibial deformity**
  - Mechanical ≠ anatomical axis → separate topic
  - Correct deformity first, or both in one operation, or two operations → long story

	Femur	Tibia
Mechanical axis	Center of femoral head → center of knee	Center of proximal tibia → center of ankle
Anatomical axis	Bisects femoral shaft	Bisects medulla
Axis difference	6° (5–7°)	~0 (1–2° ignored)
Cut angle	Valgus cut angle 5–7°, commonly 6°	Tibial cut angle 0 (except tibial deformity)
Gap affected	Distal cut → extension gap only	Tibial cut → flexion and extension

Exam Pearl

- Distal femoral cut → **extension gap** only; not flexion gap, not rotation
- Valgus cut angle → planned on long films in all cases
- Tall → valgus cut angle decreases; dwarf → increases
- Tibial cut angle 0 → except **tibial deformity**

Alignment Philosophies Ask AI

Historical Overview Ask AI

- **Traditional (mechanical) resection** → what most people do
  - Tibia below and femur above cut at 90° → two cuts parallel
- 1980s → another trend
- **Anatomic resection** → Hungerford: "I will not cut at 90°"
  - Tibial articular surface naturally 87° → normal for the patient → cut at 87°
  - Femur cut parallel above → **rectangular extension gap**
  - Appeared in the 1980s
  - Some patients → some **varus**
  - Not consistent across all varus and valgus patients
- Other approaches → **kinematic alignment**
  - **Bilateral varus** patient (bow-legged, like the footballer Aboutrika) → TKA → left in bilateral varus
  - Comfortable and functioning so for sixty years → left bowed, nothing corrected
  - Essentially **resurfacing** → surface changed only; **soft tissues** untouched
  - Angle kept as is → like **resurfacing of the knee**
  - **Unicompartmental knee arthroplasty (UKA)** → also kinematic principle
    - Varus patient → UKA → left in varus
  - These two → called kinematic
- Lecturer's practice → **mechanical alignment**; the classic approach
  - Also called **systematic** → same result in all patients
- **Anatomical alignment** → cuts at 87° and 87°; from the 1980s; still used by some
- Mechanical alignment not best option → **adjusted mechanical alignment**
  - Slight adjustment of mechanical alignment
- Kinematic problem → patient with 40–50° of varus
  - Leaving the knee so → absurd; cannot be left
  - → **restricted kinematic alignment**: deformity corrected partially, rest left

Anatomic Alignment Ask AI

Anatomic

- **Anatomic alignment** → joint line inclined 2–3°
  - Cut with 2–3° inclination relative to mechanical axis → better **loading**
  - Knee naturally inclined this way
  - Natural joint line inclined → kept inclined
  - Cutting at 87° → better for loading; 90° → not physiological
- Used for better loading

Adjusted Mechanical Alignment Ask AI

Adj. mech.

- **Mechanical alignment** not best → patient not accustomed to a perfectly straight limb
- → **adjusted mechanical alignment**
  - Existing **constitutional deformity** kept → maximum 3°
  - After mechanical alignment → knee left with slight deformity up to 3°

Kinematic Alignment Ask AI

Kinematic

- **Kinematic alignment** → newest; much debate; many users
- **Conventional TKA** → kinematic alignment not possible
  - Requires **robotic-assisted** or computer-assisted navigation
  - Most kinematic surgeons → robotic-assisted
  - By eyeballing → extremely difficult; many things unmeasurable
  - **Navigation** advantage → measures all these angles
  - Also measures **flexion and extension gaps throughout the range of motion**
- Described as **ligament-sparing total knee**
  - Ligaments untouched → **ligament-sparing** or soft tissue sparing
  - Described as **true knee resurfacing** ("a true resurfacing of the knee")
  - Described as **patient-specific**
    - Little varus → left with little varus
    - More varus → left with more varus
    - Straight knee → left straight
    - Slight valgus → left with slight valgus
- Audience question: additional varus introduced? → no; not more than the patient already had
- Objection from one surgeon → varus may have increased as the knee became **arthritic**
  - Varus for 10 or 15 years; e.g. 20° → 40° after arthritis
  - Leaving at 40° → no sense; never at 40°, accustomed to 20° all his life

Restricted Kinematic Alignment Ask AI

Restr. kin.

- **Restricted kinematic alignment** → kinematic alignment, but limited
  - Not applied to every degree of deformity
  - Very large varus or valgus → cannot be left on pre-existing alignment
  - Kinematic, corrected a little → into a safe zone, e.g. 40° → 7° or 10°
  - Kinematic, slightly corrected → within a safe zone

Evidence Ask AI

Evidence

- Best philosophy? → still completely unsettled
- **Robotic** or computer-assisted navigation → some adjustment or some restricted kinematic alignment possible
- Claimed advantages of the **kinematic technique**
  - **Faster recovery**
  - **Better functional outcomes**
  - **Better patient satisfaction**
- Literature → **not enough studies or evidence**
  - → not yet the **standard procedure**
- Recommended reading → article by **Rivière** (French)
  - One of the best; only one explaining all alignment philosophies together
  - Other publications → mechanical vs kinematic; mechanical vs anatomical
  - Others → mechanical vs anatomical vs adjusted mechanical
  - Rivière → combined all, as a **systematic review**; a very good article

Philosophy	Principle	Notes
Mechanical (traditional, systematic)	Tibia and femur cut at 90°, cuts parallel	Classic; same result in all patients; lecturer's practice
Anatomic	Cuts at 87° and 87°; joint line inclined 2–3°	Hungerford, 1980s; better loading; varus in some; inconsistent
Adjusted mechanical	Mechanical alignment + constitutional deformity ≤3°	Patient not used to a perfectly straight limb
Kinematic	Pre-existing alignment kept; true resurfacing	Ligament-sparing, patient-specific; robotic/navigation needed
Restricted kinematic	Kinematic, corrected into a safe zone	e.g. 40° → 7° or 10°

Rotational Alignment and the Flexion Gap Ask AI

Why Rotation Matters Ask AI

- Rotation important → why?
  - Not for foot pointing outward or inward
  - Aim → **balanced, equal, symmetrical flexion and extension gaps**

Tibial Cut and the Gaps Ask AI

Tibia

- **Tibial cut** → affects both flexion and extension; controls everything
- **Mechanical alignment** (not kinematic, not anatomic) → tibia cut at 90°
- Tibia usually worn more → **medial**
- Perpendicular (90°) tibial cut
  - Little removed medially, much laterally → ~2 mm vs 9 mm, or 2 mm vs 10 mm
- Femur then cut parallel to the **posterior condylar line**
  - → **flexion gap** trapezoidal → wider laterally
  - More tibia removed laterally; posterior femur cut equally both sides

External Rotation of the Femoral Cut Ask AI

Femur

- Solution
  - Not parallel to the **posterior condylar axis** → femur cut in slight external rotation
  - Femoral cutting block → rotated slightly
  - More tibia cut laterally than medially → trapezoidal gap
  - Externally rotated femoral cut → more femur cut **medially** than laterally
  - Result →

Side	Tibial cut	Femoral cut (externally rotated)
Lateral	More	Less
Medial	Less	More

- In short → tibia cut
  - Unequally → less medially than laterally
  - → femur cut the opposite way → more medially than laterally
- **Jig** (cutting guide) → rotated to 3°
  - Femur (referenced on posterior condyles) → always internally rotated relative to required cut
  - No rotation → lateral side always wide, medial side tight
  - 3° of rotation → **rectangular flexion gap**

High Yield

- Mechanical alignment → tibia cut at 90°; tibia worn more **medially**
- → less tibia removed medially (~2 mm), more laterally (~9–10 mm)
- Femur cut parallel to posterior condyles → trapezoidal flexion gap, wider laterally
- Femoral cut in 3° external rotation → **rectangular flexion gap**

The Jig Question Ask AI

Jig

- Any questions?
- Audience: rotate the jig to 3°?
  - Jig settings → 0° and 3° of **external rotation**
  - Assistant asked → jig set on 3° external rotation
  - An issue with this → a good question
  - Jig → leaves those 3 degrees in your hands; adjusted as needed
  - Jig placed posteriorly → on the **posterior condyles**
  - Posterior condyles same in all cases? → no → jig alone does not decide rotation
  - "TKA is a mechanical surgery; cutting blocks → perfect result" → those people understand nothing

Reference Lines Ask AI

Femur

- Recap
  - Tibia → **little medially, much laterally**
  - Femur → much medially, little laterally
  - Following the **posterior condylar line** → gap incorrect
  - A bit more cut medially → gap correct
- **Reference line** needed → no arbitrary cut or instrument placement
  - **Anteroposterior axis**
  - **Transepicondylar axis**
  - **Posterior condylar axis**
  - **Cut surface of the proximal tibia**

Order of the Cuts Ask AI

- Some surgeons → tibia before femur
- Lecturer's sequence → **distal femoral cut** → proximal tibia
  - → check **extension gap** → complete femoral cuts
- Others → **proximal tibia** first → remaining femoral cuts all at once
- Depends on **measured resection** vs gap balancing → separate bone cuts lecture

The Flexion Gap Question Ask AI

- Question → only aiming for a gap (opening)? → yes; flexion gap shape unknown in advance
  - Situation described → best option discussed in the **bone cuts** lecture
  - Purpose → gaps equal at the end
  - Taking more from the **tibia** → not a problem
  - Flexion gap widened too much → **flexion instability**

Femoral Rotational Landmarks Ask AI

Whiteside Line and TEA Ask AI

Femur

- Reference lines used in surgery
  - **Whiteside line** → bottom of the trochlear sulcus
    - Diathermy placed on the deepest point of the sulcus
    - → down to the top of the intercondylar notch
  - **Transepicondylar axis (TEA)** → palpate both epicondyles → draw along them
    - **Surgical TEA** vs anatomical TEA
    - Anatomical TEA → posterior point on medial epicondyle; not visible in surgery → ignored
    - Surgical TEA → palpable → used

Why the Jig Alone Is Not Enough Ask AI

Femur

- Who asked the earlier question?
- Question → does the jig set the 3 degrees by itself?
  - Jig on the red line (**posterior condylar axis**) + 3° rotation → correct rotation; but not here
  - **Valgus** knees → lateral femoral condyle hypoplastic → angle 10°, not 3°
- → rely on the **Whiteside line** and transepicondylar axis
  - Relation between them → constant → not misled by posterior condylar line
- Without Whiteside and TEA in valgus or varus cases →
  - Lost when setting the **external rotation of the femur**

Marking the Lines First Ask AI

- Audience: do you mark these lines? → yes; **marking is mandatory**
- Knee opened → lines marked first, before anything
  - Before the cuts; before even the **femoral entry point**
  - Lines first → know where you are, where you are going, where you came from
  - Mechanics go wrong → whole operation fails
  - Reliance on cutting metal pieces → can hit a wall at the end

Clinical Pearl

- Lines marked first → before cuts and before the femoral entry point
- Rely on **Whiteside line** and TEA → relation always fixed
- Posterior condylar axis → misleading in lateral femoral condyle dysplasia

Posterior Condylar Axis in Varus and Valgus Ask AI

- **Posterior condylar axis** misleading when dysplastic
  - → **dysplasia of the lateral femoral condyle**
- Relation between Whiteside line and **transepicondylar axis** → always fixed
- **Varus** knees → increased external rotation (of posterior condylar axis)
  - → those 3 degrees possibly not needed; varus lecture
- **Valgus** knees → opposite
  - → more **external rotation** needed

Knee	Posterior condylar axis	Femoral external rotation
Varus	Increased external rotation	3° possibly not needed
Valgus	Hypoplastic lateral femoral condyle → angle 10°, not 3°	More external rotation needed

Pitfall

- Valgus knee → hypoplastic lateral femoral condyle → angle 10°, not 3°
- Jig on posterior condyles alone → wrong femoral rotation

Tibial Rotational Landmarks Ask AI

Tibial Rotational Lines Ask AI

Tibia

- Tibia → steps to set correct rotation of the **tibial component**
- Lines
  - Line across the tibial plateau → transverse line
  - Line bisecting the tibia → **tibial tuberosity** slightly lateral
    - Line → at the junction of the medial third (of the tubercle)
    - Tuberosity → medial third | lateral two-thirds
    - Correct line → **centered on the tibial plateau**
- Usable lines
  - **Transverse axis**
  - Tibial **posterior condylar axis** → hard to see unless tibia fully subluxed anteriorly
  - **Largest diameter between the two plateaus**
  - **PCL insertion** → center of the plateau anteriorly
  - **Medial third of the tubercle**

Akagi Line Ask AI

Tibia

- **Akagi line** → posterior → anterior
  - Found to cross at a certain point
  - → possible slight tibial malrotation in some cases
  - Advice → **modified Akagi line** instead → between medial third and lateral two-thirds
  - End result → back to medial third / lateral two-thirds; Akagi line forgotten
- Other landmarks → on the leg

Extramedullary Alignment Ask AI

Tibia

- **Tibia** alignment assessed → deformed, valgus, varus, or correct
- Tibia and ankle position → **dorsiflexion**
  - Subluxed tibia → landmarks confused; get under it and take a position
  - Tibia can drop in any direction
  - Correct setting → foot in slight dorsiflexion
  - Extramedullary **rod** → parallel to the shin of the tibia
- Details → cuts lecture
- Second landmark → **curve to curve** on the medial side
  - Tibial plate (tray) → curve to curve medially
  - Posteromedial part slightly exposed? → possibly
  - What matters → anterior curve to curve = **parallel to the anterior cortex**
- Third → **medial third, lateral two-thirds** (of the tibial tubercle)
- Aim → **best functional outcome**
- Wrong rotation → disturbs?
  - → **patellofemoral tracking**

Summary of Tibial Rotational Landmarks Ask AI

Tibia

- **Medial third, lateral two-thirds** (of the tubercle)
- **Center of the ankle**
- **Tibialis anterior tendon**
- All → correct tibial rotation
- Center of the ankle (red line) → **not exactly between malleoli**
  - → shifted slightly medially
- Combined landmarks (tibial center, tibialis anterior tendon) → rotation right
  - Line → between medial third and lateral two-thirds
- **Keel** aligned parallel to this line → correct patellofemoral tracking

Patellofemoral Tracking Ask AI

Q-Angle Ask AI

Q-angle

- Patellofemoral tracking → **quadriceps angle (Q-angle)**
  - Between the line of the quadriceps
  - (runs outward, laterally) and the line along the patellar tendon to the center of the patella

Causes of Lateral Patellar Tracking Ask AI

Patella

- **Internal rotation of the femoral component** → patella shifts laterally
- **Internal rotation of the tibia** (tibial component) → same
- **Medialization of the femoral component** → same
- Rule → **lateralize anything except the patella**; patella always more medial
- Normal alignment → angle between femoral neck and femoral condyles
  - = **anteversion**
- Distal femur in internal rotation → patella automatically **sublux laterally**

Medialize or Lateralize Ask AI

Patella

- **Patellar button** → always medial, never lateral
  - **Medialization** → reduces Q-angle
  - More lateral → Q-angle increases
  - Q-angle increased → **patellofemoral instability**
  - Aim → Q-angle reduced back to normal
- → **medialize the patella**
- Femoral component → choice of medialize vs lateralize → **lateralize the femoral component**
- Everything lateralized except the patella → medialized
- Medialized femur → same problem → patella **maltrack**

Component	Correct position	Error → lateral patellar tracking
Patella (button)	Medialized	Lateral placement → Q-angle increased
Femoral component	Lateralized	Internal rotation; medialization
Tibial component	Correct rotation	Internal rotation

Exam Pearl

- **Lateralize anything except the patella**
- Femoral component internal rotation → lateral patellar tracking
- Tibial component internal rotation → lateral patellar tracking
- Femoral component medialization → lateral patellar tracking
- Q-angle increased → patellofemoral instability

Sagittal Alignment Ask AI

- **Sagittal alignment** → simplest plane

Femoral Entry Point Ask AI

Femur

- Entry too anteriorly →
  - → **notching**, limited flexion, recurvatum of the knee
- More posterior entry point → **flexed femoral component**
  - → incomplete extension and fixed flexion deformity

Tibial Slope Ask AI

Tibia

- Natural **posterior slope** → 7–10°
  - Most implants → **built-in slope** of 3–7°; rest built into the polyethylene
  - Excessive or inadequate → same principle; slope = 7° or implant's built-in slope
  - Reduced to **reverse slope** or zero → loss of flexion and recurvatum
  - **Exaggerated** slope → incomplete extension and fixed flexion deformity
- Sagittal alignment → the easiest

Error	Result
Femur: entry too anterior	Notching, limited flexion, recurvatum
Femur: entry too posterior → flexed component	Incomplete extension, fixed flexion deformity
Tibia: reverse or zero slope	Loss of flexion, recurvatum
Tibia: exaggerated slope	Incomplete extension, fixed flexion deformity

Built-In Slope in Practice Ask AI

Tibia

- **Lateral view** → component aligned to the axis of the femur; tibial slope correct
- Before any implant → ask the company: **built-in slope** or not?
  - Kasr Al-Ainy implant → **no built-in slope** (zero)
  - No slope cut → **inverted slope**
  - → all patients: limited flexion and recurvatum
- Always ask → read implant's **surgical technique** and details
  - Follow its designer's recommendations

Pitfall

- Implant with **no built-in slope** + no slope cut → inverted slope
- → limited flexion and recurvatum in all patients
- Read each implant's surgical technique before use

Posterior Condylar Offset Ask AI

- **Posterior condylar offset** → small portion of the posterior femoral condyle; normal offset
- Cut → implant of this shape fitted
- Overcut posteriorly (increased **posterior cut**) → offset decreases
- Decreased offset →
  - → **limited flexion**
  - → **impingement** between polyethylene and femur
  - → faster **polyethylene wear**

Restoring the Joint Line Ask AI

- Last point → **proper joint line**; restoring it → one of the goals
- Joint line restored at end of operation? →

Landmark	Joint line position
Fibular head	10–15 mm above
Patella (inferior pole)	10 mm below
Medial epicondyle	30 mm below
Lateral epicondyle	25 mm below

- Fibular head and medial epicondyle → landmark names unclear in recording
- Four numbers important → heavy cuts → within reason, or harming the patient?
- Check these numbers when drifting away from them

Joint line	Result
Elevation > 8 mm	Patella effectively **baja (infera)** → reduced range of motion, patellofemoral problems, maltracking, instability, possible fractures
Lowered	Incomplete extension, flexion instability

Must-Know

- Joint line → 10–15 mm above fibular head
- Joint line → 10 mm below inferior pole of patella
- Joint line → 30 mm below medial epicondyle, 25 mm below lateral
- **Joint line elevation** > 8 mm → patella baja (infera)
- Lower joint line → incomplete extension, flexion instability

Effects of Malalignment and Take-Home Message Ask AI

Effects of Malalignment Ask AI

- **Effects of malalignment** → any malalignment described
  - → **Accelerated wear**
  - → **Aseptic loosening**
  - → **Tibiofemoral instability**
  - → **Patellofemoral instability**
  - → **Patellar fractures**
  - → **Stiffness**
- **Soft tissue management** → varus and valgus lectures; agreed most important part

Take-Home Message Ask AI

- Now known
  - Restore **coronal alignment** of the knee
  - Restore **sagittal alignment**
  - Set **rotation of the femur and the tibia**
  - Confirm restored **normal joint line**
- **Malalignment** → stress concentration, wear, very early aseptic loosening
- Correct **bone cuts** → in detail next time

Approach for TKR

(Medial Parapatellar, Subvastus, Midvastus and Lateral Approaches)

Prof Dr Sherief Khaled

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Take-home Message 97

AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 95) As spoken 60°; standard (Insall/Garvin) quadriceps snip is an oblique proximal-lateral cut at about 45°.

Overview and Classification of Approaches Ask AI

- Session order → **approaches** → alignment → case example
- Three groups of approaches
  - Primary TKA
  - Mini approaches
  - Revision / complex primary

Primary TKA Ask AI

- **Anteromedial** (medial parapatellar) arthrotomy via midline incision
  - Used by everyone → most important → **workhorse**
- **Subvastus** — less fashionable now
  - Many surgeons extensively experienced in it
- **Lateral approach** → severe valgus, more than 15°
  - Some surgeons → anteromedial even at 15°
  - Lateral reserved for 30° valgus → very marked valgus

Mini Approaches Ask AI

- **Mini subvastus**
- **Mini midvastus**
- **Mini anteromedial**

Revision / Complex Primary Ask AI

- **More extended approaches**
- First approach still → **anteromedial**
- Modifications, in order
  - **Quadriceps snip**
  - **V-Y quadriceps turndown**
  - **Tibial tubercle osteotomy** → difficult revisions

Approaches on the Illustration Ask AI

- Standard **anteromedial** (medial parapatellar) arthrotomy
- **Subvastus** → beneath vastus medialis
- **Midvastus** → through substance of vastus medialis, not beneath
- **Trivector** → can be disregarded
- **Lateral** → limited popularity
  - Described in some books; absent from others
- Mini approaches on illustration
  - Standard anteromedial arthrotomy
  - Mini anteromedial arthrotomy
  - Mini midvastus
  - Mini subvastus
  - **Quadriceps-sparing** approach
    - Like mini anteromedial → **quadriceps tendon not opened** at all
    - Arthrotomy stops at **upper pole of patella**
    - ≈ subvastus without opening the vastus medialis split
- Mini approaches → lecturer **would not recommend**
  - Most users → **robotic assistance** or **computer navigation**
  - Reason → **very limited view** (textbook pictures shown)

Choice of Approach Ask AI

- Most important factor → **surgeon training, familiarity and preference**
  - What you do well, know and have experience in → best results → use it
- Other factors that may change the approach
  - **Previous scars** → always the **most lateral** appropriate scar
  - **Degree of deformity** → e.g. valgus 35° → may force lateral approach
  - Revision, complex primary, flexion deformity, fibrosis → larger **extensile** approach

Positioning and Preparation Ask AI

- Any approach description → starts with **position**
  - **Supine** on table
- **Landmarks** marked on limb → do not forget (X-ray shown)
- Marker over **center of femoral head**
  - Marker on X-ray = **ECG lead** (electrode)
  - Purpose → easier **intraoperative alignment check** with alignment rod
  - Metal lead palpable over drapes → easiest method
- **Tourniquet** → applied, not inflated
  - With or without tourniquet → much **debate**
  - Lecturer lately prefers operating **without tourniquet**
  - Deflation at **cementing** → some do, others not; depends
- **Lateral bolster** at side of thigh
  - Heavy limb falls to side → no assistant holding leg → very useful
- **Foot rest** under patient's foot

  - Knee flexed for tibial work → foot rests → nobody holds it

Skin Incision Ask AI

- Always **mark the skin incision** beforehand
  - No habit of palpating after draping
  - (Demonstration → palpating and marking line on patient)
- **Thick skin flaps** → do not undermine
  - Undermining → **skin edge slough** → wound problems
- Scars of previous operations → **most lateral appropriate** scar
  - Adequate distance → preserved medial **blood supply** to skin and soft tissue
- Old **transverse scar**
  - Previously used → transverse **smiley face incision**
  - Used up to ~2000 (roughly 1996–2000, lecturer's residency)
  - Plastic surgeons → **longitudinal incision at 90°** to old transverse scar is safe
  - Such scars usually old → no problem

Anteromedial (Medial Parapatellar) Approach Ask AI

- **Most commonly used** approach

Indications Ask AI

Anteromed

- **Primary**, complex primary, revision → almost everything
- **All deformities** → some experts use it routinely in these

Technique Ask AI

Anteromed

- Opening through **extensor mechanism** → starts distally beside patella
  - **Tissue cuff** left beside patella → sutured later
- Proximally → 1 cm of **quadriceps tendon** left toward vastus medialis
  - Rest of tendon → with **vastus lateralis**
- **Quadriceps tendon incision** → ~8 cm proximally
  - Thin patients → full 8 cm not needed; 8 cm is a lot
  - Obese patients → 8 cm needed; very obese → barely enough
- Key point → 1 cm beside patella + 1 cm in tendon → **extensor mechanism closure**
- After opening → excise **patellar fat pad** → evert patella

Tight Patella Ask AI

Anteromed

- Tight patella; eversion risks **patellar tendon rupture**
  - → **subluxate laterally** — patella upright, retracted laterally
  - Proceed normally → eversion not mandatory
- **Rheumatoid** patients (large number operated) → no eversion
  - Patellar tendon **may avulse**
- Case → very thin patellar tendon began to **peel off** during eversion
  - Stopped → patella **subluxated** instead of everted (dislocated)
  - First → **transosseous sutures** through tendon at peeled area
  - Purpose → avulsion not propagated; continued eversion → complete avulsion

Soft-tissue Handling Ask AI

Anteromed

- Joint opened anteriorly → **ACL** excised
- First removed → **anterior horns** of medial and lateral menisci
- **Do not skeletonize** proximal tibia at the start
  - Some colleagues → strip tibia completely bare; do not follow
  - **No release** at the beginning
  - Cuts first → then assess whether release needed
  - Only small area cleared → **periosteum** or **deep MCL** elevated
- After ACL or PCL excision → **Hohmann retractor** behind tibia
  - Likely purpose → sublux tibia forward

> **Pitfall**
> - Tight patella forced into eversion → **patellar tendon rupture**
> - Rheumatoid knee → **patellar tendon avulsion** → subluxate, do not evert
> - Peeling tendon → **transosseous sutures** before continuing
> - Proximal tibia → **no skeletonizing**; release only after cuts

Insall Variation Ask AI

- No 1 cm medial cuff → incision **directly over patella**
  - Rationale → better if **patellar resurfacing** planned → no soft tissue on patella
- Lecturer does not use it → **closure very difficult**
  - With cuff → soft tissue to soft tissue closure → easy
  - Over patella → **transosseous sutures** needed → more complex
  - Hence → not very popular

Subvastus Approach Ask AI

- **Entire extensor mechanism intact**

Advantages and Drawbacks Ask AI

Subvastus

Advantages	Drawbacks
Faster recovery	Probably needs greater experience/exposure in thick limbs
Improved **patellar tracking**	Probably very difficult in obese patients
—	Exposure probably inferior; standard approach more universal

- **Not usable in revision** → unlike anteromedial (note this point)

Indications Ask AI

Subvastus

- **Primary TKA** → easy primaries
- **Unicompartmental** knee replacement

Technique Ask AI

Subvastus

- Incision → **parallel to tibial tubercle**, medial to patella → as anteromedial
- At vastus medialis → cut **proximal → distal**
  - **Parallel to VMO attachment** to tendon → extensor mechanism preserved
  - Cut angle ~50–60°
  - Key → whole **VMO preserved proximally**
- Patella difficult to subluxate laterally → **MPFL release**
  - Still not subluxating → **suprapatellar release**

Patient Selection Ask AI

Subvastus

- Lecturer never tried it; some surgeons like it
- Users (skilled and less skilled) → strict **selection criteria**
  - **Very easy case** only
  - Not stiff
  - Patient not short
  - Not a tight knee
  - Probably not bilateral (unclear in recording)
  - **No flexion deformity**
  - Many conditions must be met

Mini Approaches Ask AI

- Illustrations → surgeon's actual view through each mini approach
- **Not recommended** without **computer-assisted navigation** or robotic assistance
  - Small exposure → not ideal
- **Mini subvastus** → same limited view after opening
  - Hard to feel comfortable or confident doing TKA through it
  - Everyone wants to see the field
- Require **specific training** → learn with a surgeon doing mini approaches

Lateral Approach Ask AI

- Same concept as medial approach → used in **valgus**

Indications Ask AI

Lateral

- **Main indication** → valgus deformity
- Not used in every valgus knee
  - Some → anteromedial for valgus less than 30°; lateral for more than 30°
  - Others → anteromedial for all valgus knees

Advantages Ask AI

Lateral

- **Direct attack** on side of deformity
  - Site of **bony problems** and soft tissue problems
- Valgus → **MCL very lax**
  - MCL left **undisturbed** → avoids further release
  - Direct attack on **contracted lateral tissue**

Drawback: Closure Ask AI

Lateral

- Release + valgus correction → **lateral soft tissue and capsule shortage**
  - → closure difficult
- **Keblish → fat pad** technique → closure achieved (valgus lecture)
  - Shown with closure

Technique Ask AI

Lateral

- Standard **midline incision → lateral parapatellar** opening
- 1 cm of quadriceps tendon left toward **vastus lateralis** (not medialis)
  - Rest → toward medialis
- 1 cm left **lateral to patella** → proximally follow this line

Closure Ask AI

Lateral

- Closed **in flexion**
- After release → **lateral defect** always remains → closure unclear
  - Distally → **iliotibial band**
  - Proximally → **vastus lateralis**
  - Middle → partly **capsule**, partly **fat pad** → Keblish's reason
- **Fat pad preserved** → not excised at opening
  - Used at end → closes the defect (small or not so small)
- First → **transosseous suture** of iliotibial band into femur
  - → soft tissue approximation before closure starts

Clinical Pearl

- Lateral approach → **preserve fat pad** at opening
- Fat pad + capsule → close mid-lateral defect (**Keblish**)
- **ITB transosseous suture** into femur → before closure
- MCL in valgus → lax → left undisturbed

Closure in Flexion Ask AI

- **All TKAs closed in flexion** → anteromedial, lateral or mini
- Reason → **flexion achieved** from day one postoperatively
- Knee at 90° during closure → **soft tissues lax**

Extensile Approaches Ask AI

- **Quadriceps snip** → the classic one
  - Medial parapatellar extension + proximal cut
- Then → **V-Y quadriceps turndown**
- Illustrations → one quadriceps snip variant; others all V-Y turndowns
  - Classic V → **Coonse-Adams** → nobody uses it now
  - **Modified Coonse-Adams** → long limb + short limb at 45° → flap turned down
  - Distal extension variant → also a quadriceps snip

Quadriceps Snip Ask AI

Quad snip

- Indications
  - **Stiff knees**
  - Severe fibrosis
  - Revisions with **non-evertible patella**
- After usual anteromedial incision → cut into quadriceps tendon proximally
  - Direction → **parallel to vastus lateralis fibers** → VL not cut
  - Split as far up as needed
  - Vastus lateralis split as high as required
- Proximal portion → arthrotomy extended **obliquely**, parallel to VL fibers
- Cut angle 60° = angle of **VL insertion into quadriceps tendon** *[Q1 · As spoken 60°; standard (Insall/Garvin) quadriceps snip is an oblique proximal-lateral cut at about 45°.]*
  - → exposure extended

V-Y Quadriceps Turndown Ask AI

V-Y

- Current form = **modified Coonse-Adams**
  - Long incision → limb at 45° → flap turned down
  - **Midline** opening
  - **Revisions only**
- **Classic Coonse-Adams** → no longer performed
- Most used currently → quadriceps snip + V-Y turndown
- Medial limb → from **distal tibial tubercle** (tuberosity)
  - Distally → up quadriceps tendon to top → turn at 45°
  - Down **across quadriceps tendon** and vastus lateralis
  - **VL origin** + lateral retinaculum detached
  - Whole flap turned down → entire knee exposed without patella
- Closure
  - Flaps approximated → limb lengthens slightly → V limbs brought together
  - Closed as a **Y** → hence "V-Y quadriceps turndown"
  - Same **V-Y principle** as plastic surgery → quadricepsplasty, skin, flaps

Feature	Quadriceps snip	V-Y quadriceps turndown
Cut angle	60° → VL insertion angle	45° limb off long limb
Cut direction	Parallel to VL fibers	Across quadriceps tendon and VL
Indications	Stiff knee, severe fibrosis, revision with non-evertible patella	Revisions only
Basis	Medial parapatellar extension	Modified Coonse-Adams
Closure	—	Closed as Y → lengthening

Exam Pearl

- Quadriceps snip angle 60° → **VL insertion angle**
- V-Y turndown limb at 45° → **modified Coonse-Adams**
- Classic **Coonse-Adams** V → no longer used
- V-Y closure → **Y shape** → limb lengthens

Tibial Tubercle Osteotomy Ask AI

- Last approach → **very important in revisions**, especially difficult ones
- **Technical tricks** → once mastered → very easy

Dimensions Ask AI TTO

- **Incision** → midline, extended distally 8–10 cm
  - Because **osteotomy length** → 7–10 cm
- Length in sources
  - Some books → 5–7 cm; others → 8–10 cm
  - Lecturer → average → 7–10 cm
  - Most books → 7 cm and above, mostly 8 cm; one book → 5–7 cm
- **Thickness** → 1 cm
- **Distal end** → rounded/tapered, not square-ended
  - → avoids **stress riser**

Technique Ask AI TTO

- **Micro sagittal saw** (small saw) → medial → lateral
  - 1 cm deep → 1 cm of tibial tuberosity left
- **No lateral breakthrough** with the saw
- **Osteotome** → completes lateral break → fragment hinged over
- Rationale
  - Entire **periosteal sleeve** preserved
  - **Lateral soft tissue attachments** preserved → fragment vascularized
  - Probably aids healing once osteotomy closed
- **Adequate length** → avoids failure (picture)
  - Picture → labelled 10 cm, measures ~1.5 cm → too short
  - Unclear whether author meant it as long

Closure Ask AI TTO

- 3 **drill holes** → posterior part of osteotomy bed
  - **Medial** → **lateral** (picture B) → wires passed through
- After **implant** insertion → wires tightened → osteotomy closed

Problems Ask AI

TTO

- Revision with **anteriorly placed stem** → wires may not pass
  - → **pre-pass wires** beforehand or during stem preparation
  - → know whether stem needs **offset** or slight rotation
- Do not close until the end → **patella baja** may be found → needs raising
  - Raise → fragment moved **proximally**
  - Proximal translation → **proximal bone excised**
  - Otherwise → **bony prominence** in front of knee
  - → always excise with proximal translation

High Yield

- Osteotomy length 7–10 cm; thickness 1 cm
- Tapered distal end → no **stress riser**
- Saw medial → lateral; **osteotome** completes lateral hinge
- Anterior stem → **pre-pass wires**
- Patella baja → proximal translation + **proximal bone excision**

Take-home Message Ask AI

Setting	Approach
Everything → primary and revision	**Anteromedial**
Primary → alternative	**Subvastus**
Valgus	**Lateral**
Mini approaches	Specific training required first
Revision	Anteromedial still **workhorse** + quadriceps snip, V-Y turndown, tibial tubercle osteotomy

- Revision options → many
  - **Quadriceps snip**, V-Y turndown, **tibial tuberosity osteotomy** → very important

Total knee Arthroplasty

(Bone Cuts, Instrumentation and Patellar Preparation)

Prof Dr Sherief Khaled

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

- **C1** (p. 110) “1 mm to 9 mm valgus” → “1° to 9° valgus”
- **C2** (p. 114) “distal femoral condyles last” → “posterior femoral condyles last”
- **C3** (p. 123) “rotated 3° internal rotation” → “3° external rotation”
- **C4** (p. 127) “cannot do the left side” → “right side not done”

Overview of the Surgical Technique Ask AI

Scope and Bony Steps Ask AI

- Workshop system → **Johnson & Johnson** knee
  - Images → from its surgical technique manual
  - Slides = what the hands-on program shows
- **Total knee replacement** → two components
  - Diseased joint surfaces → replaced by prosthesis
  - Ligament balancing → restore normal **kinematics**
- Bony steps
  - **Three femoral cuts** → distal, anterior, posterior
  - **Chamfer cuts**
  - **Notch (box) cut**
  - **Patellar** preparation
- Cutting methods
  - Conventional instruments
  - **Computer navigation**
  - **Robotic assistance** → separate technique
  - **Patient-specific instrumentation (PSI)**
    - Concept sound; lecturer's group → PSI thesis years ago
    - Not used in Egypt → needs 3D-printed cutting block, unavailable then

Balancing Theories Ask AI

- Two classic theories → both established
  - Performable with all current instrument sets, any company
  - Each → own advantages and disadvantages
- **Hybrid technique** → advantages of both, disadvantages of neither
- Classic two
  - **Measured resection**
  - **Gap balancing**

Measured Resection Ask AI

Principle Ask AI MR

- **Measured resection** → bone cuts of known thickness = component thickness
  - Soft tissues ignored at this stage
  - Resected bone → replaced by **prosthesis** of same thickness
- **Femoral implant rotation** → from anatomical landmarks (previous lecture's lines)
- PCL requirement
  - Theoretically → **PCL** must be intact
  - PCL sacrificed (**posterior-stabilized, PS**) → true measured resection very difficult
    - Requires separate dedicated technique
  - Originally designed for **cruciate-retaining (CR)** knees
  - Current instruments → measured resection possible in PS too

Femoral Rotation Landmarks Ask AI MR

- Anatomic landmarks
  - **Transepicondylar axis (TEA)**
  - Anteroposterior axis → **Whiteside line**
  - **Posterior condylar axis (PCA)**
  - Images → from the **Alignment & Mechanics** lecture
- PCA → most accessible; **posterior femoral condyles** always palpable
  - Most instrument systems → rely on them for rotation and **femoral sizing**
- Normal condylar anatomy → 3° **external rotation** to PCA
  - → Posterior cut parallel to **TEA**
  - 3° → compensates normal knee anatomy
  - Applied to posterior femoral cut → balanced, **rectangular flexion gap**
  - Omitted → unbalanced flexion gap
  - Goal → symmetric, balanced flexion gap

Advantages Ask AI MR

- **Joint line** preserved
  - 10 mm femur removed → 10 mm prosthesis; same for tibia
  - → Joint line restored at original level
- **Patellar tracking** preserved
  - Joint line raised or lowered → tracking disturbed
- Correct **alignment** → fixed, measured cuts
- Easily **reproducible** → fixed cuts, no "art"

Disadvantages Ask AI

MR

- Dependence on **anatomical landmarks** → hard to palpate
  - Exposure needed → remove synovium to feel **medial femoral epicondyle**
  - **Epicondylar axis** → not easily identified
- **PCA** → depends on native anatomy
  - **Condylar hypoplasia** → altered PCA
  - **Condylar erosion** in valgus → altered PCA
  - → Inaccurate femoral rotation; landmark unreliable intra-operatively
  - Reference intact, surface deformed → not independent of **deformity**
- **Soft-tissue releases** → only after all bone cuts
  - After both **extension gap** and flexion gap made
  - Releases → disrupt flexion gap symmetry → **flexion gap imbalance**
  - Lecturer's view → not a good idea
- **Condylar lift-off**
  - Varus/valgus → one femoral condyle separates from tibial insert → re-contacts
  - → **Eccentric loading** → accelerated polyethylene wear
  - → Rapid joint wear
  - Instability and lift-off → at 0°, 30°, 60° and 90° flexion

High Yield

- Measured resection → cuts = implant thickness, soft tissues ignored
- 3° external rotation to PCA → posterior cut parallel to TEA
- Releases after all cuts → flexion gap imbalance
- Condylar lift-off → at 0°, 30°, 60°, 90° → eccentric loading → polyethylene wear

Gap Balancing Ask AI

Principle Ask AI GB

- **Gap balancing** → exact opposite of measured resection
  - Measured resection → cuts first, soft-tissue check after
  - Gap balancing → gaps (soft tissues) first → cuts decided accordingly
- Sequence
  - **Tibial cut** first
  - Spacer/block in extension → flexion gap cut to same thickness
  - **Femoral rotation** → not referenced to anatomical lines
- Steps
  - Symmetric **ligament tension** in extension
  - Femoral implant rotation → from symmetric flexion gap
  - E.g. extension gap 15 mm → flexion gap cut to 15 mm
- Cuts parallel to **proximal tibial cut** → fixed reference
  - Once made → never changed
  - Extension gap → **distal femoral cut**
  - Flexion gap → **posterior condylar cut**
  - Flexion gap matched to extension gap → regardless of condyle size
  - Regardless of TEA, **Whiteside line** → hence "gap balancing"
- Designed for **PCL-resected** knees → flexion gap balanced with extension gap before bony resection
  - → Considered easier for **PS** knees
- Cut → according to gap obtained

Prerequisites Ask AI GB

- Start with soft tissues, not bone
  - Remove **osteophytes**
  - Address **ligamentous contracture** or attenuation
- Correct measurement + correct resection → correct gaps (theoretically)
- Many conditions → just to set femoral rotation correctly
- All addressed from the start; osteophytes removed

Tibia-first Approach Ask AI GB

- Surgeons start with **proximal tibial cut**
  - Show of hands → about four attendees' surgeons do this
- Rationale → proximal tibial cut fixed → reference; everything matched to it
- Aim → better **flexion gap balance**, gap symmetry
  - No reliance on landmarks or fixed 3°
  - Fixed 3° → gap wider one side, narrower other → per varus/valgus

Advantages Ask AI GB

- Flexion gap well balanced → always **rectangular**, never trapezoidal
  - Measured resection contrast → rotation from lines
  - 3° external rotation often inexact → gap slightly wider medially or laterally
- No **lift-off** → no eccentric loading → no accelerated polyethylene wear
  - → No **prosthetic loosening**

Disadvantages Ask AI GB

- **Joint line elevation**
  - Tibial cut → controls everything
  - Large deformity → increased resection to widen gap
  - E.g. tibial cut → 2 cm extension gap → distal femoral cut matched to 2 cm
  - More distal femur resected → joint line elevated
  - → Patella relatively lowered → like **patella baja**
- **Mid-range instability**
  - **Posterior capsule** may be tight
  - Distal femoral cut increased → gap created against tight capsule → mid-flexion problem
  - Testing at full extension + 90° only → 45° forgotten
  - Show of hands → only two attendees test mid-flexion
  - High-standard surgeons → attention to **mid-range flexion stability**
  - Tight posterior capsule → laxity or tightness at 45°; insert may lift forward
- **Non-anatomic femoral rotation**
  - No anatomical lines → femoral cuts set by extension gap size only
- Loss of **posterior femoral condylar offset**
  - Large distal femoral cut → wide extension gap → large posterior condyle resection
  - → Reduced posterior condylar offset → reduced flexion
  - Deep flexion → posterior tibia **impinges** on posterior femur

Measured Resection vs Gap Balancing Ask AI

Feature	Measured resection	Gap balancing
First step	Bone cuts of measured thickness	Soft-tissue (gap) balancing
Femoral rotation	Anatomical landmarks (TEA, Whiteside, PCA)	Symmetric flexion gap
Fixed reference	Implant thickness	Proximal tibial cut
Designed for	CR (intact PCL)	PCL-resected (PS)
Joint line	Preserved	May be elevated → patella baja
Flexion gap	Risk of imbalance (releases after cuts)	Rectangular
Lift-off	At 0°, 30°, 60°, 90°	None
Other risks	Landmark error (hypoplasia, valgus erosion)	Mid-range instability; reduced posterior offset

Hybrid Technique Ask AI

Concept Ask AI

Hybrid

- Proposed → combine advantages, avoid disadvantages of both
- Combines
  - Accurate anatomical **component position** → coronal, sagittal, axial → as measured resection
  - **Joint conformity** (balanced gaps) → as gap balancing
- Avoids
  - Non-anatomic positioning → of gap balancing
  - Gap imbalance → of measured resection
- Preferred by most surgeons; all current **instrument systems**
- **Bony landmarks** + soft tissues → simultaneously
  - Not cuts first, balance later
  - Not balance first, cuts later
- "Femur first" or "tibia first" → both possible; same **concept**
- Simultaneous attention
  - Cuts, lines, femoral rotation, resection amount
  - **Soft-tissue tension**
- Steps alternate → soft tissue → measured cuts → soft tissue → cuts + **gap adjustment**
- Focus of this lecture

Sequence Ask AI

Hybrid

- All cuts as measured resection → measured resection, not hybrid
- Order
  - **Distal femoral cut** → intramedullary femoral alignment rod
  - **Tibia** → ligaments balanced (extension gap)
  - Complete femur → **flexion gap** → ligaments balanced
  - Flexion assessed → ligaments balanced with **trial implants**
- Tibia first or distal femur first → either fine
- Whole femur first → **measured resection** (ligaments ignored)
- Hybrid → distal femur → proximal tibia → complete femur

Pitfall

- Cutting the entire femur first → measured resection, not hybrid
- Hybrid order → distal femur → proximal tibia → remaining femur
- Tibia or distal femur first → either acceptable

Exposure and Distal Femoral Cut Ask AI

Know the Implant's Technique Ask AI

- **Every implant has its own surgical technique**
  - New company → request surgical technique; instrument demonstration first
- Example → **Smith & Nephew** workshop
  - Instruments reviewed → **Sawbones** case → technique review → then real case
- Unfamiliar instruments or joint system → very **dangerous**
- Techniques differ in details → review separately for each implant

Osteophyte Removal Ask AI

- After opening → remove **osteophytes** first
  - Distort landmarks and tensions in all directions
  - **Posterior osteophytes** → limit flexion (stops at ~90°)
  - **Tent the ligaments**
- Examples later → cases applying these points

Approach Ask AI

- **Medial parapatellar arthrotomy**
- **Evert** or sublux the patella
  - Tight → cut **lateral patellofemoral ligament**
- Excise **fat pad**; remove ACL
- First step → **femoral preparation**

Femoral Entry Point Ask AI

- Lines drawn → **Whiteside line** + TEA; others imagined
- Femoral canal → slightly medial to midline of distal femur
- Entry → 5–10 mm anterior to **notch**
  - Alternative reference → cruciate ligament attachment (as heard)
  - Ligament attachment → inconsistent between individuals
  - Notch → constant **bony prominence**
  - Osteophytes over notch → remove → true notch
- Slightly medial to **midline** → e.g. 2 mm medial
- Entry → **drill**

Distal Femoral Cut Ask AI

- Drill → remove → insert **intramedullary rod**
- Rod → guide for **distal femoral valgus cut**
- Angle → 5°, 6° or 7° → per patient
  - = Difference between **mechanical axis** and anatomical axis
  - Pre-op **long-limb films** → measure axes → set angle
- **Johnson & Johnson** guide → 1° to 9° valgus *[C1 · transcript: "1 mm to 9 mm valgus"]*
  - 9 notches; 3, 5, 7 labelled
- Guide on rod → setting pressed (5, 6, 7) → two **posts** on top
- Cutting block → slides onto the two posts
- **Pins** in "0" holes → cut through slot
- Block → removes 10 mm **distal femur**
- Average knees → 10 mm; usual angle 5–7°
  - Angle → from **full-length weight-bearing AP X-ray**
- **Saw** in slot → 8–10 mm from most prominent condyle

Cutting Accuracy and Instrument Problems Ask AI

- Beware **sclerotic bone** + blunt saw blade
  - Theory → new blade every case; rarely in Egypt
  - Sclerotic side → medial in varus, lateral in valgus
  - Saw deflection → **inaccurate cut**
- **Check the cut** → flat instrument across surface → catches or drops
- Original instruments → exact **slot**; no deviation possible
- Worn/swapped instruments → slot wider → cut error
  - → Joint won't close; component won't seat flush
- Choose company → good instruments or replacement policy
- Accepting worn instruments → continued supply
  - Refusal to operate → forces replacement
  - Requires everyone refusing; otherwise case goes elsewhere

Modifying the Distal Femoral Cut Ask AI

- Situations
  - **Fixed flexion deformity (FFD)** → frequent, marked in small-built, short males
  - **Recurvatum**
- Adjustment

Situation	Distal femoral resection	Effect
FFD	Extra 3 mm	Joint line raised 3 mm
Recurvatum	Less (e.g. 7 mm vs 8–10 mm)	Joint line lowered → patella relatively raised

- **Distal femoral cut** → extension gap
- **Posterior cut** → flexion gap
- Remaining femoral cuts → later

Proximal Tibial Cut and Extension Gap Ask AI

Proximal Tibial Cut Ask AI

- After distal femoral cut → **proximal tibial cut** → controls extension gap
- Landmarks drawn before **alignment device**
  - **Tibial crest**
  - Point at end of tibial crest → guide follows crest
    - Crest absent 10–15 cm above ankle
  - **Second metatarsal** line
- **Extramedullary alignment device** → parallel to tibia
- Resection
  - 8–10 mm → from least affected (**intact**) side
  - ~2 mm → from **affected** side
- Cut → perpendicular to **mechanical axis**
  - Tibia → mechanical and anatomical axes coincide
  - → 90° cut → 0° **varus/valgus**
  - Femur → 5–7° difference between axes
- **Stylus**
  - Diseased side → 2 mm
  - Intact side → 8, 9 or 10 mm → company-specific → read manual
- Guide options → **slotted** or non-slotted
  - Saw through slot or over guide surface

Exam Pearl

- Tibial cut → 90° to mechanical axis → 0° varus/valgus
- Tibia → mechanical = anatomical axis
- Femur → 5–7° between axes → distal femoral valgus cut

Tibial Alignment Landmarks Ask AI

- Line → junction of medial third / lateral two-thirds of **tibial tuberosity**
  - Centre of guide → along tibial crest → toward second metatarsal
- Rod → on **second metatarsal** with ankle at 90° dorsiflexion
  - Foot turned or plantarflexed → reference changes

Tibial Slope Ask AI

- **Posterior slope** of tibial cut → 0°, unless CR
  - Native slope → not 0 (~7–10°, as heard)
- **CR** → slope cut needed
- PS → not needed → 0° per company technique
- **PS insert** → slope built in
- CR design → no built-in slope → cut into bone
- Increasing slope → adjust distal part of **extramedullary rod**
  - Each forward advance → ~1° more slope
- Audience → no CR users → cut as is

Assessing the Extension Gap Ask AI

- After tibial cut → assess **extension gap**
  - Soft-tissue work here → hence hybrid
- **Spacer blocks** → company-specific
  - Johnson & Johnson → **blue spacer block** (extension)
  - **Motion block** (flexion) → 1 mm smaller
  - Principle → both spacers should be equal
  - Likely cause → larger femoral component **anteroposterior dimension**
  - Blue block for both gaps → error → know the system
- **Laminar spreaders**
  - Not in orthopaedic sets → must request
  - Two, opposed
  - **Osteoporotic** bone → spreader crushes, sinks medially → place at rim
- **Coronal (mediolateral) alignment**
  - Second rod (alignment rod)
  - **ECG electrode** → radiopaque marker over palpated hip centre
  - Distal → second metatarsal
  - Varus/valgus → detected

Extension Gap Balancing Ask AI

- Correct **distal femoral valgus cut** → rod aligned
- Varus/valgus → one side lax → **soft-tissue releases**
  - Reassess gap after releases
- Leg down → spacer inserted; simple releases if varus/valgus
- Spacer won't enter → increase **distal femoral cut** or tibial cut
  - Distal femur → +2 mm only, then stop
  - Reassess extension gap → limited releases → reassess **flexion gap**
- Repeated 2 mm femoral recuts → **joint line elevated** + flexion gap tight
  - Easier resection → **tibia**
- Instrument allowance
  - Distal femur → +2 mm
  - **Proximal tibia** → +2–4 mm
  - More → not normally needed
- Very tight knees → proximal tibia +5–6 mm and distal femur +4 mm
  - Small-built, short males with **flexion deformity** → most difficult

> **Pitfall**
> • Tight extension gap → distal femur +2 mm only, then stop
> • Repeated femoral recuts → joint line elevated + tight flexion gap
> • Further resection → from the tibia

Trial Components and Balancing Ask AI

Femoral Rotation and Sizing Ask AI

- Next → rotation and **femoral sizing**
- **Femoral sizing guide** → stylus on highest point of **lateral anterior femoral cortex**
  - Size read from guide
- Pegs at 3° → fixed with **pins** → guide rotated
- **4-in-1 cutting block** → four cuts
  - Anterior, posterior
  - Anterior and posterior **chamfers**
- **Notch (box) cutting guide** → box cuts

Trial Reduction Ask AI

- **Femoral trial** + tibial trial + thin insert → range of motion
- Problem → **soft-tissue releases** → balance flexion + extension together
  - Assess at 90° and mid-flexion (45°)
- Basics only → varus and valgus → separate **lectures**

Basic Releases Ask AI

Varus

- Primary knee, tight medially
  - **Osteophytes** → tibial + femoral, medially
  - **Periosteal sleeve** → ~2–3 cm below joint line
- **PCL** → released in both varus and valgus
- Still tight → **posteromedial capsule** + posterior oblique ligament (POL)
  - Cut or **pie-crusted**
- Finally → **semimembranosus**
- Varus lecture → more options, e.g. **medial plateau** reduction osteotomy (as heard)

Basic Releases Ask AI

Valgus

- Remove **PCL** → PS design
- **Pie-crust** posterolateral capsule
- Then → iliotibial band or popliteus first → per surgeon

Surgeon background	Release first	Release last	Reason
Sports injuries	Iliotibial band	Popliteus	Popliteus → important structure
Arthroplasty	Popliteus	Iliotibial band	Popliteus release → no concern

- Tight **iliotibial band** → pie-crusting
- Details → valgus lecture, incl. bony cuts

Gap Imbalance Troubleshooting Ask AI

- Balanced in flexion + extension → case complete
- Think each situation through
- Tightness from under-resection (recording unclear) → **long distal femur**

- **Loose in flexion, tight in extension**
  - **Upsize the femur** → fills flexion gap
  - Upsizing → increases AP dimension
  - + Resect more distal femur → extension gap
- Alternative → **posterior augments** behind femoral component → same effect as upsizing
  - May need anterior component shift → minor
  - Many companies → augment only on **stemmed femoral component**
  - Some companies → stemmed femur only in **revision** systems
    - CCK, rotating hinge → stemmed femur
    - Primary system → stemmed tibia only
  - → Read the implant's technique manual
- **Tight in flexion, balanced in extension**
  - Downsize femur or increase posterior cut
  - Increased posterior cut → reduced **posterior offset**
  - Options
    - **Downsize femoral component** → AP dimension only
      - Other dimensions unchanged
    - Release **PCL** → already done in PS
    - Increase **tibial slope** → re-mount extramedullary guide → recut
      - Front raised 7 mm → ~1° posterior slope
    - Resect more **posterior femoral condyle**
      - → Reduced posterior femoral offset → reduced flexion
    - Release **posterior capsule**
- Release before recut
  - PCL + posterior capsule → **soft-tissue releases** first
  - Recut unavoidable → minimize additional resection
  - Next → **tibial side** (slope)
  - Last → posterior femoral condyles → preserve offset *[C2 · transcript: "distal femoral condyles last"]*
- **Loose in flexion, balanced in extension**
  - Extension balanced; flexion opens up
  - **Upsize the femur**
    - Same size, no thicker insert → flexion stays loose
    - Adjusting for extension → spoils balanced extension
  - Or posterior femoral **augments** → harder if stemmed component required
- **Loose in flexion and extension** → build up tibia
  - Thicker **polyethylene insert**
  - Metal tibial augments
- Problem-solution table → understand problem first; don't memorize
  - E.g. problem → over-resection of a structure

Situation	Solution
Loose in flexion, tight in extension	Upsize femur + more distal femoral resection; or posterior augments
Tight in flexion, balanced in extension	Downsize femur; PCL/posterior capsule release; ↑ tibial slope; posterior condyle resection last

Situation	Solution
Loose in flexion, balanced in extension	Upsize femur; or posterior femoral augments
Loose in flexion and extension	Thicker insert or metal tibial augments

Clinical Pearl

- Soft-tissue releases before changing bone cuts
- Posterior condyle resection → last option → preserves posterior femoral offset
- Posterior femoral augment → often requires stemmed femur → revision systems only

Tibial Component Rotation Ask AI

- **Keel** preparation → three landmarks
  - Central → medial third of tibial tuberosity
- **Keel punch** placed → impacted; check rotation
- Rotation toward posterolateral or medial → both **unacceptable**
- **Medial overhang** → irritates medial ligament → pain
  - Alters patellofemoral tracking
- Excess **external rotation** → same effect

Patellar Management Ask AI

Resurfacing: Yes or No Ask AI

- **Patellar resurfacing** → long-standing controversy
- Audience poll (~40) → few resurface
- Advantages
  - Less **anterior knee pain**
  - Better **patellofemoral function**
- Opponents → complications 5% to 60%
- Source author → resurfaces in all cases
  - Less anterior knee pain + better PF function
  - His **complications** → 5–10%, never 60%
- Mandatory resurfacing (even for non-resurfacers)
  - Very advanced **patellofemoral osteoarthritis**
  - **Inflammatory diseases** → rheumatoid arthritis, systemic lupus, ankylosing spondylitis
- Routine resurfacers → one exception → discussed next

Exam Pearl

- Resurfacing → less anterior knee pain, better patellofemoral function
- Mandatory → very advanced patellofemoral OA
- Mandatory → inflammatory/rheumatological disease

Patellar Cut Ask AI

- Cut parallel to articular surface → **tilted** cut
- Cut **medial more than lateral**
  - Medial → below articular surface
  - Lateral → at articular surface level
  - Otherwise → tilted
- **Patellar button** → medial
  - Prevents **laxity** and maltracking
  - **Femoral component** → lateralized
  - Patellar **ridge** → more medial than lateral, not central
  - Button over thicker part; other region lacks bony support
  - Slightly medial → reproduces ridge
- Cut depth → per **button thickness**

Patellar Thickness Ask AI

- Target **total patellar thickness** → 20–23 mm (some → ~23 mm)
  - Button 8.5 mm (thinnest) + 12–13 mm residual bone
- 20 mm patella − 8.5 mm → 11.5 mm → insufficient
- **Minimum residual thickness** → 12–13 mm
  - Some books → up to 16 mm
- Too thin → **fracture** during cementing or through cement holes
- **Sizer** → measures cut; stylus available
- Preparation → per button shape
  - Three **peg holes**
  - Drill bits → **stopper**
  - No stopper → perforation + fracture (thick drill)
- **Cement** → button compressed with clamp

Patelloplasty Ask AI

Patella

- Common non-resurfacing practice → **circumferential "circumcision"** with diathermy only
  - Wrong → insufficient
- **Patelloplasty** → defined steps
  - Remove **osteophytes**
  - Pits/irregularities → smoothed with **file**
  - **Lateral facetectomy** → part or all of lateral facet
  - Sclerotic bone → **drilling**
  - Circumcision (circumferential diathermy) → **denervation**
  - Excess fat pad/synovium → **excised**
- No resurfacing → complete **patelloplasty**, not circumcision alone

Patellar Tracking and Q-angle Ask AI

- Preserve **Q-angle**
- ↑ Q-angle → poor tracking → **tendency to sublux**
  - **Internal rotation** of femoral component → patella lateral
  - **Medialization** of femoral component → patella lateral
  - **Internal rotation of tibial component** → patella lateral
  - **Lateralization of patellar button**
- **No-thumb test** on closure
  - **Tourniquet** tethers quadriceps → test inaccurate
  - Edges held with **towel clip** or one stitch
  - Some → no aids (lecturer, in a few cases)
  - Very tight cases → patella tends to **subluxate**
  - One stitch in **retinaculum** → test tracking (lecturer hasn't used)
- Any of the four errors → patella lateral
  - → ↑ Q-angle → poor tracking → **patellofemoral instability**
- Key → **component rotation** + mediolateral position
- Rule → **lateralize everything except the patella**
  - Femur → lateral
  - Tibia → lateral
  - Patella → **medial** → original vertical ridge medial

Must-Know

- Lateralize everything except the patella
- Patellar button → slightly medial → reproduces the ridge
- ↑ Q-angle → femoral IR, femoral medialization, tibial IR, lateral button
- Patellar cut → medial more than lateral
- Minimum residual patella → 12–13 mm

Cementing, Closure and Radiographic Evaluation Ask AI

Final Checks and Cementing Ask AI

- Final trials; rod re-inserted → reaches **ECG lead**
- Then → **cementing**
- Bone surfaces → washed well → **pulsatile lavage** or saline
  - Dried → gauze
  - Optional → drill sclerotic bone
- Cement → **component undersurface**
  - Middle + anterior/posterior undersurface
  - Avoid posterior condyles of component → extruded cement irretrievable
- Same sequence → J&J workshop

Joint Line Reminder Ask AI

- **Joint line** references (previous lecture, as heard)
  - ~10 mm from fibular head
  - ~30 mm from medial epicondyle
  - 10–15 mm relative to patella
- Elevated → poor **patellofemoral tracking** + mid-flexion instability
- Lowered → **instability** likewise
- Closure → post-operative X-rays
- Audience question → insert first?

Knee Society Radiographic Evaluation Ask AI

- **Knee Society** radiographic evaluation system
- Few attendees use it → only notching checked
- **Component alignment** angles
  - **Alpha angle** = distal femoral angle
    - Depends on valgus cut → 5° cut → 85°
  - **Beta angle** = proximal tibial angle → 90° if correct
  - Lateral → **theta angle** + sagittal angle → femoral component aligned with femur
  - Lateral tibial slope → PS 90°; CR → some slope per design
    - Advanced **extramedullary guide** → e.g. 89°
- **Eccentric component placement**
  - Medial/lateral, superior/inferior
  - Subluxation, dislocation
- **Tibial surface coverage**
- **Patellar problem list**
- **Radiolucencies** → 7 tibial zones (AP)

Radiolucency score	Interpretation
4 or less, non-progressive	Probably insignificant
5–9	Close follow-up for progression
10 or more	Likely significant → possible/impending failure

- Zones

Component	View	Zones
Patella	—	5
Tibia	Lateral	3
Tibia	AP	7

- Image in all books → evaluate any TKA on **AP + lateral**

Take-home Message Ask AI

- Balancing techniques of TKA
  - **Measured resection**
  - **Gap balancing**
  - **Hybrid technique**
- End of lecture

Sawbones Workshop: Bone Cuts Ask AI

Setup Ask AI

Workshop

- **Demonstration on phantom (Sawbones)** → shown on screen
- Five-minute break → session filmed

Femoral Landmarks and Entry Point Ask AI

Workshop

- **Whiteside line** → diathermy from deepest trochlear groove → top of notch
- Lateral epicondyle → medial epicondyle → **TEA**
- Two constant lines; **posterior condylar axis** ignored
- **Entry point** → 5–10 mm above (anterior to) notch
  - Model femur → unlike real; cut off at top
  - Bowed model → entry through **notch**

Distal Femoral Cut Ask AI

Workshop

- Valgus guide → 6° (bowed model)
  - Flat against femur
  - Pins in "0" holes (assistant **Bassem**)
  - Zero holes → marked with a **square**
- Distal femoral cut first → **valgus cut** at 6°
- Guide disassembled; rod removed
- Noted → big problem in the **technique**

Tibial Cut Ask AI

Workshop

- Guide → roughly parallel to tibia (lower region)
  - Points toward **second metatarsal**
  - Mark second metatarsal + **tibial crest** beforehand
- Mounted around **centre of ankle**
- PS → no slope → guide parallel → zero
- Front advanced up to 7 mm → ~1° more **slope**
- **Tibial tuberosity** marked at medial third / lateral two-thirds
  - Aligned with guide line → some call it modified Akagi line (as heard)
- Stylus 2 mm → **diseased side**
  - Lateral → 8 mm in this system
  - 8 mm system → **bone-preserving**
  - Model → no worn side → 2 mm medial + 2 mm lateral
- Guide → half **pins** (advantage)
- Pins → **zero slot** (square)
- Guides → +2 and +4 mm **holes**
- **Angel wing** → confirms cut position
  - Always remember it, every step
- Half (cut-out) arch → avoids **patellar tendon**
  - Complete arch (most companies) → presses on tendon

Tibial Defects Ask AI

- Small (contained) → **cement**
- Larger (as heard) → **graft**
- **Wedge (augment)** → sizes and heights vary by company
- Extent → third, half or full of **plateau**

Extension Gap and Alignment Check Ask AI

Workshop

- Extension gap → **blue block**
- Flexion → **motion block** (smaller)
- Rod relative to **ECG lead** over hip → confirms valgus cut
- 6° correct → mechanical–anatomical difference in this **knee** = 6°
- Then → **femur**
- Marked varus/valgus → **minimal releases** → tibial + femoral osteophytes
- **Medial release of tibia**
  - Medial periosteum elevated 2–3 cm below articular surface
  - Releases **deep MCL**
  - **Superficial MCL** → intact

Femoral Sizing and Rotation Ask AI

Workshop

- **Femoral sizer** → anterior or posterior referencing
  - Most instruments → both
  - Each → own pros and cons (not discussed)
- Deformity (e.g. valgus, condylar deficiency) → **anterior referencing**
  - Posterior referencing → misleading
- Usual → sizer on condyles, 3° external rotation, **anterior stylus** on lateral anterior cortex
- **Valgus** → posterior condyles inaccurate
- J&J external rotation → 3°, 5° or 7° → per **posterior femoral condyles**
- Model measured size 6 → **size 3** used (6 unavailable)
- 3° pegs locked → guide rotated 3°
- Peg 5 → 5° **external rotation**

4-in-1 Block and Box Cut Ask AI

Workshop

- **4-in-1 cutting block** → four cuts
  - Anterior, posterior
  - Anterior chamfer, posterior chamfer
  - **Chamfers** → corners of curved component edges
- Square slot at 0 → cuts exactly per book
- Block shifted anteriorly → saw cuts more **posteriorly**
- **Tight flexion, balanced extension** → block on +2 (lower hole)
  - → 2 mm more from **posterior femoral condyle**
  - → Wider flexion gap
- Mind your **hand**
- Measured 6, used 3 → whole anterior part removed (not **notching**)
- **Anterior cut first** → otherwise obstructs chamfers
- Some instruments → smaller blade for chamfers → read technique
- **Narrow saw blade** → box cut for femoral component

Sawbones Workshop: Trials, Patella and Closure Ask AI

Femoral Trial Ask AI

Workshop

- Box cut → **protect the tibial cut**
- **Flexion gap** formed → fine-tuning possible
- Trial size 3; **impactor**
- Uncut posterior chamfer remnant → no **free-hand cuts**
- Recut → size 4
- **Introducer** + sliding hammer → insert/extract femur

Tibial Trial and Rotation Ask AI

Workshop

- **Tibial trial**
- **Curve-to-curve**; fragment fell into defect
- Don't force → **posterolateral overhang** undesirable
- Size 2.5 → small → 2 used
- Medially **line-to-line** → correct rotation
  - No posterolateral or medial overhang
- Trial reduction on existing pins → **punch**
  - **Cancellous bone** → punch sinks normally
- Drill in **reverse** → compacts bone
  - Plastic bone → forward
  - Stop → limits depth
- **Tower** with stop → impact keel per size

Tibial Defect Graft Ask AI

- **Graft** fixed → two headless screws → cement → tibial tray

Patellar Preparation Ask AI

Workshop

- Patella prepared before cementing → cement opened once
- First instrument → **sizer for patellar thickness**
  - Patella < 12–13 mm → cannot resurface
- **Clamp** → holds patella → non-tilted saw cut
  - Design acceptability → varies by company
- Residual after cut → ~8.5–11 mm (8–11 mm) → **system-specific**
- Button thickness → depends on patella amount
- Button → **slightly medial** (ridge medial), slightly superior
- **Three-peg drill** → stopper at end of slot
- Thin patella → cement drill **perforated** to other side
- Trial patella → size 32 → in three peg holes
- Real → cement → clamp closed → **locks** → held until cement sets
- Cut → more **medial** than lateral
  - Parallel to articular surface → slanting

Closure and Peri-articular Injection Ask AI

- Small **posterior osteophytes** in notch → removed with small osteotome (unavailable here)
- Finish femur → recheck **patellar tracking** → close retinaculum
- **Drain** → most don't; optional
- Before closure
  - **Tranexamic acid** 1 g → intra-articular
  - **Peri-articular injection** → Marcaine + Xylocaine in saline + ketorolac + 8 mg dexamethasone
  - → Real difference in **pain control**
- Capsule, subcutaneous, skin → closed with **knee flexed**
- **Crepe bandage**
- **Weight-bearing as tolerated** + range-of-motion exercises

Questions and Answers Ask AI

Femoral Rotation in Valgus Knees Ask AI

- Methods → posterior referencing or **external rotation** options
- Lecturer → **anterior referencing**
- Valgus → **femoral condyles** deficient → difficult
- **Valgus knee**
  - Lateral **posterior condyle** → hypoplastic, eroded
  - → Small lateral, large medial condyle
  - **PCA** tilted 10° to TEA (not usual 3°)
  - Sizer on posterior condyles → rotated → everything wrong
- → Ignore posterior condyles
  - **Anterior referencing** → Whiteside line + TEA
- **Posterior cut parallel to TEA**
- Set 3° on condyles → actually 10° **internal rotation**
  - 3° external rotation → still 7° internal rotation *[C3 · transcript: "rotated 3° internal rotation"]*
- J&J → 5° or 7° options
- **Hybrid technique** → relies on Whiteside line + TEA, never posterior condyles

Pitfall

- Valgus knee → hypoplastic, eroded lateral posterior condyle
- PCA → 10° to TEA, not 3°
- 3° ER on condyles → still 7° internal rotation
- Valgus → anterior referencing; posterior cut parallel to TEA

Residual Cartilage in Valgus Ask AI

- Valgus distal cut → intramedullary rod, no lateral release
  - Residual lateral **cartilage** → left
- Drill **sclerotic bone**
- 2 mm cartilage left → restore **joint line**
- Don't increase **distal femoral cut** → joint line raised
  - Valgus → lateral condyle barely touched
- Cartilage → **file** or drilling at end
- Distal femoral cut +2 mm needed → cartilage removed; never beyond 2 mm
- No routine constrained construct (**CCK**)
- Depends on **degree of valgus** → 15° ≠ 50°
- 50° valgus with severe combined **deformity** → different
  - E.g. 8 mm medial distal femur cut; lateral condyle 2 cm short
  - Resect 2.5 cm distal femur? → No → **compensate** difference
  - Depends on deformity → valgus lecture

Tibial Component Rotation Ask AI

Q&A

- Questioner → **curve-on-curve**
  - Tight → slightly posterolateral, not posteromedial
- Answer → curve-on-curve alone **insufficient**
  - Medial third of **tibial tuberosity**
  - Line through medial part of tibial tubercle → **black line** on instrument
  - Curve-on-curve
  - **Rod** parallel to tibia → on second metatarsal
- Combined → avoids **internal rotation**
- Correct placement → posteromedial or posterolateral tibia uncovered
  - Divide exactly; slight shift → not a problem
- Uncertain medial vs lateral → 1 mm **lateral**
- All lines together → tuberosity third, rod on second metatarsal, curve-on-curve

Hybrid Technique Recap and Reading Advice Ask AI

Six Goals of the Hybrid Technique Ask AI

- **Hybrid technique** → advantages of gap balancing + measured resection; problems avoided
  - Accurate **bone cuts**
  - Correct **rotation**
  - Accurate **anatomical lines**
  - Correct **soft-tissue balancing** → medial + lateral ligaments
  - Accurate **flexion and extension gaps** → per patient anatomy
  - **Restore the joint line**
- → Six goals achieved
- Relies on **soft-tissue balance** + anatomical lines + measured resection

Reading the Literature Ask AI

- Many concepts → audience confused → pace kept slow
- Check when reading
  - **Cruciate-retaining** or cruciate-sacrificing
  - **Femur-first** or tibia-first
    - J&J → either
  - **Measured resection** or gap balancing → makes a difference
  - **Distal femur first** or proximal tibia first → step order
  - **Primary** or deformity (varus/valgus)
  - Primary **knee OA** vs post-traumatic, inflammatory, rheumatological
- Confusion → reading **retaining** techniques (slope cut, guide advanced, PCL kept)
  - Tight flexion → PCL **snips** to lengthen → check rollback
- **References** → ~12 books; names posted to group

Clinical Cases Ask AI

Case 1: Bilateral OA with Raised Inflammatory Markers Ask AI

- 60-year-old **female**
  - Bilateral **knee OA**
  - Bilateral **genu varum**
  - Bilateral **flexion deformity**
  - CRP 10, ESR 60–80 (first hour) → **worrying**
- Rheumatological condition → all joints affected post-op
- **AP view** → presented June 2016
- **Skyline view** → patellar involvement → resurfacing option
- **Long film** → angles
- **Rheumatology** referral → no rheumatological disease
- Left knee
  - **Distal femoral valgus cut** → 6°
  - **NexGen** (Zimmer), tibia size 3
  - LPS-Flex **insert** → 10 mm
- High **acute phase reactants** → always synovial biopsy + culture
  - Surprises → once **tuberculous synovitis** (TB knee)
    - Doing well 6 months post-op
    - No biopsy → post-op secondary bacterial infection → **septic knee**
    - **Tuberculous granuloma** → TB test positive → chest diseases → treated
- This patient → **chronic non-specific synovitis**; culture → no growth
- Flexion deformity → **posterior capsular release** + osteophyte excision
- Post-op X-ray → poor → request **true lateral view**
- True lateral → **posterior osteophytes** removed
  - Good posterior femoral offset → full flexion
  - Sizing good; 10 mm insert good
  - Everything lateralized, except **patella**
- Two months later → **DVT** left lower limb + patellar clunk
  - **Vascular** referral → DVT medication
- Eight months after first operation → **anticoagulants** finished → right knee
- Right → patellar involvement too
- **Synovium** → very abnormal
- Right knee → **synovial biopsy**; 6 mm cut; sizes differ
  - Right tibia fit more accurate; left slightly overhanging
  - → Rationale for **gender-specific** implants or PSI → improved cuts
- Poor patella → **all-polyethylene patella** 32 mm (Zimmer), 8.5 mm thick
  - Cement holes visible
  - Insert 12 mm; **posterior osteophytes** removed
- **Patellar button** → correct centre and rotation, slightly medial
  - Lateral → trouble
- Three months post-op → persistent **swelling**
- Rheumatologist → **antiphospholipid syndrome**

  - Present since 2016 → undetected → should have been treated from start
- Difficult **diagnosis**
- **Fall on knee** → nine months post-op
  - **Extensor mechanism** → intact
  - Crushing injury → patella completely crushed; very thin shell
  - Loose **fragments** → removed
  - Patellar augmentation device (term as heard) → not available in Egypt
  - Patella only addressed
- Trial → **mid-range instability**
  - → Flexion deformity
  - Insert 12 mm → 17 mm
  - **MCL plication** at 45° flexion
  - Third **synovial biopsy** → confirm antiphospholipid syndrome
- Pathology → sometimes **"rheumatoid synovium"** with good treatment response
- **Follow-up** → 4 years left, 3.5 years right
  - Small patellar remnant; walking normally; extensor mechanism intact
  - Right 17 mm insert vs left 10 mm
- TKR → **life-changing procedure**; complications possible
- **Rheumatological conditions** → careful planning

Clinical Pearl

- Raised acute phase reactants → synovial biopsy + culture in all cases
- Synovial biopsy → once revealed tuberculous synovitis → avoided septic knee
- Rheumatological conditions → careful planning; don't forget synovial biopsy

Case 2: Varus Knee, Stemmed Tibia Ask AI

- **Medial** and tight → medial release
  - **Basic releases** → medial osteophytes, deep MCL, periosteum
  - No semimembranosus, no POL, no **augments**
  - **Sclerotic bone** drilled medially
- **NexGen** with tibial stem
  - Stem → 15 mm diameter, 75 mm length
  - Stem + tibial keel → 120 mm (12 cm)
  - LPS-Flex **insert** → 10 mm
- **Stem indications**
  - Varus > 15°
  - Obesity
  - Osteoporosis → present here
- **Pre-emptive analgesia** pre-op
- Post-op → **peri-articular injection**; tranexamic acid
- Patient from Esna → next-day discharge → **no drain**; PCA post-op
- **Alignment** → lateral view pre vs post → varus improved
- **Kinematic alignment** → deformity left as is
- **Restricted kinematic alignment** → partial correction; not left markedly deformed
- **Haemoglobin** → 9.7

- Work-up → cardiac abnormality → **cardiology** → catheterization
  - → **Open heart surgery** → following year, late 2022
- **Double anticoagulants** + very high HbA1c → not optimized → right side deferred
- Last seen 2024 → 4 years post-op; still **medical problems**
- 64 years at left TKA → now 67–68 → right not done → **medical comorbidities** *[C4 · transcript: "cannot do the left side"]*
- → **Optimization of patients** (previous lecture)
- Lessons
  - **Pre-operative optimization** → crucial
  - **Multimodal analgesia**, pre-emptive → beneficial
  - **Flexion deformity** → posterior releases
  - **Medial tightness** → release in sequence
  - **Stems** → varus > 15°, obesity, osteoporosis
  - After **open heart surgery** → postpone TKA ≥ one year (as heard)
  - After **stents** → 6 months

High Yield

- Stem indications → varus > 15°, obesity, osteoporosis
- After open heart surgery → postpone TKA
- After stents → postpone 6 months
- Restricted kinematic alignment → partial deformity correction

Case 3: Varus with Posterior Osteophytes Ask AI

- 58-year-old **female**
  - **Advanced OA**
  - Varus + **posterior osteophytes**
- **PFC Sigma** (Johnson & Johnson)
  - Distal femoral valgus cut → 5°
  - Femur 2.5, tibia 2.5
- **Half sizes** (2 → 2.5 → 3) → preferable
  - Here → size 2 small, 3 large
- **Stem** → 60 mm long, 13 mm diameter
  - Short stem, always **cemented**
- Insert → 10 mm

Case	Implant	Valgus cut	Sizes	Insert	Stem
1 (left)	NexGen (Zimmer)	6°	Tibia 3	LPS-Flex 10 mm	—
1 (right)	NexGen; all-poly patella 32 mm × 8.5 mm	—	—	12 mm → 17 mm (revision)	—
2	NexGen	—	—	LPS-Flex 10 mm	Tibial, 15 × 75 mm (120 mm total)
3	PFC Sigma (J&J)	5°	Femur 2.5, tibia 2.5	10 mm	60 × 13 mm, cemented

Case 4: Gunshot Injury of the Knee Ask AI

- 31-year-old man → **firearm (machine-gun) injury** (Upper Egypt)
- November 2019 → **open tibia** (external fixation)
  - Open wound managed
- **Tibial plateau** defect; femur fractured; defect below plateau
- Best options → above-knee **amputation** or arthrodesis
  - Hard to accept for Egyptian patients given means
- Referred as "difficult **tibial plateau**"
- **Reconstruction** very difficult → open reduction pointless
- Age 31 → open reduction + return to work as driver if possible
  - Else → **arthrodesis** once bone stock sufficient
- Single bullet, through-and-through → debridement
  - Two fixators → **delta frame**
- **Machine-gun (high-velocity) wounds** → usually not heavily contaminated
  - → Left open; drained
- Presented at **AO Davos course** (2019)
  - Karl-Heinz Frosch → approaches
  - Cong-Feng Luo; Schatzker; Sean Nork; Matt Graves
- Opinion → no return to work

- Luo → **three-column classification** of tibial plateau fractures → no guns in China
- Eastern European approach → **Ilizarov** + bone stock restoration → arthrodesis
- **Masquelet technique** → cement
- Patient **refused amputation and arthrodesis**
  - Plan → bone stock for TKA or arthrodesis
  - → Ilizarov + **cement spacer**
- Cement in void → **Masquelet technique**
- **Ilizarov** applied → image six weeks post-op
- **Cross-knee** frame into femur (December 2019)
- Three months later → cement spacer removed
  - → Two **femoral heads** as allograft
- Follow-up (~2/2020, as heard) → some **bone stock**
- October 2020 → **rings above knee** removed
  - Walks with stiffness; limb shorter
  - Tibia not centred → second procedure → centred under **tibial plateau**
- Later picture (June 2021, as heard)
- **Ilizarov removed** (September 2022, as heard)
  - Alignment centred; nearly arthrodesed (some space remains)
  - → **Extension stiffness**
  - Patient wanted more → return to work
- **Limb-length discrepancy** → limp; short leg
- **Total knee** decided (February 2022/2023, as heard)
- Post-op → **offset stem** + proximal screws (prevent fracture)
  - **Fibular ostectomy**
- Complication
  - **Image intensifier** at every step → not after final implantation
  - **Stem outside** → between tibia and femur
  - Next day → back to OR → tibia redone
  - **Tibial plateau fracture** → stem exited through it
  - Intra-op AP + lateral stem images → none after final **implantation** (October 2022)
- February 2023 **follow-up**
- December 2022 → **union** complete
  - Patella → nothing done
  - February 2023 image
- August 2023 → ~1 year → last **image**
- Last visit (August 2023) → **walking**

Pitfall

- Image after final implantation, not only intermediate steps
- Missed → stem exited through tibial plateau fracture → next-day revision

TKA tips and tricks

(An Operative Sequence from Assessment to Closure)

Prof Dr Mostafa Shokry

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 142) “slope of 7 mm” → “7° slope”

C2 (p. 156) “pillow under the knee” → “pillow under leg, not under knee”

Queried - kept as spoken, but looks wrong

Q1 (p. 157) as spoken; conflicts with one-week stop earlier; LMWH usually stopped 12-24 h preop

Scope and Operative Steps Ask AI

- Lecturer's own practice and years of operating → all with one system
- Follows operative sequence → from **history** and examination onward
- Applied practice points only → not textbook, not every point of view
- Steps covered, in order
  - **Preoperative assessment**
  - Medication given with induction
  - Positioning
  - Skin preparation
  - Tourniquet
  - Draping
  - Surgeon and assistant position
  - Skin incision
  - Deep tissue dissection
  - Distal femur preparation
  - Proximal tibia preparation
  - Extension gap assessment
  - Sizing
  - Rotation of the femoral component
  - Finishing the femoral cuts
- Remaining steps → following slides
- Scope: uncomplicated **primary total knee**
  - May include varus deformity + mild soft-tissue contracture
  - Excludes major problems needing stems or augments
- System: **Zimmer NexGen fixed-bearing** total knee

Preoperative Assessment Ask AI

History Ask AI

- Most patients **female**
- Age: formerly elderly only → now fifties, forties, perhaps younger
- **Weight** → see and estimate before operating
- Female patients → thigh size often missed without proper examination
- Leg gives **false impression** of thigh size → examine and assess thigh
- Post-lockdown → large patients expected as routine surgery resumes
- Ask about
  - Old joint injury or **trauma**
  - **Previous knee injection** → very important, almost never asked
    - Hyaluronic acid or steroid injection → may cause **infection**
- Medication
  - All **anticoagulants** stopped one week before surgery
  - Bridge with **Clexane** → subcutaneous injection

Examination Ask AI

- Look at the patient as a whole
- **Male manual worker** → large muscle bulk + OA → affects approach choice
- Usual approach: **subvastus** → the one described here
  - Muscular patient → wary; muscle bulk obstructs view
- Also assess
  - **Previous scars**
  - Effusion
  - Thigh size
  - **Varicose veins** → affect operation, postoperative bleeding, dissection
  - Existing deformities → varus magnitude → tibial cut needed
- Detailed in Dr. Mahmoud's lecture → not repeated
  - Also note **flexion deformity** and ischaemia

Investigations and Imaging Ask AI

- Usual labs, including **PT and PTT** (coagulation profile)
- **Urine analysis** → often omitted, but important
  - UTI often present → postoperative problems
- ECG and **echocardiography**
- Imaging: **standing AP** and lateral views
  - No full-length standing X-ray in Egypt → **CT scanogram** instead
  - **Skyline view** → patella and patellar tracking
  - **CT of distal femur** (sometimes) → transepicondylar axis + rotation angle for later

Induction, Positioning and Theatre Setup Ask AI

Medication at Induction Ask AI

- Remind anaesthesia team → antibiotic + Cyklokapron
- **Cyklokapron** → routine in these cases
- Antibiotic: **Fortum** (ceftazidime) 1 g with induction → strongly preferred
- **Tranexamic acid** (Cyklokapron) 1 g
- Both **IV** → drip or direct injection
- Timing: **before tourniquet inflation** → must reach circulation first

High Yield

- Fortum 1 g + Cyklokapron 1 g IV at induction
- Both given before tourniquet inflation

Positioning Ask AI

- Europe/developed countries → dedicated **posts**; here → Egypt, most hospitals
- Large **IV fluid bottle** → post (bolster) for the leg
  - Directly under bend of knee → ideal place, always used, recommended
  - Used for opening, releases, exposure
- Some add second bolster at 90° flexion
  - Helps at start, before full flexion achieved
- **Side stand** (side support) → supports leg, prevents excessive movement
  - Usually at tourniquet level, knee flexed
- Patient **supine**
- Other devices elsewhere → this is the hospital's setup

Skin Preparation and Tourniquet Ask AI

- **Shaving on the table** → never earlier
  - Earlier shaving → small wounds → infection
  - Requires thorough cleaning of field
- **Tourniquet** → well padded, as high on thigh as possible
- Exsanguinate with **Esmarch** → inflate with knee flexed
- Why flexed? → lengthens **quadriceps** before clamping
  - → easier dissection
  - → proper intraoperative patellar tracking assessment
- After inflation → wash with **povidone-iodine** (Betadine) in warm water
  - Ensures shaved and tourniquet areas properly prepared

> **Clinical Pearl**
> • Tourniquet inflated with knee flexed → quadriceps length preserved
> • → easier dissection + true intraoperative patellar tracking

Draping Ask AI

- Cloth drapes, in order
  - **Screen** first
  - Big sheet under leg + equally big sheet over leg
  - Small sheets around tourniquet → one above, one below
  - Sterile **compression bandage** over leg → foot still palpable
    - Reason: **second metatarsal** palpated later for tibial rotation
  - **Split sheets** → one longitudinal, one transverse
- Split sheet → flex leg to desired position, then extend
  - → draping defects show on later flexion
- Wipe skin again with **alcohol** → connect suction + diathermy → ready
- **Disposable TKA sheets** → same method, easier, better

Surgeon, Assistants and Theatre Layout Ask AI

- Right-handed surgeon
  - Stands on **patient's right side** → right or left knee
- **Scrub nurse** → beside surgeon, on his right
- Second assistant → opposite
- Layout diagram: surgeon, nurse on his right, patient's foot, patient's head
- **First assistant** → opposite
- Company assistant → beside first assistant
- Extra assistant → surgeon's left, just before the screen
- Layout shows nurse's table + company table and instruments

Subvastus Approach and Joint Exposure Ask AI

Skin Incision Ask AI

- Lecturer's practice, Nasr City hospital
- **Anterior midline incision** → always, in primary TKA
  - Other incisions exist, advocated by some
- Made with knee flexed → skin stretched → **shorter incision** on closure
- Extent: 5 cm above upper patellar edge → 1 cm below tibial tuberosity
- Knee flexed → best, **recommended position** for skin incision

Subvastus Approach: Patient Selection Ask AI

Subvastus

- **Subvastus** in most cases → fat or thin
  - Excellent postoperative rehabilitation
  - Very little pain
  - Rapid mobilisation → patients up and moving pain-free
  - Very good results
- Exception: **muscular patient**
  - Difficult to slide patella with whole muscle
  - → larger incision, wider opening, more effort
  - Lecturer does not use it in muscular patients
- **Fatty patient** → bulk is fat outside muscle → muscle mobilised easily

Skin Flap and Deep Dissection Ask AI

- Skin elevated sharply, medially → **full-thickness flap**
  - → avoids skin necrosis
  - Especially full thickness on lateral side
- Otherwise → **black patch** in wound → sloughing → later problems
- Sharp medial dissection → end of **vastus medialis** → medial intermuscular septum
- At septum → open white **fascia** over quadriceps
- Peel quadriceps off **medial intermuscular septum** → proximally, as far as possible
  - **Bluntly with finger** → no sharp dissection; muscle slides off
- **Index finger** under muscle → already inside the knee
- Mark incision line
  - Scalpel or diathermy → **superficial line** 0.5 cm medial to patella
  - Leaves tissue cuff for later repair
  - Carried down gently beside patella
  - Superficial only → like a pen mark
- Confirm line correct + adequate distance from patella
  - Also adequate distance from **patellar tendon**
- Deepen → diathermy or scalpel
  - Line first → then completed to the end
- Muscle turned over with patella
  - **No eversion** of patella
  - Patellar tendon not rotated
  - Patella **slid into lateral gutter**
- → whole joint exposed
- Quadriceps, patella, tendon → **one piece** → easier postoperative rehabilitation

Clearing Anterior Femur and Fat Pad Ask AI

- Remove **suprapatellar synovium** → anterior femur, just above knee
  - Anterior surface needed later for femoral sizing → clean and clear
- Remove **patellar fat pad** (beneath patellar tendon) → fat cleared above and below
- Dissect **medial soft-tissue sleeve** beside patellar tendon → toward MCL
  - Not sharp → **periosteal elevator**, lifted as one sleeve
  - Limit: joint midline, 5–8 cm downwards → safe
  - No further release at this stage
- **Hohmann** placed in this area

Capsule, Menisci and Cruciates Ask AI

- Capsule still attached to **medial meniscus** above
- Anterior tip of medial meniscus → left on capsule, cut with it
- Capsule dissected off meniscus → medial capsule lifted to ~midline
- Meniscus left in place → **midline exposed**
- Remove visible obstructing parts now → menisci, **ACL**, PCL
  - Not far posteriorly → easier to retrieve later
- **PCL-sacrificing** procedure → daily practice; PCL not preserved
- Hohmann between patella and **distal femur**
  - Retracts patella + quadriceps
- Flex knee → patella slides into gutter, **not everted**
- → distal femur in view

Femoral Entry Point and Distal Femoral Cut Ask AI

Femoral Entry Point Ask AI

- **Intramedullary entry point** → whole operation built on it → place exactly
- First clear **intercondylar notch** osteophytes → locate entry accurately
- Remove medial and lateral osteophytes as far as possible
- → femoral condyles exposed
- Location, knee at 90° flexion
  - 8–10 mm above (anterior to) **PCL origin**
  - Slightly medial to apex of notch
- Roughly the point giving medullary entry
- **Enlarge** drill hole → later passage for IM rod
  - Not made exactly rod size
  - → rod engages canal, takes own correct position
  - → does not shift downward or backward
- Image: left femur → entry point + PCL origin
  - Just medial, 8–10 mm in front of PCL origin
  - Medial to centre of **intercondylar notch**
  - Osteophytes → removed first
- After drilling → **wash medullary canal** → prevents fat embolism on rod insertion

Valgus Angle Ask AI

- Small details → difference for the patient later
- Canal washed → valgus angle set → rod introduced

Primary case, no deformity	Valgus angle
Male	5°
Female	6°

- Males taller → **smaller valgus angle**; females shorter → larger
- Any male case → 5° preferred (lecturer's experience)
- Agrees with Dr. Mahmoud's lecture
- → **intramedullary rod** placed

Distal Femoral Cut Ask AI

Femur

- IM rod at set valgus angle
- **Distal femoral cutting jig** mounted → headless pins in "0" hole on top
- **Zero** → 9 mm resected from distal femur = implant thickness
- Jig pinned → **IM rod removed** → distal femoral cut
- Saw blades often not new → problems
- **Middle-entry technique**
  - Saw enters middle → to far end
  - Takes part of lateral + part of medial condyle → two equal pathways
  - Then complete laterally → then medially
  - → two sides equal
- Reused **saw blade** → flexible
  - → cuts medial less than lateral
  - Medial condyle more sclerotic than lateral
- Middle less sclerotic → entering there evens both sides
- After cut → remove jig, **leave headless pins**
  - Additional distal femoral cut may be needed
    - If extension gap tight
- Jig fixed with headless pins → does not move
  - Pins at zero → cut made
- **Double check**: measure resected piece → 8 or 9 mm → cut correct

Pitfall

- Reused flexible saw blade → medial condyle undercut
- Medial condyle more sclerotic → blade deflects
- Fix: enter middle first → complete lateral → complete medial

Proximal Tibial Cut Ask AI

Exposure and Extramedullary Guide Ask AI

Tibia

- After distal femur → straight to proximal tibia → **extension gap first**
- **Extramedullary device** → IM guide not used
- In flexion → Hohmanns to sublux tibia
  - Lateral side of tibial condyle
  - Posterior to tibia, in the notch
  - → tibia subluxed forward → whole condyle exposed
- Hohmann toward **MCL** → protects it during cut
- Zimmer device scale
  - Forward → **increases slope**
  - Backward → decreases slope
- Built-in 7° **posterior slope** → Zimmer fixed-bearing system
- Advantages of EM guide
  - Reduces **fat embolism** risk → tibial canal not opened
  - Bypasses tibial deformity
- Adjustment in two parts
  - 1) **Level** (depth of cut) → set first
    - **Stylus** → 2 mm and 10 mm settings
    - 2 mm on lowest point of tibial plateau → device locked
  - 2) **Rotation**
    - 7° slope + guide off midline/mechanical axis → posterior cut deformity → spoils everything *[C1 · transcript: "slope of 7 mm"]*
    - Align device with **tibial crest** → correct rotation
    - Also check junction of medial 1/3 and lateral 2/3 of **tibial tuberosity**
- Rotation confirmed → insert pins
- **Lateral pins first**
  - Lateral condyle softer
  - Medial condyle more sclerotic → pin slides, bends, causes problems
- **Screw** also fixes device
- **Angel wing** → preview of cut → extra verification
- Hohmann positions
  - Posterior → sublux tibia forward
  - Lateral → retract patella + patellar tendon
  - Medial → protect **MCL**
- Sequence: stylus on lowest point → fix device → rotation on tibial crest → fix jig

Making and Checking the Tibial Cut Ask AI

- Protector between **patellar tendon** and saw → always
  - Oscillating saw's side can cut tendon without entering it
- Cut complete to far end → remove jig, **leave pins** → re-cut or fine-tuning later
- Lift cut bone with **osteotome**
- Hold with company **grasper** (strong; used for headed pins) → pull
- **Diathermy** releases piece → medial to lateral
  - Posterior soft tissue released while pulling → whole piece out
- **Double check**: thickness of thick and thin parts of resected piece
- Bone out + knee **semi-flexed** → better posterior view
  - Remove medial/lateral meniscal remnants, PCL remnants, posterior debris
  - Easier at this point
- Cuts done → distal femoral + proximal tibial
- → assess **extension gap**

Extension Gap Balancing Ask AI

Finger Test Ask AI

- Knee fully extended → **assistant** extends + applies traction
- **Index finger** transversely in lateral and medial spaces
  - Tension felt → medial tighter than lateral? By how much?
  - Both tight → **bony cut** needed
  - Medial only tight → soft-tissue problem
- Finger feel differs from the **block** → lecturer's experience, useful with time

Extension Gap Block and Medial Release Ask AI

Release

- Usual method → **extension gap block** → does it enter?
  - 9 or 10 used most
- Enters laterally, not medially → **medial soft-tissue contracture** → release
- **Release sequence** → stepwise, re-examine after each
  - 1) Medial tibial condyle **osteophytes** → cause MCL tenting → removal lengthens MCL
  - 2) Still slightly tight → upper MCL + **pes anserinus** insertion with periosteum
    - Periosteal elevator, osteotome or curette
    - Widen → re-examine
  - 3) Still tight → posterior (deep) MCL → small portion in **posteromedial corner**
  - 4) More needed → **posteromedial capsule corner**
- Gradual, in order → until medial tension = lateral in extension gap
- **No bony cuts** at this stage
  - Exception: finger or smallest block won't enter → problem lies in cuts
- Re-cut → **femur**
  - Femur responsible for extension gap
  - Tibial re-cut → also alters flexion gap
- Result: extension gap **equal and rectangular**
- Image: extension gap block in place
- Best blocks → 9 and 10

- Block → final **insert** size
  - 10 enters → usually 12 insert
  - 9 enters → usually 10 insert
- **Rotation** check with block in position → rod shows axis + tibial rotation
  - Diathermy mark possible

Must-Know

- Medial release: osteophytes → upper MCL + pes → deep MCL → posteromedial capsule
- Re-examine after every step
- Smallest block won't enter → cut problem, not balance
- Re-cut the femur → femur controls extension gap

Femoral Sizing, Rotation and Finishing Cuts Ask AI

Femoral Sizing Ask AI

- After extension gap → back to femur → **femoral component size**
- NexGen (hospital) → size by **AP diameter**, not mediolateral
- ML dimension added in later designs, e.g. **Gender knee** → width varies within a size
  - e.g. size D → D1, D2 … → larger ML size
- **Posterior condylar referencing** → device flush on posterior condyles
- Varus knee → medial condyle eroded/**hypoplastic** → posterior referencing problem
- Standard: flush with posterior condyles, resting on **distal cut**
  - **Stylus** on highest point of anterior femur → measures distance
- Exact fit → that size
- **Between two sizes** → larger, in most cases
  - Larger → can be downsized later
  - Too much bone resected → cannot be undone
  - → larger or nearest size

Femoral Rotation Ask AI

Femur

- Rotation device within sizing device → **external rotation** of femoral component
  - → rectangular flexion gap
  - → more bone from medial posterior condyle
  - → patellar tracking
- 3° → routine in **primary straightforward TKA**
  - With posterior condylar referencing → no major change needed
- Other rotation references
  - **Whiteside line**
  - Transepicondylar axis
  - Tibial cut
- Several references together → confirm 3° → greater **accuracy**
- Stylus on distal femur, over posterior condyles → 3° jig → pins
- **Angel wing** → check anterior cut
- No **notching** → step in anterior femoral surface → weakens it
  - → postoperative **periprosthetic fracture**, sometimes intraoperative
- Angel wing → posterior condylar resection → more medial than lateral
- Posterior resection 8 or 9 mm → suits joint, work correct
- Size → communicated to company
- **External rotation jig** placed + confirmed
  - With reference lines + built-in posterior condylar referencing
- Pins inserted → all removed → jig completing distal femoral cuts placed
- Image (right knee): tibial cut, posterior condylar line, transepicondylar line, 3°
  - → rectangular cut → **balanced flexion gap**
- Finishing jig mounted on these two pins

Positioning the Finishing Block Ask AI

- Central, medial or lateral?
  - Goal → correct **patellar tracking**
  - Not pushed to an extreme → **lateralised**
    - As far as possible
- **Two lines** on top → patellar component position/track
  - → block as lateral as possible → fix
- **Medial pins first**
  - Medial pin hammered → jig shifts laterally → desired direction
  - Lateral pin first → pushes jig → alters intended position

Checking the Anterior and Posterior Cuts Ask AI

- **Angel wing** → check anterior + posterior cuts and resection amounts
- Notching, or less anterior/more posterior condylar resection → **entry point** wrong from start
- After angel wing check → **remove pins**
  - Forgotten → saw hits them during chamfer cuts
- Block fixed by its **spikes** (pegs) + headed pins
- Anterior cut → **piano sign** (Dr. Mahmoud's lecture)
- Some assess **flexion gap** after anterior + posterior cuts
  - Before chamfers → size change or correction still possible
  - Lecturer → completes cuts first → flexion gap after

Order of Femoral Cuts Ask AI

Femur

- Pins removed → order
  - 1) **Anterior cut**
  - 2) Posterior cut
  - 3) Anterior chamfer
  - 4) Posterior chamfer
- Posterior chamfer → protect **tibia** from descending saw + MCL medially

Trochlear Recess Ask AI

- **Anterior chamfer bone** → kept, folded, set aside for later use
- **Recess** cut → seat of femoral component trochlea
- Two lines (patellar track) → correspond to component **trochlea**
- Lines scored with dedicated saw
- Saw unavailable in hospital → **osteotome** + oscillating saw (not reciprocating)
- Remove block → finish with osteotome + **file** → smooth

Box (Notch) Cut Ask AI

PS

- **Notch** block → PCL-sacrificing design only
  - CR design → no notch → more bone preserved (CR **privilege**)
  - Lecturer → PCL-sacrificing in all cases
- Jig **spikes** → into previous spike holes + screws
  - Flush with anterior surface + distal cut
- Headed pins → saw medial, lateral, then central → **bone block** out
  - → seat of **cam** engaging the post
    - **Posterior-stabilised fixed-bearing** design
- Real-case images: jig, protruding pins for removal, screws, angel wing, notch jig
- Anterior chamfer bone → **plug femoral canal** (medullary hole)
  - Two pieces impacted into hole opened at start
  - → **bone bridge** + bone stock between the two condyles
  - Middle otherwise empty, weak → fractures easily
  - → prevents medullary bleeding after tourniquet release
- Image: **notch cutting guide** → notch done

Posterior Capsule Release and Flexion Gap Ask AI

Posterior Capsule Release Ask AI

- After femoral cuts → **posterior capsule release** → very important
- Tight posterior structures → **flexion contracture** (extension lag) → cannot extend
- **Curved** instrument → widen posterior area
- Remove **posterior osteophytes**
- Careful entry → watch vessels
  - → **neurovascular bundle**
- **PCL remnants** act here → removal reduces tightness
- Done before assessment
- Remove meniscal remnants + tight capsule → without injuring anything

Flexion Gap Assessment Ask AI

- Textbook image → before chamfers; here → **after chamfers**
- Check **balanced** medially and laterally → further releases?
- Same block as extension gap? → flexion gap = extension gap?
- Mismatches (tight flexion/loose extension, reverse, both) → separate lecture
- Normally → gaps equal; small difference → no difficulty
- Flexion slightly tighter than extension → does not matter much
- Key → **balanced** medial to lateral
  - One-sided stress → increased **wear**
- Assessed at 90° → index finger usable too

Trial Components and Tibial Preparation Ask AI

Femoral Trial Ask AI

- All femoral cuts done → **femoral trial component**
- Confirm with company rep → size requested, size handed
  - **Left or right** → mistakes do happen
- Never placed and hammered at once
- Introduce gently **by hand** onto pegs → peg holes in bone, pegs on trial
- Key area → middle, the **notch**
- Finger in middle → feel medial/lateral bone blocking seating
- **Bony incongruity** → clear with saw before hammering
- Hammered anyway → **lateral condyle fracture**
  - Component lateralised → toward lateral side
  - Lateral condyle smaller than medial
  - Has happened, happens, will happen
- Always feel around notch by hand → excess bone blocking trial?
  - Check above and below → never hammer while sensing a problem
- Seat by hand beyond bone → hammer **gently**
  - Bony problem → pushed aside, enters evenly, no fracture
- **Fractures** common → feel for excess bone first
- Impact with **pusher** → assess
  - Lateral side
  - Medial side
  - **Overhanging**
  - Patella sitting comfortably
- Some leave trial while doing tibia → lecturer: test, clean, remove

Pitfall

- Femoral trial hammered over bony incongruity → lateral condyle fracture
- Seat by hand first → feel the notch → hammer gently
- Posterior Hohmann drifting laterally → tibial condyle fracture

Hohmann Placement for the Tibia Ask AI

- **Assistant's** responsibility
- Hohmanns → posterior
  - Lateral + medial → tibial condyles fully visible
- **Posterior Hohmann** → common assistant disaster
  - Drifts laterally → condyle injury, sometimes fracture
- Keep **central**, wedged → neither medial nor lateral
- Condyles in most females → **osteoporotic**
- Box cut weakens area → never lever against condyles
  - → **fracture**, or fragment pushed in (shown)
- Image: anterior chamfer bone plugging canal → less **bleeding**, bone stock
- Plug after removing trial
  - Trial left in → crowded field, difficult work
- Trial removed → **retractors** placed with care

Tibial Sizing and Rotation Ask AI

Tibia

- Knee flexed, Hohmanns set → field ready
- Tibial sizes matching femur, e.g. 4 or 5 → trial both
  - → which gives **tibial plateau coverage**
- **Tibial plate** (template) placed
  - Base for subsequent cuts and steps
- Usually two matching sizes (one above, one below) → choose appropriate
- **Lateral side** placed first and set
  - Medial matters less → bone adjustable later
- **Rotational alignment** → three methods
  - 1) Coverage → **curve on curve**
  - 2) Plate midline → junction of medial 1/3 and lateral 2/3 of **tibial tuberosity**
    - Marked at start or now
  - 3) **Extramedullary rod** → hole in long handle
    - Projection → in line with tibial crest → onto **second metatarsal**
- **Foot** position must be good → sometimes rotated outward
  - Or inward → correct → all methods must confirm alignment
- Mobile bearing → one extra, most important step (below)
- Satisfied → **pins** → lateral side first
  - Softer → pins enter easily
  - Medial first → hammering moves plate → position changes
  - Two lateral pins → confirm position → hammer medial freely

High Yield

- Tibial rotation 1 → curve on curve coverage
- Tibial rotation 2 → midline at medial 1/3–lateral 2/3 of tuberosity
- Tibial rotation 3 → EM rod along tibial crest to second metatarsal
- Foot must not be rotated during the check

Mobile-Bearing Rotation Check Ask AI

Mobile

- Extra move → chosen **insert** placed after plate
- Femoral + tibial trials + insert → **full extension** → rod in handle → check rotation
- Mobile bearing → **insert follows femur**, independent of tibia
- Tibia rotated unlike femur → insert ends 15° rotated to tibial plate
  - → **prominent** → later arthrofibrosis, many problems
- Mobile bearing not currently used → mentioned and shown only
- Full video with Dr. Mohamed planned

> **Exam Pearl**
> • Mobile bearing → insert follows femur, not tibia
> • Tibial rotation mismatch → insert 15° rotated → prominent → arthrofibrosis

Tibial Keel Preparation Ask AI

- Pins → lateral first → **sclerotic medial**
- **Central drill guide** → keel preparation
- Drill in **reverse**
  - Most patients osteoporotic → weak bone
  - Reverse → compacts surrounding bone, does not drill
- **Punch** → keel area → preparation complete
- Image: **tibial plate** and drill

Full Trial and Insert Size Ask AI

- Femur + tibia ready → **full trial**
- Tibial trial size known → try **insert** sizes (plastic)
- Gap block used at start → approximate guide
  - Block 10 → 10 or 12 insert, and so on

Insert needed	Interpretation
10 and 12	Ideal → cuts, balance, releases restored normal anatomy + joint line
9 (10 won't enter)	Borderline → tight gaps → cut problem or release problem
14	Some excess (laxity)
17 or above	One gap larger → compensation; or very poor cuts, compensated through insert

- Size 17 → mounts on tray + fixed with **screw**
  - Smaller sizes → mount directly on **tray**
  - Extra company step → warn company in advance
- Master the **system** in use

Implantation and Closure Ask AI

Opening the Implants Ask AI

- Components and trials checked → company opens specific sizes
- While company opens → trials removed
- Removing **tibial tray** pins → canal open → pins may fall in
  - Cover canal with **finger or gauze** → nothing falls in
  - Pin in canal → disaster
- Image: tuberosity mark (medial 1/3–lateral 2/3) aligned with tray middle

Patella Ask AI

Patella

- Routine → **no patellar button** (no resurfacing)
- Timing → cuts done, wide field, patella easy to evert while company prepares
- Remove **patellar osteophytes** all round
- Clean + **reshape** articular surface
- Diathermy held transversely → around whole patella
  - **Denervation** ("circumcision") → less postoperative anterior knee pain

Lavage and Local Tranexamic Acid Ask AI

TXA

- During implant opening → very thorough **wash**
  - Cancellous bone → clear, white
  - Pat dry with towel → completely dry field
- Extra step → another 1 g **Cyklokapron**
- Two 60 cc syringes with needles → each 1 g Cyklokapron in **saline**
- **First syringe** → injection
  - Posterior capsule medially + laterally → not middle (neurovascular bundle)
  - Medial + lateral gutters, soft tissues
- **Second syringe** → intra-articular irrigation
  - Left ~2 minutes while company mixes **cement**
  - Then suctioned → field dried very thoroughly

Cyklokapron dose	Route	Timing
1 g	IV	At induction, before tourniquet inflation
1 g	Local soft-tissue injection	After lavage
1 g	Intra-articular irrigation	~2 minutes, then suctioned

Cementation Ask AI

Cement

- Dry field → implantation **in flexion** → tibia first
- Hohmanns → posterior, lateral, medial → field visible
- **Tibial component**
  - Cement → tray undersurface + keel
  - Seat gently by hand → confirm position → **impactor**
  - Remove excess cement all around, including posteriorly
- Tibia fully seated → **femoral component** → by hand first
  - Pegs engaged first → confirm seating on cuts → impactor
  - Remove cement medially, laterally + in **intercondylar notch**
- No excess cement in notch → insert **post** sits there
  - Cement contact → **early loosening**
- Trial insert → **same size as trial**, not upsized
  - Some upsize for impaction → not done here
- Trial insert in → **full extension** with pressure
  - Held until cement sets
  - Role of the **assistant** here
- Cement set → examine with trial insert
  - **Varus-valgus stability** medially and laterally
  - Ligaments
  - Flexion and extension
  - Patella in place + tracking
  - Muscle intact
- In flexion → varus-valgus stability, releases, collaterals checked
- All fine → company opens **insert** of that size → trial removed

Pitfall

- Excess cement in intercondylar notch → contacts insert post → early loosening
- Tibial tray pins fall into open canal → cover with finger or gauze

Final Insert Ask AI

- Hohmanns again → posterior, medial, lateral
  - Posterior Hohmann → **gauze** over components → no damage
- Wash + dry **tray** very thoroughly
- Insert tilted posteriorly → gun-like pusher → **click** → locked
- Reduce → re-examine collaterals + **patellar tracking**
  - Flexion + extension

Wound Closure Ask AI

- Layer by layer, inside out → **knee flexed** ~70°
- Closed in extension → easier, but **stitches** tight on flexion
  - Closed at 70° → easier flexion and recovery
- **Drains** → two
  - Intra-articular, under quadriceps → exits proximally
  - Medial gutter (elevation off medial intermuscular septum) → exits proximally
- Why proximal exit? Distal exit → removal as patient mobilises → blood + **serous fluid** leak
  - → patient anxious, thinks knee is bleeding
- Soft tissue closed
- Subvastus preserved quadriceps → **medial tissue sleeve** visible
  - Reattached at this angle with rest of patellar tendon
  - Muscle completely intact → simple closure → easy recovery
- End of presentation → questions

Questions and Answers: Bleeding, Bone Cuts and PCL Ask AI

Drains Ask AI

- Q: Good haemostasis → skip drains?
  - **Drains in all cases**
  - Hip → sometimes no drain; knee → never tried without
  - Knee → **bloody operation** → bleeds after tourniquet release + haemostasis
  - Bleeding → joint + wound tension → less movement → avoidable problems
  - Cyklokapron (1 g induction, 1 g local, 1 g irrigation) → drain output 300–400 mL
  - Without Cyklokapron → up to 1 litre
- Q: Removal timing? Clamped at night?
  - Routine **removal** at 48 hours
  - Still heavy bleeding at 48 h (does not happen) → kept → avoids **haemarthrosis**
- Q: Vacuum with closure?
  - Formerly → drain **filter** collected blood for reinfusion → excellent
  - Now → drain on its own **suction** → removed at 48 hours
  - With Cyklokapron → very good results

High Yield

- Cyklokapron ×3: 1 g IV induction, 1 g local, 1 g irrigation
- Drain output → 300–400 mL vs 1 litre without
- Drains always in knee → removed at 48 hours

Tibial Cementation and Overhang Ask AI

Tibia

- Q: Tibial cement → base only, or keel too?
  - **Base + proximal keel** → not whole keel
  - Side **flanges** in soft bone → malposition (tracking) + rotation on insertion
  - Impacted cement → spreads downward
- Q: Tibial overhang acceptable?
  - **Medial overhang** → tenting, as with osteophytes → spoils release
  - Continuous **soft-tissue irritation**
    - → reduced ROM, **arthrofibrosis**, severe joint tension
    - → good cuts, balance, implantation, yet very poor mobility
  - Tibia → **overhang completely unacceptable**
  - Femur → not acceptable either, but may be tolerated → no soft tissue interferes

Distal Femoral Recut Ask AI

Recut

- Q: Distal femoral jig on plus or minus → when?
  - Distal femoral cut → affects **extension gap**
  - Distal femoral + tibial cuts done
  - Block very tight → not even smallest (9) enters
  - → cut error, not balance → no bilateral releases
  - **Homogeneous tightness** in extension gap → bone cut incorrect
  - Pins left at zero → +2 or +4 → more distal femur → wider extension gap

Why Not Complete the Femoral Cuts First? Ask AI

- Q: Complete femoral cuts before tibial cut and gap assessment?
  - Frequently asked
  - **Extension gap** tested first → balance + soft tissue → then more bony cuts
  - Size taken + bone removed → irreversible; not easily adjusted
  - One of the available methods
    - Gaps completed and assessed → direction seen → later problems accounted for
  - Some complete first → may not know how to manage later difficulties

Osteophyte Removal Ask AI

- Q: Femoral osteophytes → before or after cutting?
  - Femur → intercondylar notch (entry hole) + medial and lateral condyles
    - → actual femoral size seen → no **overhanging**
  - Sizing + ML positioning with osteophytes present → overhang after removal
  - Removal → **real ML size** of femur
  - Tibia → osteophytes must go → **MCL tenting**
    - Also → set flexion and extension gaps

Tranexamic Acid and the Drain Ask AI

TXA

- Q: Inject TXA into the drain? → **No**
- Q: Drain clamped then opened, or left open?
  - Induction dose → **IV**
  - Local dose → soft-tissue injection, not fluid in joint → not sucked out by drain
  - Like **local anaesthesia** infiltration
    - Posterior capsule medially + laterally → away from neurovascular bundle
    - Medial + lateral gutters → soft tissue around bony cuts
  - **Irrigation** → solution covers bony cuts while company opens → suctioned out
    - Not left in → no concern about loss through drain

Hypoplastic Condyle and Fat Embolism Ask AI

- **Hypoplastic posteromedial condyle** → upcoming lectures
- Q: Fat embolism and IM rod
  - Rod inserted immediately after opening canal → ↑ **intramedullary pressure**
  - Before rod → **suction and irrigation** → vigorous flush
    - Fat globules seen coming out
    - → no fat remains → rod does not raise fat embolism risk

PCL Sacrifice versus Retention Ask AI

PCL

- Egypt → most cases **late**
  - Few patients with sport/self-care culture presenting at early OA
  - Patient comes reluctantly, sometimes under pressure → knee fully **degenerative**
- PCL policy → lecturer's own **philosophy**, not a general rule
- Two grounds
  - Degenerate knee → PCL healthy enough to rely on for **stabilisation**?
    - Later rupture → whole procedure fails
  - Will it withstand the knee afterwards?
- Pro-retention → intact PCL **replicates anatomy** and biomechanics → keep it
- Contra → diseased PCL → knee works normally without it
- Lecturer → **sacrificing** route
  - Worked very well for him and others → no problems
  - Patients late → **soft tissues** in poor condition
- Personal philosophy → both approaches correct

Questions and Answers: Aftercare, Patella and Pin Order Ask AI

Postoperative Rehabilitation Ask AI

Rehab

- Q: Maximum postoperative flexion? Postoperative protocol?
  - Immediate weight bearing with **walker**
  - Subvastus → **pain-free** movement
    - vs medial parapatellar/quadriceps-cutting approaches → large difference in pain + movement
    - Pain varies between patients
  - Walks from second day → own legs, walker
  - Walker → two weeks
  - Stitch removal + wound healing → **physiotherapy** (centre or physiotherapist)
  - First two weeks → **nothing prohibited**
    - No flexion restriction
    - Walker only → balance + stability not yet adjusted

Night Splint for Flexion Deformity Ask AI

- Preoperative flexion deformity → **night splint** strongly preferred
  - Corrected intraoperatively, but **body image** retains it
  - → at night knee returns to slightly flexed resting position
- Alternative → **pillow** under leg (not under knee) → knee fully extended *[C2 · transcript: "pillow under the knee"]*

Tibial Stylus with a Large Defect Ask AI

Tibia

- Large **defect** → augment anticipated medially
  - Varus knee → rely on **lateral condyle** → 10 mm stylus on high side
  - Stylus on lowest point of defective medial condyle → very large proximal tibial cut
- Rule → reference the **normal side** when augment or large medial defect

Patellar Resurfacing Ask AI

Patella

- Q: Patellar components in primary cases → role?
  - Many surgeons use **patellar button** → most in Egypt don't
  - Lecturer → never used; no patellar revision or procedure so far
  - Button → reduces **maltracking** rate
    - e.g. patella dislocates on flexion, rotation problem, medial muscles not tracking
    - → button placed **medially and superiorly** → solves problem
  - **Anterior knee pain** without resurfacing → not seen so far

Femoral Notching Ask AI

Notching

- Q: Notch found on femoral trial insertion → management?
  - Anterior **notching** → apparent during the cut → don't wait
  - Anterior cut → watch saw path
  - About to notch at exit → stop, withdraw saw
    - Complete remaining cuts
    - Then make the **step** deliberately → osteotome or saw
  - Notching occurred → some add **stem** → fracture risk high
    - Or redo everything → **stemmed femoral component**

Tourniquet Release Ask AI

- Q: Deflate before or after cement? Gradual or at once? After closure?
  - After **cementation** → deflate
    - Bleeding → barrier between cement + cancellous bone → poor adherence
  - **Gradually** → significant tourniquet hyperaemia
    - Many gauzes on wound + hand compression
    - Release gradually → fully off
    - Lift hand, remove gauzes → inspect → **haemostasis** as needed

Tranexamic Acid and Anticoagulation Ask AI

TXA

- Q: Anticoagulant/Clexane stopped preop + Cyklokapron → thrombotic events?
  - **No** → not seen
  - Usual **anticoagulant protocol** → stopped 6 hours before surgery → restarted immediately after *[Q1 · as spoken; conflicts with one-week stop earlier; LMWH usually stopped 12-24 h preop]*
  - Cyklokapron → acts only during surgery and just after
    - Not repeated or continued
    - Stops bleeding → does not increase **coagulation**

Order of Component Cementation Ask AI

Cement

- Q: Femur first or tibia first?
  - Femur first → blocks view beneath → **crowded field**
  - Tray placement, insert, **curve on curve** check → very difficult
  - Needs flexion beyond 110° → tibia clear of femur → very difficult
  - Everyone → **tibia first**, then femur
  - Femur in place → thigh elevated → tibia implantable even when tight
  - Femur in place with tibia beneath → sometimes impossible
    - Especially **obese** patient → posterior fat, large leg → insufficient flexion to clear tibia

Order of Pin Insertion Ask AI

Pins

- Q: Lateral headless pin always first, except posterior femur?
  - Exception = **femoral finishing block** → medial pin first
  - Lateral pin hammered → shifts block slightly
  - Block deliberately **lateralised** for patellar tracking
    - Pushed laterally → fine
    - Pushed medially → spoils patellar tracking
  - Tibial tray → medial **sclerotic** (late presentation)
    - Company short pins weak → deviate → change tibial rotation
    - Or pin bends → lifts tibial component unevenly when fully hammered
  - Easiest method found → no problems at all
  - Lateral first → always **soft** → pin enters quickly
  - At least two lateral → prevent rotation, set component → medial hammered comfortably
  - Same for the **proximal tibial jig**
    - Medial sclerotic, lateral not
    - Medial pins deflect, not **pre-drilled** → hammering lifts whole jig out
    - Other way → sometimes causes problems

Instrument	Pins first	Reason
Proximal tibial cutting jig	Lateral	Sclerotic medial → pin slides, bends, deflects, lifts jig
Tibial plate/tray	Lateral (at least two)	Soft → quick entry; medial first → plate moves, rotation changes
Femoral finishing block	Medial	Hammering pushes block laterally → desired lateralisation

Lateral Release for Patellar Tracking Ask AI

- Q: Tracking problems needing release → how?
  - Sometimes → all fine except patella tracks poorly on flexion
    - Cannot flex fully, or marked **tension** in full flexion
  - Standard **lateral release** of retinaculum → muscle not cut
    - Start with perforations → tattooing (pie-crusting)
    - Responds → done; if not → incision extended, unsutured → medial side comes along

Questions and Answers: Tourniquet, Bearings and Approach Ask AI

Operating Without a Tourniquet Ask AI

- Q: Newer papers → no tourniquet?
  - Very obese, tourniquet impossible → operate **without tourniquet**
  - Some inflate only at **cementation**
    - Not available in Egypt → no such sterile tourniquet
    - Only sterile Esmarch + compression at cementation
    - Nearly impossible here
  - "Shawarma" thigh → tourniquet impossible or reaches patella → operate without
    - Very troublesome, especially at **cementation**

Cement Type Ask AI

- Q: Antibiotic or plain cement?
  - Lecturer → **plain cement** in ordinary circumstances
  - Many prefer antibiotic-loaded cement → prophylaxis
  - No intra-articular antibiotic either

Order of Femoral Cuts: Zimmer versus Lecturer Ask AI

Femur

- Q: Attendee → Zimmer recommends a different order

Step	Lecturer	Zimmer (per attendee)
1	Anterior cut	Anterior cut
2	Posterior cut	Anterior chamfer
3	Anterior chamfer	Posterior cut
4	Posterior chamfer	Posterior chamfer

- Different schools → lecturer's order gave good results so far
- Chamfer order difference → matters little afterwards
- Many **NexGen** models → refers to fixed-bearing model in Egypt/hospital
  - Later models → jigs changed shape, differ from description

Postoperative Click Ask AI

- Q: Postoperative click?
  - Possible → especially **patella** clicking on anterior femoral component
  - **Mobile bearing** > fixed → insert very free, tibial rotation poorly set
    - Sound at end of movement

Mobile versus Fixed Bearing Ask AI

Bearing

- Q: Difference?
  - **Fixed bearing** → insert fixed on tibial tray, does not move
  - Mobile bearing → formerly for **younger patients**
    - Insert mobility → bearing point shifts → less **wear**, longer life
  - Later: correctly placed fixed bearing → lasts as long
    - Balanced ligaments medially + laterally, correct insert pressure
  - Difference → only in **implantation technique**
- Rotating platform/mobile bearing → tray in correct tibial rotation, insert placed after
  - → insert **follows femur** and its rotation, not tibia
  - → does not bring tibia under femur
  - Insert + femur → one unit; tibia in a different position
- Hence extra step → **tray, insert, femur** assembled together
  - Rotation set together on tibia
  - Diathermy mark first → tray placed on mark, not tibial tuberosity
  - Tuberosity mark (medial 1/3–lateral 2/3) not required → components aligned together
- To be shown on video

	Fixed bearing	Mobile bearing
Insert	Fixed on tray, no motion	Follows femur and its rotation
Rationale	Correct placement + balanced ligaments → same longevity	Shifting bearing point → less wear
Tibial rotation reference	Curve on curve, tuberosity line, EM rod	Tray + insert + femur assembled; diathermy mark
Click	Less	More, if insert very free and tibial rotation poorly set

Medial Release Ask AI

Release

- Q: Medial release needed? → **nearly all cases**
  - 1) Medial **osteophytes**
  - 2) **MCL** released from joint line toward insertion, top-down
    - Not exceeding 8 or 10 cm → avoid complete detachment; ~8 cm
    - Part of pes anserinus may be released
  - 3) Tension remains → **deep MCL**
    - In posteromedial corner → released
  - 4) Still tense → whole **capsular corner** peeled off bone
    - → tibia externally rotates
- TKA → very large topic; today → **non-complex primary**; next → complications

Postoperative Anticoagulation Ask AI

- Q: Postoperative anticoagulant?
  - 14 days in TKA → extended if additional **risk factors**
  - In hospital → **injection**; after discharge → oral

Bilateral TKA Ask AI

- Q: Bilateral in the same session?
  - Lecturer → 3 times so far
  - Single surgeon + one assistant → **exhausting**
  - **Team** on both sides simultaneously, sets alternated → saves time + effort
  - Alone → about double the time → burden on surgeon and patient
    - **Intensive care** admission
    - Very high postoperative bleeding
    - **Ultra-major trauma** → but feasible

Advantages of the Subvastus Approach Ask AI

Subvastus

- Q: Advantages over the usual approach?
  - **Quadriceps** and extensor mechanism untouched
    - → remains in one piece
  - Postoperatively → better movement, faster rehab, less pain, less **bleeding**
  - **Medial parapatellar** → easier, much wider exposure
    - But patient cannot/afraid to flex → feels stitches will tear
  - Subvastus → much easier for patient and surgeon postoperatively
  - Other approaches → all described, used without problems
  - Lecturer's preference → very good results

	Subvastus	Medial parapatellar
Extensor mechanism	Untouched, one piece	—
Exposure	Harder in muscular patients	Much wider, easier
Postoperative flexion	Pain-free, nothing prohibited	Cannot/afraid to flex; stitches feel tight
Recovery	Faster rehab, less pain, less bleeding	—

Click and Patellar Maltracking Ask AI

- Q: Postoperative click → patella-related? Management?
  - **Maltracking** → muscle strengthening + physiotherapy
  - Conservative failure → re-exploration ± **patellar button**

Full Flexion Ask AI

- Q: When is full flexion allowed?
  - **No flexion restriction**
  - Full flexion not reached immediately → nothing prohibited in positioning
  - Whatever the patient can do → he does
  - Only requirement → walker, later combined with physiotherapy
  - Approach keeps **muscle intact** → no problem